MAMMA ADA CENTRE

In Collaboration with FACM ITALY

UG 2nd Year

Learn Build Apply Innovate

GAIO

Global AI Olympiad

Think | Explore | Create | A Better Tomorrow

From Data to Decisions for a Smarter Future

6 KEY LEARNING AREAS

1. **Learning from Data**
   - Making Machines Smarter
   - Data to Intelligence
2. **Program with AI**
   - Data into Decisions
   - Build with Python
   - Prompt with Purpose
3. **AI for Smarter Business**
   - AI for Engineering Solutions
   - Real-world Applications
4. **Pick Your AI Path**
   - Portfolio to Profession
   - Career Opportunities
   - Industry Readiness
5. **AI Data Studio**
   - Smart Automation Tools
   - Hands-on Projects
6. **Fair AI for the Future**
   - Ownership in the AI Age
   - Ethics • Privacy
   - Social Impact

Knowledge | Innovation | Problem Solving | Collaboration | Global Perspective | Responsible AI

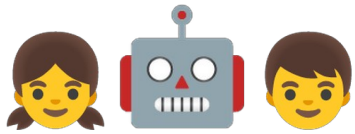
This book belongs to

Name: _____

Class & Section: _____

School: _____

📖 PREFACE



AI is transforming every profession, from engineering and medicine to commerce, law and media. As a university student, you must understand how AI works, judge its outputs critically and apply it responsibly in your discipline. This book builds exactly those capabilities.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of UG 2nd Year students, with university-level depth.

Every topic opens with an in-depth "Concept Explained" chapter, followed by a Concept Deep Dive, an AI Story, a case study, a lab activity and a Python Code Lab, with worksheets, quizzes, challenge problems and university-style exam preparation.

Responsibility runs through every chapter: you will examine AI's limits, bias, errors, privacy risks and impact on work, and the ethical duties of everyone who builds or uses it.

🤝 A note for teachers and parents

Faculty can use the Concept Explained chapters for lectures, the Code Labs and lab activities for practical sessions, and the exam-prep pages for assessment. Encourage students to verify AI outputs, cite sources and reflect on ethics. An answer key is provided at the end of the book.

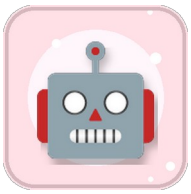
We hope this book helps you think critically, build confidently and lead responsibly in an AI-powered world. Your ideas today can change tomorrow!

With best wishes,

The GAIO Academic Team

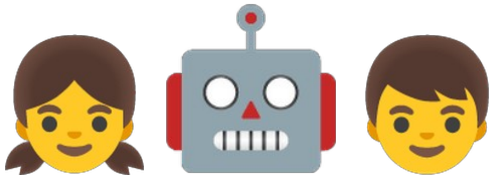
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Innovators!



MEET BOLT!

Welcome to university-level AI! I am Bolt, your AI guide. This year you will write real Python, use AI as a coding partner, study AI in your profession and practise exam-style answers. Let's build your AI edge!



Look for these pictures in your book:

CONCEPT EXPLAINED Concept explained in depth

CONCEPT DEEP DIVE Detailed explanation & code

SCENARIO Real-life scenario

CASE STUDY Analyse a real case

LAB ACTIVITY Hands-on lab

WORKSHEET Practice exercises

PUZZLE & DEFINE Puzzles & definitions

QUIZ TIME MCQ practice

AI STORY An AI story to think about

CHALLENGE PROBLEMS Higher-order thinking

CODE LAB Python code lab

EXAM PREP University-style exam prep

My 6-Month AI Journey

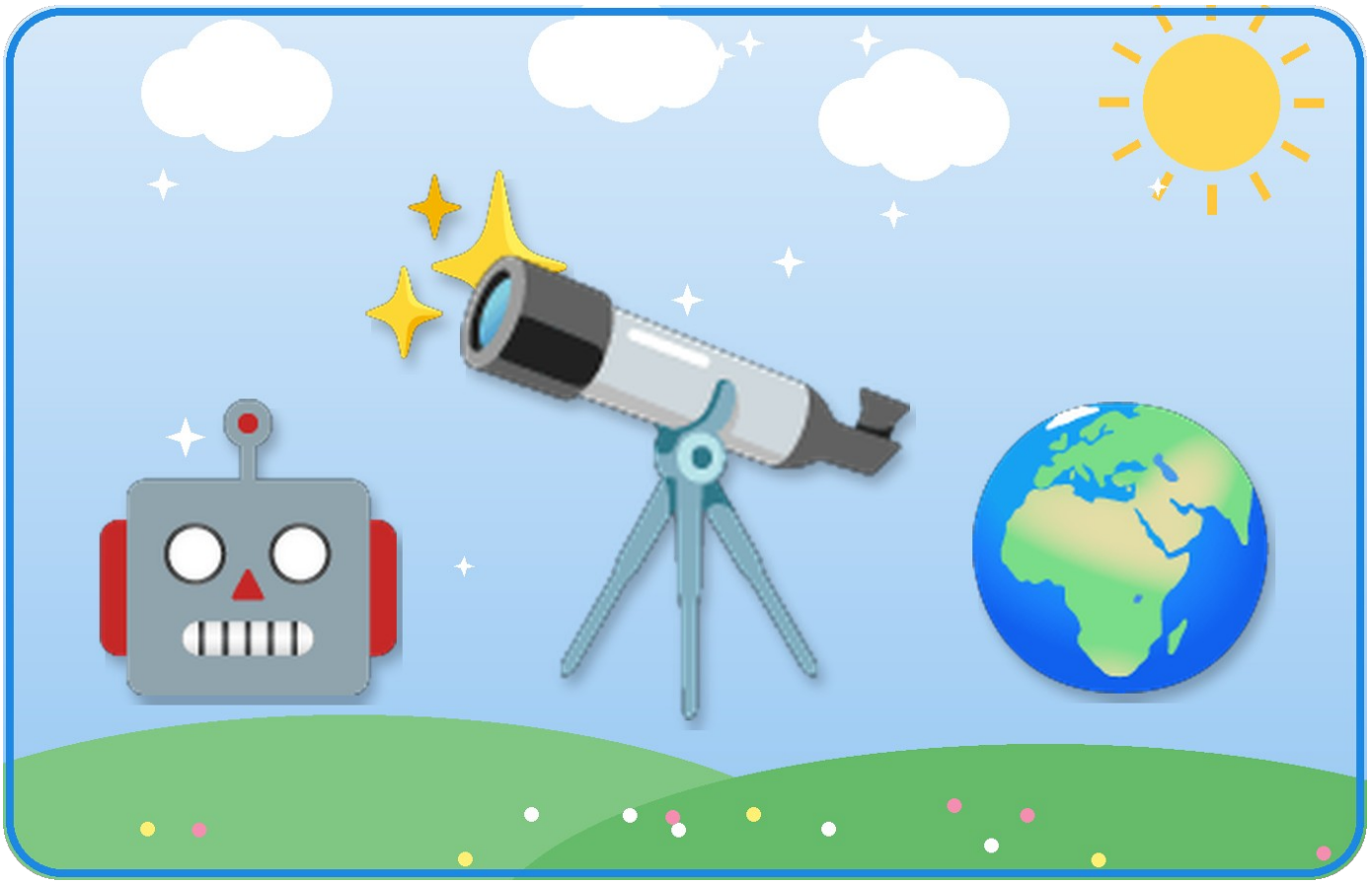
Follow the path. One month at a time!

MONTH 1 **AI DISCOVER** AI Basics	**Learning from Data • Making Machines Smarter**
MONTH 2 **AI CONNECT** Communicate with AI	**Program with AI • Data into Decisions**
MONTH 3 **AI SOLVE** AI Applications	**AI for Smarter Business • AI for Engineering Solutions**
MONTH 4 **AI RISE** AI & Career	**Pick Your AI Path • Portfolio to Profession**
MONTH 5 **AI CREATE** AI Tools	**AI Data Studio • Smart Automation Tools**
MONTH 6 **AI CARE** Responsible AI	**Fair AI, Fair Future • Ownership in the AI Age**

MONTH 1

AI DISCOVER

See • Understand • Explore



Area: AI Basics

In this unit, you will explore:

Topic 1

Learning from Data

Topic 2

Making Machines Smarter

TOPIC 1 • MONTH 1 • AI BASICS

Learning from Data

LET'S LEARN

- Good models start with good data and good questions.
- Explore first: distributions, outliers, correlations.
- Bias–variance trade-off governs generalisation.

The data-to-model pipeline

Collect

Clean

Explore (EDA)

Model

KEY TERMS

EDA Exploratory data analysis

Bias–variance Underfit vs overfit

CV Cross-validation folds

📖 CONCEPT EXPLAINED: Learning from Data

💡 In this chapter

In your first year you trained simple models. In the second year the focus shifts to a harder truth: a model is only as good as the data and the evaluation behind it. This chapter teaches you how professionals explore data, prepare it carefully, understand the bias–variance trade-off, and validate models so that their results can be trusted.

🔍 1. Exploratory Data Analysis (EDA)

Before building any model, analysts explore the data to understand its shape and its problems. EDA begins with basic questions: How many rows and columns are there? What type is each column? How many values are missing? Are there duplicate records? What are the minimum, maximum, mean and median of each numeric feature?

Next come visual checks: histograms show distributions, box plots reveal outliers, scatter plots and correlation heatmaps show relationships. EDA often uncovers surprises, such as ages recorded as 250, a 'gender' column with eight spellings, or a feature that suspiciously predicts the target perfectly. Finding these early saves weeks of confused modelling later.

🖐 TRY IT

For any dataset you have, write down five EDA questions you would ask before modelling.

🧹 2. Cleaning and preparing data

Missing values can be removed (if few and random), filled with the mean or median (simple imputation), filled with the most frequent category, or predicted from other features. Each choice changes the data, so record what you did. Adding a 'was_missing' indicator column can preserve useful information about missingness.

Categorical features must be encoded as numbers, commonly with one-hot encoding. Numeric features on very different scales (income in lakhs, age in years) should often be standardised so that models like kNN, SVMs and neural networks treat them fairly. Outliers must be investigated, not blindly deleted: some are errors, while others are the rare cases you most need to detect.

✏️ **WORKED EXAMPLE**

Ages: [21, NaN, 23, 22, NaN]. Median of known ages = 22. After median imputation: [21, 22, 23, 22, 22]. A was_missing column records [0, 1, 0, 0, 1].

⚖️ 3. The bias–variance trade-off

Every model makes two kinds of error. Bias is error from overly simple assumptions: a straight line fitted to curved data will miss the pattern, giving poor scores on both training and validation data (underfitting). Variance is error from excessive sensitivity to the training data: a very deep decision tree memorises noise, scoring brilliantly on training data but poorly on new data (overfitting).

As model complexity increases, bias usually falls and variance rises. The goal is the sweet spot with the lowest total error on unseen data. Learning curves, which plot training and validation scores as data or complexity grows, help you diagnose which problem you have.

🤔 **PAUSE & THINK**

Train accuracy 98%, validation 72%. Is this high bias or high variance? Name two fixes.

🔁 4. Validation done right

A single train/test split gives one noisy estimate. k-fold cross-validation splits the data into k parts, trains k times, each time validating on a different part, and reports the mean and standard deviation of the scores. A small standard deviation means the model's performance is stable.

The most dangerous mistake is data leakage: information from the test set or the future sneaks into training. Examples include scaling the full dataset before splitting, including the same patient in both train and test sets, or using a feature that is only known after the outcome. Leakage produces wonderful scores that collapse in the real world. Using scikit-learn Pipelines, which fit preprocessing only on training folds, prevents many leaks.

✏️ **WORKED EXAMPLE**

5-fold CV scores: 0.80, 0.82, 0.78, 0.84, 0.76 → mean 0.80, range 0.08. Report: 'Accuracy 0.80 ± 0.03 (std).'

✅ Key takeaways

- EDA reveals data problems before they ruin models.
- Imputation, encoding and scaling choices must be deliberate and recorded.
- Bias = underfitting; variance = overfitting; aim for the balance.
- k-fold CV gives more reliable estimates than one split.
- Prevent leakage with pipelines and careful feature choice.

❓ Quick check: answer in one or two lines.

1. List four checks you would perform during EDA.

2. Explain the bias–variance trade-off in two lines.

3. Give two examples of data leakage.

🔬 CONCEPT DEEP DIVE: Learning from Data

🔷 EDA: summary statistics, distributions, missing-value maps, outlier checks and correlations before any modelling.

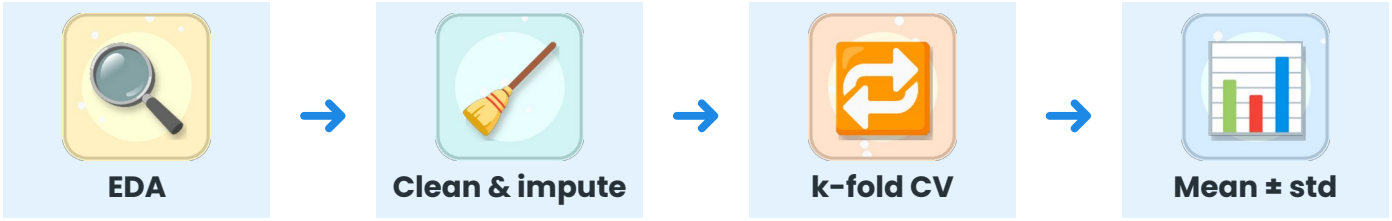
🔷 Imputation strategies: drop, mean/median, most-frequent, or model-based; always record what you did.

🔷 Bias–variance trade-off: simple models underfit (high bias); complex models overfit (high variance).

🔷 k-fold cross-validation trains k times, each time validating on a different fold; report mean ± std of scores.

🔷 Data leakage (e.g., scaling before splitting, future info in features) gives falsely optimistic results.

🔄 How it works



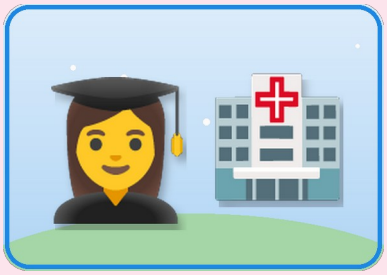
💻 Cross-validation in scikit-learn

```
from sklearn.model_selection import cross_val_score
from sklearn.linear_model import LogisticRegression
s = cross_val_score(LogisticRegression(max_iter=500), X, y, cv=5)
print(s.mean().round(3), s.std().round(3))
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Hospital Data Mystery

Characters: *Meera (B.Sc Data Science student), Dr. Prakash (hospital quality head), her teammate Ali.*



SCENE 1

Meera's team got an exciting internship task: predict which patients would be readmitted within 30 days. Within a day, their model scored 97% accuracy. Ali was ready to celebrate. Meera felt uneasy. 'Nothing is this easy,' she said.

Pause & predict: Why might a 97% score be suspicious?

SCENE 2

She began EDA properly. Thirty percent of ages were missing. Some patients appeared twice, once in training and once in testing. Worst of all, a column called 'followup_visit_scheduled' was filled in only after readmission. The model was peeking at the answer.

Pause & predict: Name the data problems Meera found.

SCENE 3

They removed the leaking column, de-duplicated patients, imputed ages with the median and added a was_missing flag. Then they used 5-fold cross-validation inside a pipeline. The new score was 78%, much lower, but stable across all folds.

Pause & predict: Why is a stable 78% more valuable than a leaky 97%?

SCENE 4

Dr. Prakash reviewed the work. 'Last year a vendor sold us a 95% model that failed within a month,' he said. 'Yours is honest. We can trust it.' The hospital piloted Meera's model to schedule follow-up calls for high-risk patients.

⏸ **Pause & predict:** What did Dr. Prakash value most in Meera's work?

🌟 The big idea

Honest data preparation and validation beat impressive but leaky numbers.

💬 Discuss with your classmates

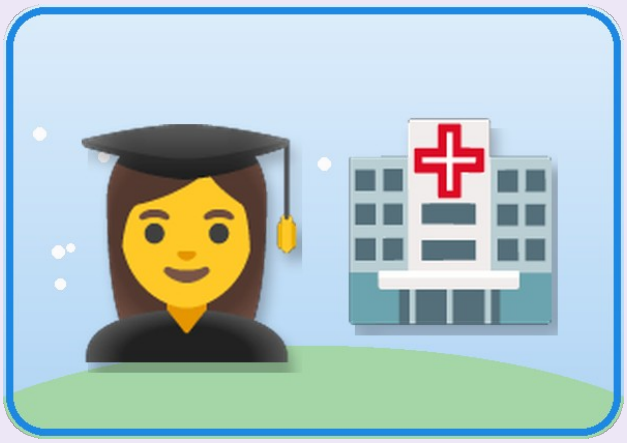
1. Have you ever seen a result that looked 'too good to be true'? What happened?

2. How would you explain data leakage to a non-technical manager?

SCENARIO: Learning from Data

Read the scenario. Think about the choices made at each step.

1



Meera's team predicts hospital readmissions.

2

EDA reveals 30% missing ages and duplicate patients!

3

They clean, impute and use 5-fold cross-validation.

4

Model is stable across folds. Ready for review!

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Learning from data everywhere

Hospital readmission risk

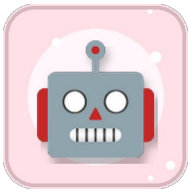
Basket analysis

Yield prediction

Credit scoring

Load forecasting

Student success analytics



BOLT SAYS:
Garbage in, garbage out. I can only be as good as the data you feed me!

FUN FACT!
Data scientists often report spending most of their time cleaning and preparing data!

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: Google Flu Trends Overshoots

CASE FILE

Google Flu Trends estimated flu levels from search queries. It worked well at first, but in the 2012–13 season it predicted roughly double the doctor visits reported by the US CDC. Changes in search behaviour and media coverage had shifted the data, and the model had overfit to patterns that no longer held. The service was later discontinued.

🔍 Analyse the case

1. Which data problem caused the overestimate?

2. How might regular validation against CDC data have helped?

3. What lesson does this teach about 'big data' models?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: EDA Detective

You need:

Colab

Messy CSV

Teams of 3

Steps:

1. Load a messy student-survey CSV with pandas.
2. Find missing values, duplicates and outliers.
3. Plot 3 distributions and 1 correlation heatmap.
4. Write 5 findings and 2 data-quality warnings.

THINK & TALK

Which finding would have misled a model most?

Record your observations & results:

💻 CODE LAB: Cleaning & Cross-Validating

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import pandas as pd, numpy as np
2  df = pd.DataFrame({'age': [21, np.nan, 23, 22, np.nan],
3                     'marks': [78, 65, 90, 90, 55]})
4  df['age'] = df['age'].fillna(df['age'].____())
5  print(df['age'].tolist())
6  print(df.duplicated().sum())
```

COMPLETE **T1. Fill the blank to impute with the median.**

PREDICT **T2. What age list is printed?**

PREDICT **T3. How many duplicate rows are counted?**

EXTEND **T4. Add a column 'z' = standardised marks.**

WORKSHEET: Learning from Data (Part A)

Name: _____ Date: _____

1. Match the data issue to its fix.

Missing values ●

● Standardise / normalise

Duplicates ●

● Impute or drop

Different scales ●

● Investigate, cap or remove

Outliers ●

● drop_duplicates()

2. High bias or high variance? Tick ✓.

	Bias	Variance
Train 62%, validation 60%		
Train 99%, validation 71%		
Linear model on curved data		
Very deep unpruned tree		

WORKSHEET: Learning from Data (Part B)

Name: _____ Date: _____

1. 5-fold CV scores: 0.80, 0.82, 0.78, 0.84, 0.76. Use the word box.

WORD BOX: 0.80 ✿ 0.84 ✿ 0.08

Mean score = ____

Best fold = ____

Range (max - min) = ____

2. Circle the examples of data leakage.

Scaling on full data before split

Splitting before any preprocessing

Using 'discharge date' to predict admission

Cross-validation with pipelines

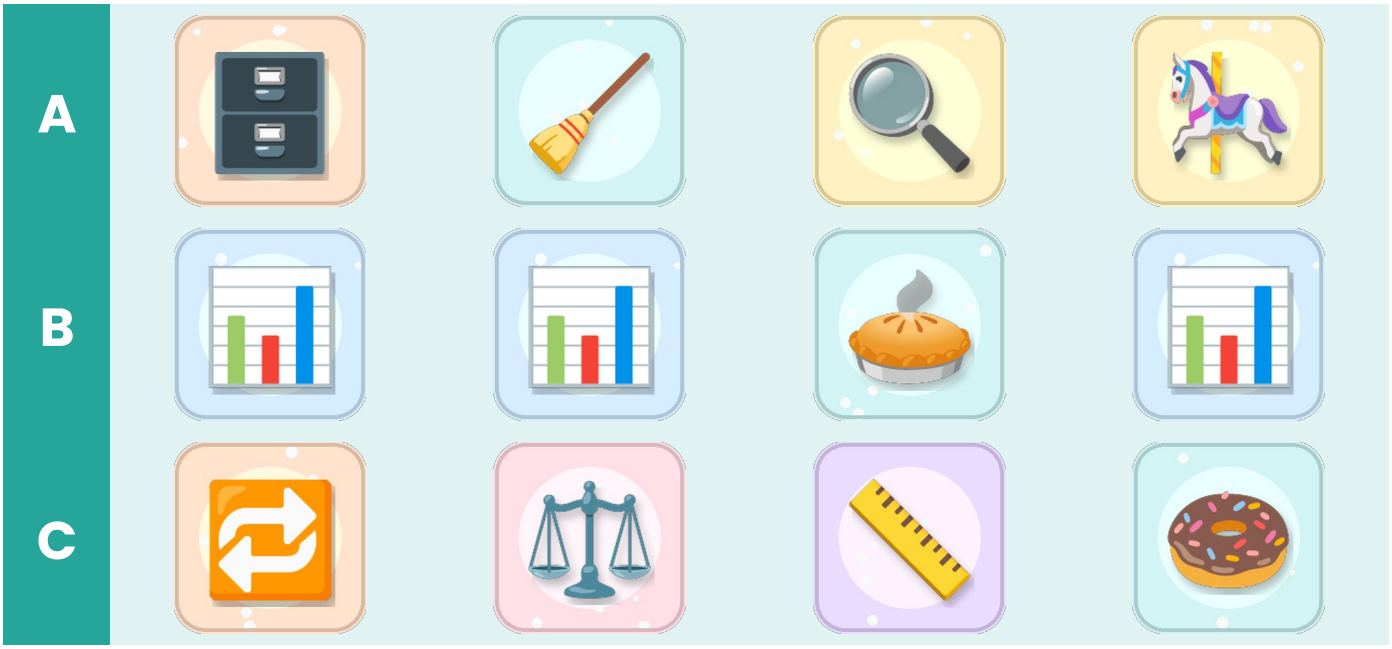
Same patient in train and test

Fixed random seed

PUZZLE & DEFINE: Learning from Data

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

EDA

Meaning: _____

Example: _____

Bias–variance

Meaning: _____

Example: _____

CV

Meaning: _____

Example: _____

? QUIZ TIME: Learning from Data

Circle the right answer!

Q1. EDA stands for exploratory data...

a) Analysis	b) Art	c) Audio

Q2. High variance usually means...

a) Underfitting	b) Overfitting	c) Perfection

Q3. k-fold CV rotates the...

a) Colours	b) Labels	c) Validation fold

Q4. Scaling features helps models like...

a) kNN / SVM	b) Dice	c) Rules

Q5. Outliers should first be...

a) Always deleted	b) Investigated	c) Ignored

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: Learning from Data

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	EDA should come before modelling.		
	Scaling before splitting is best practice.		
	k-fold CV gives several performance estimates.		
	High bias means underfitting.		
	Missing values never affect models.		
	Duplicate records can cause leakage.		

HOME CONNECT

Collect 2 weeks of a household metric (e.g., electricity units). Do a mini-EDA: mean, max, any outliers?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Learning from Data

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why can scaling data before the train/test split cause leakage?**

COMPARE **Q2. Compare a hold-out split and 5-fold cross-validation.**

APPLY **Q3. CV scores: 0.80, 0.82, 0.78, 0.84, 0.76. Mean? Is the model stable?**

EVALUATE **Q4. Is mean imputation always acceptable?**

CREATE **Q5. Design an EDA checklist of 6 items for any new dataset.**

📝 EXAM PREP: Learning from Data

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is EDA? List three EDA steps.

2. Define the bias–variance trade-off.

3. Explain k-fold cross-validation.

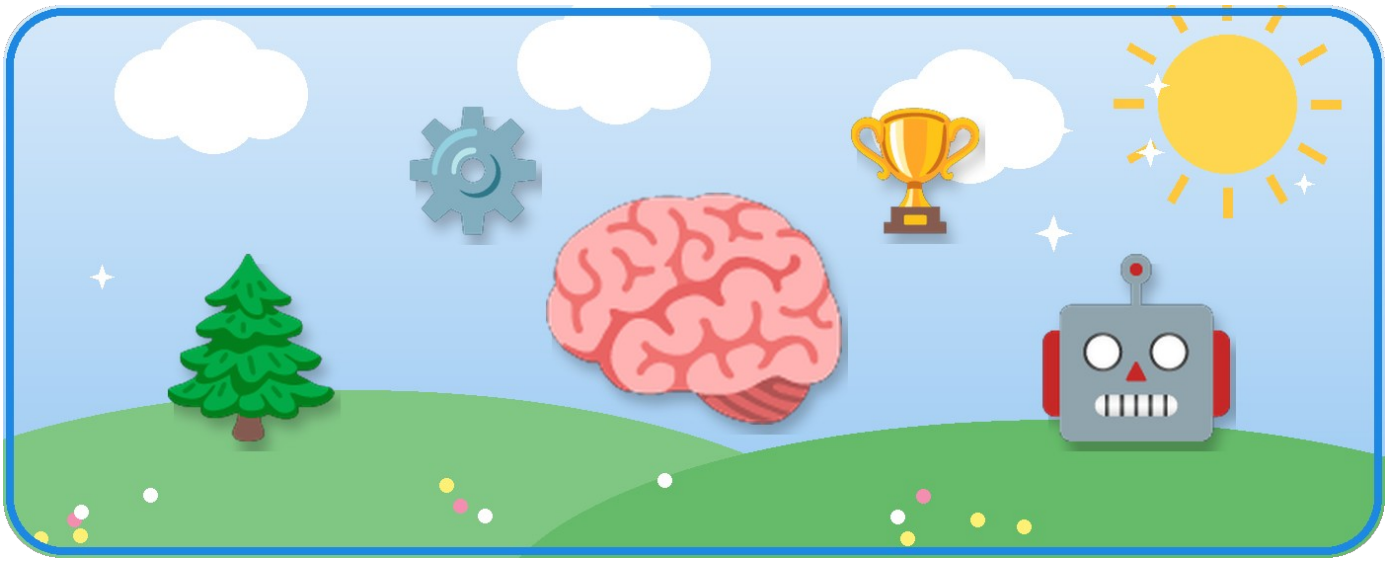
4. What is data leakage? Give one example.

PART B • Long answer (5 marks)

5. Describe a complete data-preparation workflow for a messy dataset and explain how you would evaluate a model reliably.

TOPIC 2 • MONTH 1 • AI BASICS

Making Machines Smarter

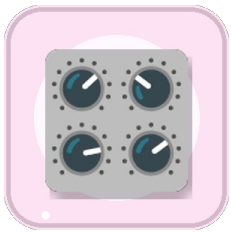


LET'S LEARN

- Smarter models: better features, tuning, ensembles.
- Regularisation stops models memorising noise.
- Ensembles (forests, boosting) beat single models.

Ways to improve a model

Feature engineering



Hyperparameter tuning

Ensembles

Regularisation

KEY TERMS

Random forest
Many voting trees

Boosting Fix previous errors

Regularisation
Penalty on complexity

📖 CONCEPT EXPLAINED: Making Machines Smarter

💡 In this chapter

Once your data is clean and your evaluation is honest, how do you make a model genuinely better? Professionals rely on four levers: smarter features, tuned hyperparameters, regularisation to control complexity, and ensembles that combine many models. This chapter explains each lever and when to pull it.

🧪 1. Feature engineering

Features are the inputs a model learns from, and better features often beat better algorithms. Feature engineering creates new, more informative inputs from raw data: ratios (debt-to-income), differences (marks this term minus last term), counts and aggregates (number of late payments in six months), date parts (day of week, festival season) and text features (word counts, keywords).

Domain knowledge is the secret ingredient. A loan officer knows that a steady salary matters more than a single large deposit; a doctor knows that a trend in blood sugar matters more than one reading. Working with domain experts to design features is one of the most valuable skills a data scientist can have.

🖐 TRY IT

Design three new features to predict whether a student will drop a course.

🎛 2. Hyperparameters and tuning

Parameters, such as the weights of a regression, are learned from data. Hyperparameters are settings chosen before training: the depth of a tree, the number of trees in a forest, the learning rate of boosting, the k in kNN. Different values can change performance dramatically.

Grid search tries every combination from a list; random search samples combinations and is often more efficient; Bayesian optimisation learns which regions look promising. Whatever the method, tuning must use cross-validation on training data only, keeping the test set untouched for the final check.

✏️ **WORKED EXAMPLE**

Grid: n_estimators → {100, 300}, max_depth ⨯ {4, 8, None} ⨯ 6 combinations. With 5-fold CV, that is 6 × 5 = 30 model fits.

🧲 3. Regularisation: controlling complexity

Regularisation adds a penalty for complexity, nudging models toward simpler solutions that generalise better. In linear models, L2 regularisation (Ridge) shrinks all weights smoothly toward zero; L1 regularisation (Lasso) can push some weights exactly to zero, effectively selecting features.

Other models have their own forms: limiting tree depth or minimum samples per leaf, dropout in neural networks, and early stopping, which halts training when validation error starts rising. The regularisation strength is itself a hyperparameter to tune.

🤔 **PAUSE & THINK**

Your Lasso model set 12 of 30 feature weights to zero. What does this tell you about those features?

🌲 4. Ensembles: the wisdom of many models

Bagging trains many models on random bootstrap samples of the data and averages their predictions. Random forests add randomness by letting each split consider only a subset of features, making the trees less similar. Averaging many de-correlated trees reduces variance dramatically.

Boosting trains models sequentially: each new model focuses on the errors of the previous ones. Gradient boosting libraries like XGBoost and LightGBM are among the strongest methods for tabular data. But ensembles cost more compute and are harder to explain, so the 'best' model must balance accuracy with speed, cost and interpretability.

✏️ **WORKED EXAMPLE**

Seven trees vote: spam, ham, spam, spam, ham, ham, spam → 4 spam vs 3 ham → prediction: spam, with 57% agreement.

✅ Key takeaways

- Domain-driven features often matter more than algorithms.
- Hyperparameters are tuned with CV, never on the test set.
- L1 can select features; L2 shrinks weights; early stopping helps too.
- Bagging reduces variance; boosting reduces bias.
- The best model balances accuracy, cost and explainability.

❓ Quick check: answer in one or two lines.

1. Differentiate parameters and hyperparameters.

2. How is a random forest different from a single decision tree?

3. Compute total fits: 3 × 4 grid with 5-fold CV.

🔬 CONCEPT DEEP DIVE: Making Machines Smarter

🔷 Feature engineering: ratios, aggregates, date parts, encodings (one-hot, target) often matter more than the algorithm.

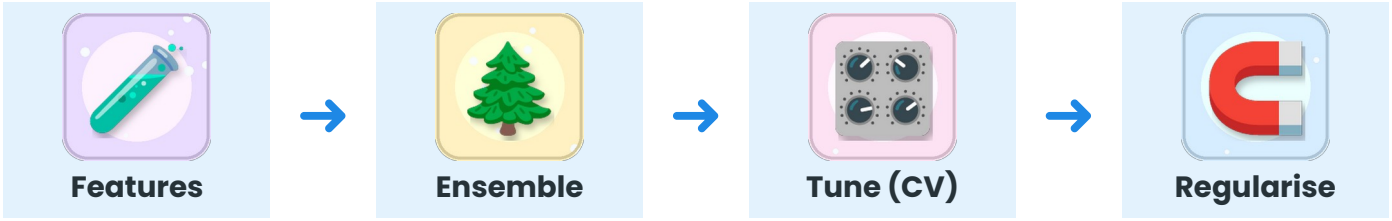
🔷 Bagging (random forest) trains many trees on bootstrap samples and random feature subsets, then averages votes.

🔷 Boosting (AdaBoost, gradient boosting, XGBoost) adds models sequentially, each correcting residual errors.

🔷 Regularisation: L1 (Lasso) can zero out weights; L2 (Ridge) shrinks weights; trees use depth & leaf limits.

🔷 Hyperparameter tuning: grid search, random search or Bayesian optimisation, always with cross-validation.

🔄 How it works



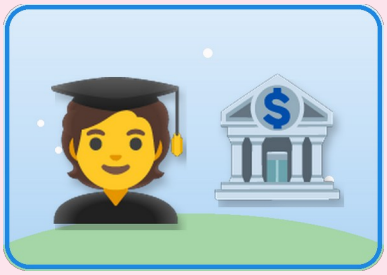
💻 Tuning a random forest

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
grid = {'n_estimators': [100, 300], 'max_depth': [4, 8, None]}
gs = GridSearchCV(RandomForestClassifier(), grid, cv=5).fit(X, y)
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Loan Model Tournament

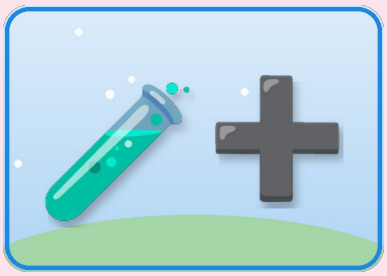
Characters: *Rohan (B.Tech student), Ms. Sharma (credit manager at a cooperative bank), his classmate Divya.*



SCENE 1

A cooperative bank asked Rohan's class to predict loan defaults. Rohan's single decision tree reached 74% on validation. Divya's logistic regression reached 72%. Neither impressed Ms. Sharma. 'We lose lakhs every month on defaults,' she said.

Pause & predict: What could Rohan try first to improve his model?



SCENE 2

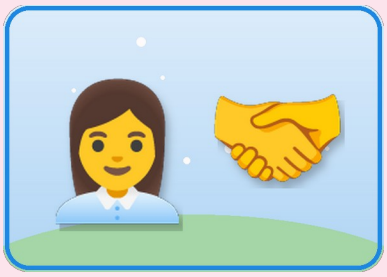
Rohan visited the bank and listened. Ms. Sharma explained that regular salary credits and the ratio of EMI to income mattered most. Rohan engineered features: debt-to-income ratio, months of steady income, number of late payments. His tree jumped to 79%.

Pause & predict: Why did talking to the credit manager help so much?

SCENE 3

Next he built a random forest and a gradient-boosting model, tuning them with cross-validated grid search. Boosting reached 85%. But it was slower and harder to explain. Divya added L1 regularisation to her logistic regression: 81%, with only eight clear features.

Pause & predict: What trade-offs exist between Rohan's and Divya's models?



SCENE 4

Ms. Sharma chose a combination: boosting to flag risky applications and Divya's simple model to explain decisions to customers. 'Our customers deserve reasons,' she said. The class learned that the best model depends on the whole situation.

⏸ **Pause & predict:** Why did the bank use two models?

🌟 The big idea

Better features, careful tuning and ensembles make models smarter, but explainability and context decide what is used.

💬 Discuss with your classmates

1. When would you choose a simpler model over a more accurate one?

2. What features would you engineer to predict exam success?

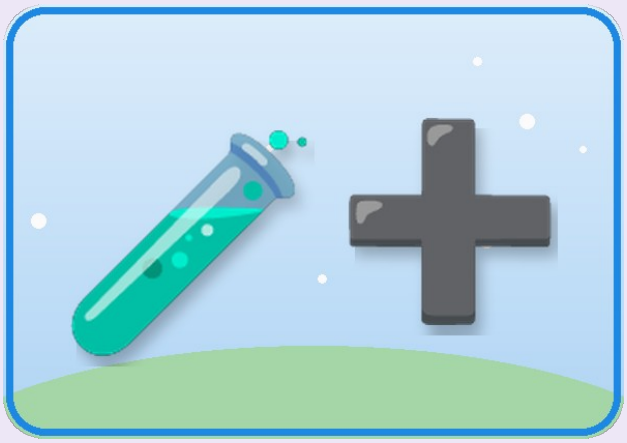
SCENARIO: Making Machines Smarter

Read the scenario. Think about the choices made at each step.

1

Rohan's single decision tree predicts loan defaults: 74%.

2

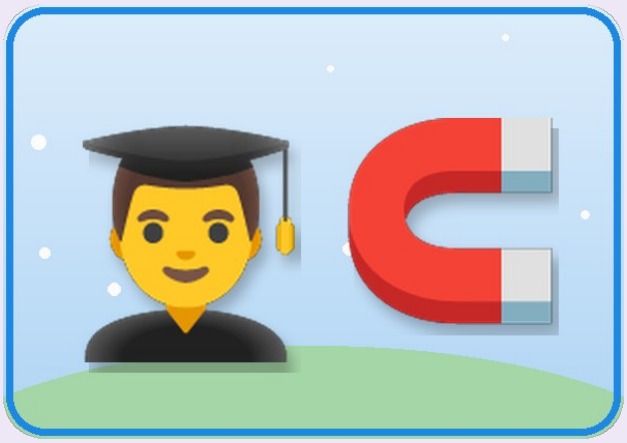


He engineers features: debt-to-income ratio, payment streaks.

3

A random forest reaches 81%; gradient boosting 85%.

4



He tunes regularisation to keep validation scores stable.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Smarter models in industry

Gradient boosting for credit

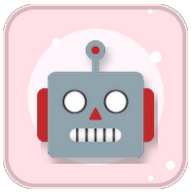
Ensembles for recommendations

Tuned models for diagnosis

Demand forecasting

Fraud scoring

Search ranking



BOLT SAYS:

One tree can be fooled. A forest of 500 trees voting together is much wiser!

FUN FACT!

Gradient-boosted trees (e.g., XGBoost) have won a large share of tabular-data competitions!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: The Netflix Prize Ensemble

CASE FILE

In 2009, team BellKor's Pragmatic Chaos won Netflix's US$1 million prize by improving rating predictions by just over 10%. Their solution blended hundreds of models. Netflix later said it did not deploy the full winning ensemble: the extra accuracy was not worth the engineering complexity, and its business had shifted to streaming.

🔍 Analyse the case

1. Why did blending hundreds of models improve accuracy?

2. Why might a company reject a more accurate model?

3. What does this teach about 'best model' vs 'best solution'?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Wisdom of the Crowd

You need:

Jar of beads

Guess slips

Whole class

Steps:

1. Each student guesses the number of beads (no talking).
2. Compute the class average guess.
3. Compare: average error vs typical individual error.
4. Link to bagging: many weak guesses → strong estimate.

THINK & TALK

When would the crowd average fail? (Hint: shared bias)

Record your observations & results:

💻 CODE LAB: Bagging by Hand

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  from collections import Counter
2  votes = ['spam', 'ham', 'spam', 'spam', 'ham', 'ham', 'spam']
3  pred = Counter(votes).most_common(____)[0][0]
4  conf = votes.count(pred) / len(votes)
5  print(pred, round(conf, 2))
```

COMPLETE **T1. Fill the blank to get the top vote.**

PREDICT **T2. What is printed?**

MODIFY **T3. If one 'spam' flips to 'ham', what changes?**

EXTEND **T4. Weight each vote by tree accuracy [0.9, 0.6, ...].**

WORKSHEET: Making Machines Smarter (Part A)

Name: _____ Date: _____

1. Match the technique to its idea.

Bagging ●

● Try every hyperparameter combo

Boosting ●

● Train on bootstrap samples, vote

L2 regularisation ●

● Each model fixes previous errors

Grid search ●

● Penalise large weights

2. Majority vote of 5 trees. Use the word box.

WORD BOX: Spam ✿ Not spam ✿ 3 of 5

Votes: S, S, N, S, N → prediction: ____

Votes: N, N, S, N, S → prediction: ____

Spam votes needed to win: ____

WORKSHEET: Making Machines Smarter (Part B)

Name: _____ Date: _____

1. Grid-search arithmetic. Use the word box.

WORD BOX: 6 ✿ 30 ✿ 5

Combos: {100, 300} × {4, 8, None} = ____

Folds in cv = 5 → fits per combo = ____

Total fits = ____

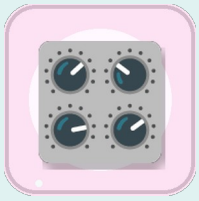
2. L1 or L2 regularisation? Tick ✓.

	L1 Lasso	L2 Ridge
Can set some weights exactly to zero		
Shrinks all weights smoothly		
Useful for feature selection		
Handles correlated features gently		

PUZZLE & DEFINE: Making Machines Smarter

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Random forest

Meaning: _____

Example: _____

Boosting

Meaning: _____

Example: _____

Regularisation

Meaning: _____

Example: _____

? QUIZ TIME: Making Machines Smarter

Circle the right answer!

Q1. Random forest is an example of...

a) Bagging

b) Guessing

c) Cleaning

Q2. Boosting trains models...

a) Randomly

b) Sequentially

c) Never

Q3. Regularisation reduces...

a) Data

b) Speed

c) Overfitting

Q4. Settings chosen before training:

a) Hyperparameters

b) Weights

c) Labels

Q5. Creating new input columns is...

a) Painting

b) Feature engineering

c) Deleting

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: Making Machines Smarter

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Random forests average many trees.		
	Boosting trains models in parallel independently.		
	L1 can make some weights zero.		
	Hyperparameters are learned during training.		
	Feature engineering can matter more than the algorithm.		
	Netflix fully deployed the winning prize ensemble.		

HOME CONNECT

Ask 10 people to estimate a jar's coins. Compare the average with individual guesses.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Making Machines Smarter

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does a random forest reduce variance compared to one tree?**

COMPARE **Q2. Compare bagging and boosting.**

APPLY **Q3. Grid: n_estimators {100, 300}, max_depth {4, 8, None}, cv = 5. How many fits?**

EVALUATE **Q4. Is a 0.5% accuracy gain worth 10× training cost?**

CREATE **Q5. Engineer 4 new features for predicting student dropout.**

📝 EXAM PREP: Making Machines Smarter

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Differentiate bagging and boosting.

2. What is regularisation? Name two types.

3. What are hyperparameters? Give two examples.

4. Why does feature engineering matter?

PART B • Long answer (5 marks)

5. Explain how random forests and gradient boosting work, compare their strengths, and describe how you would tune them.

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. Scaling before splitting causes...

a) Data leakage | b) Speed | c) Colour

Q2. Train 99%, validation 71% =

a) High bias | b) High variance | c) Balanced

Q3. Random forest uses...

a) Boosting | b) Cleaning | c) Bagging

Q4. L1 regularisation can...

a) Zero some weights | b) Grow weights | c) Shuffle data

Q5. 2 × 3 grid, cv = 5 → fits:

a) 6 | b) 30 | c) 11

Q6. CV mean of 0.80, 0.82, 0.78:

a) 0.82 | b) 0.78 | c) 0.80

WORD SEARCH: AI DISCOVER

Find these words. They go across → or down ↓.

EDA	CV	BOOSTING	LEAKAGE	FOREST	LASSO

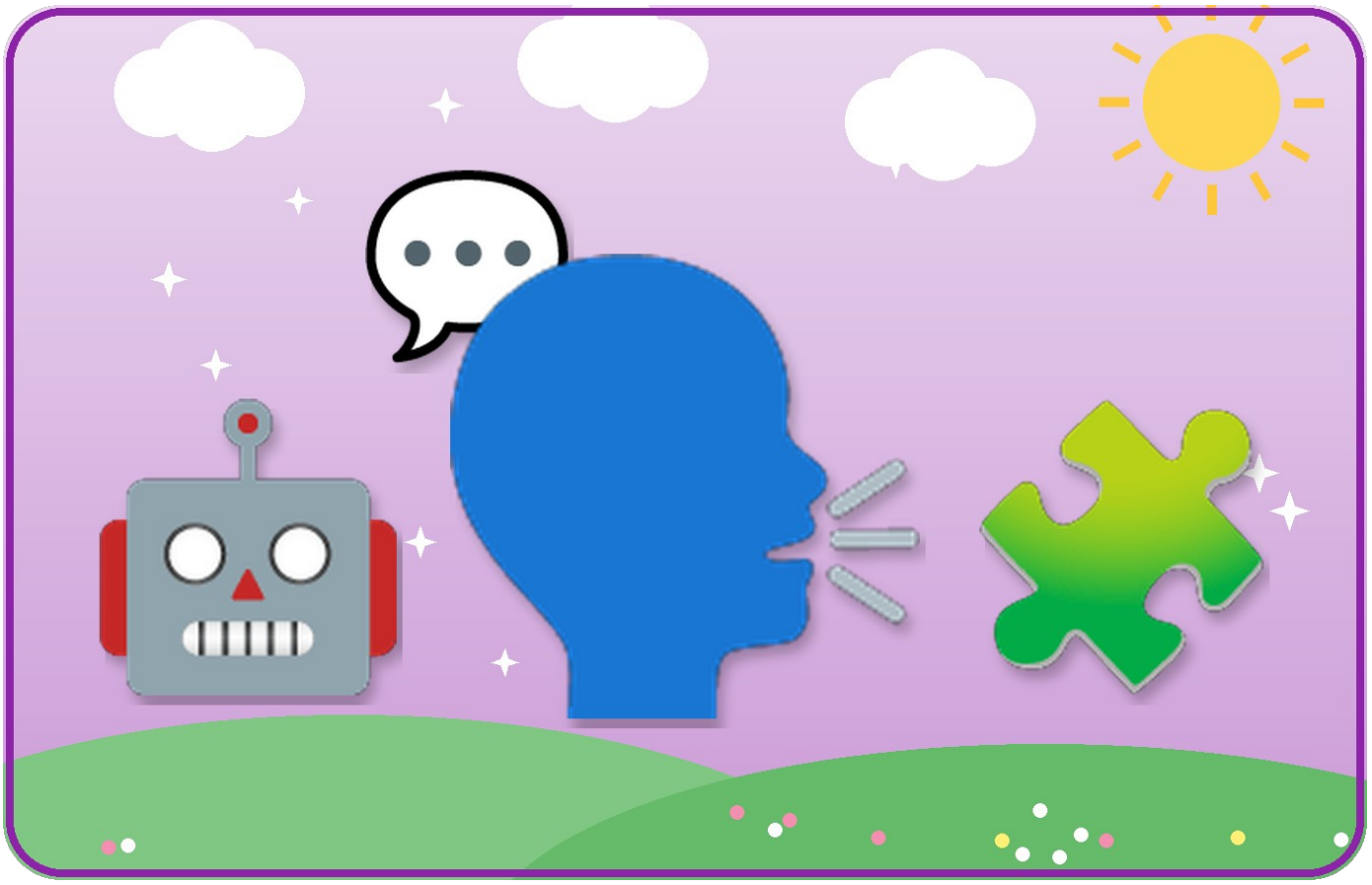
L	S	O	S	G	K	K	U	R	S
N	B	O	O	S	T	I	N	G	U
F	B	R	H	N	O	F	M	T	M
O	C	P	S	L	A	S	S	O	D
L	C	F	S	N	M	R	A	R	B
D	V	K	W	U	U	N	F	F	S
H	A	G	F	O	R	E	S	T	T
T	H	N	L	E	A	K	A	G	E
E	D	A	S	M	U	M	P	I	T
W	A	N	S	E	S	T	G	O	B

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code



Area: Communicate with AI

In this unit, you will explore:

Topic 3

Program with AI

Topic 4

Data into Decisions

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

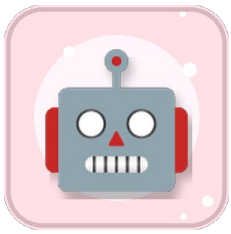
Program with AI

LET'S LEARN

- AI pair-programmers speed up drafting and refactoring.
- Specs + tests make AI-generated code trustworthy.
- Use APIs to call AI models from your own programs.

Programming with AI

Specify



Generate

Test

Refactor

KEY TERMS

TDD Test-driven development

API key Secret access token

Refactor Improve code structure

📖 CONCEPT EXPLAINED: Program with AI

💡 In this chapter

In second year you move from writing small scripts to building programs that call AI models and are written with AI assistance. This chapter covers how to specify work so AI can help, how test-driven development keeps AI code honest, how LLM APIs work, and how to manage security, cost and reliability.

📝 1. Specifications: telling AI what 'done' means

AI coding assistants perform best when the task is precisely specified. A good specification describes the function's purpose, inputs and their types, outputs, edge cases (empty inputs, invalid values), constraints (performance, libraries allowed) and examples of correct behaviour.

Vague requests ('make a grade calculator') produce plausible but unreliable code. Specific requests ('grade(marks: int) → str; 90–100 A, 75–89 B, 40–74 C, below 40 F; raise ValueError outside 0–100') produce code you can actually verify.

🖐 TRY IT

Write a full specification for a function that converts temperatures between Celsius and Fahrenheit.

🧪 2. Test-driven development with AI

Test-driven development (TDD) follows a cycle: write a failing test that describes desired behaviour (red), write the minimum code to pass it (green), then improve the code's structure without changing behaviour (refactor). With AI, you write or review the tests, and the assistant drafts code to pass them.

Tests turn vague trust into evidence. They catch hallucinated logic, protect against regressions when you change code later, and serve as living documentation. Boundary values, such as 39, 40, 89, 90 and 100 for a grading function, are where bugs most often hide.

✏️ WORKED EXAMPLE

def test_grade(): assert grade(90) == 'A'; assert grade(89) == 'B'; assert grade(40) == 'C'; assert grade(39) == 'F'. Plus: with pytest.raises(ValueError): grade(101).

🔌 3. How LLM APIs work

An API (Application Programming Interface) lets your program send a request to an AI model running on a server and receive a response. A typical request includes the model name, the messages or prompt, and parameters like temperature and maximum tokens. The response contains generated text plus usage information (tokens consumed).

Your code must handle failures: network errors, timeouts, invalid keys (HTTP 401), rate limits (HTTP 429) and server errors (HTTP 500). Robust programs retry with exponential back-off, validate the model's output and log what happened.

🤔 **PAUSE & THINK**

Why might retrying instantly after a 429 'too many requests' error make the problem worse?

🛡️ 4. Security, cost and responsibility

API keys are like passwords: store them in environment variables or secret managers, never in code or public repositories. Validate user inputs to prevent prompt injection and never execute code or commands generated by a model without checks.

API usage is billed by tokens, so costs scale with traffic. Cache repeated requests, keep prompts concise, choose smaller models when sufficient and set budget alerts. Finally, disclose AI assistance in academic work as required, and make sure you understand every line you submit.

✏️ **WORKED EXAMPLE**

1,000 requests × 800 tokens = 800,000 tokens. At ₹0.0005 per token, cost ≈ ₹400. With 90% cache hits, billed cost ≈ ₹40.

✅ Key takeaways

- Precise specs make AI-generated code verifiable.
- TDD: red → green → refactor; test boundaries carefully.
- LLM APIs need error handling, retries and output validation.
- Keep keys secret; never run unchecked model output.
- Manage token costs with caching and concise prompts.

? Quick check: answer in one or two lines.

1. What are the three steps of the TDD cycle?

2. What do HTTP 401 and 429 mean?

3. Give two ways to reduce LLM API cost.

🔬 CONCEPT DEEP DIVE: Program with AI

🔷 AI coding works best with precise specs: inputs, outputs, constraints, edge cases and examples.

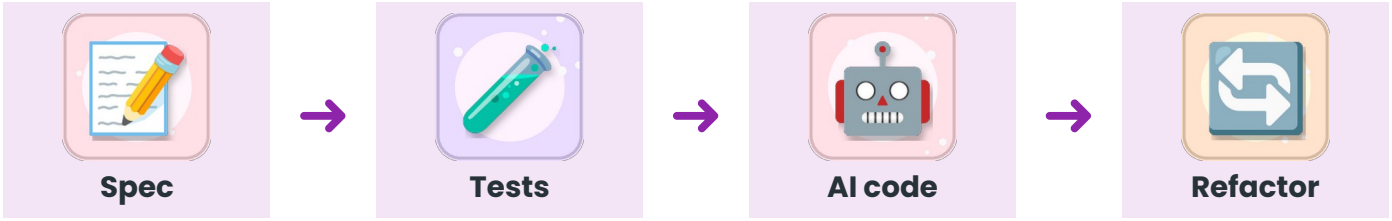
🔷 Test-driven development: write failing tests, generate/implement code, refactor while keeping tests green.

🔷 LLM APIs: send messages + parameters (model, temperature, max tokens); parse responses; handle errors & rate limits.

🔷 Security: keep keys in environment variables, validate inputs, avoid executing untrusted model output.

🔷 Cost & latency: tokens cost money; cache repeated calls, batch requests, and log usage.

🔄 How it works



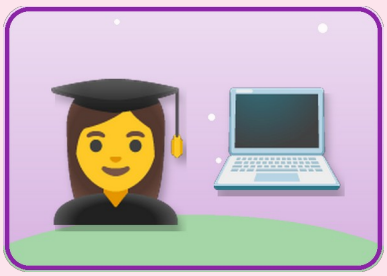
💻 Calling an LLM API safely

```
import os, requests
key = os.environ['LLM_API_KEY']          # never hard-code
r = requests.post(URL, headers={'Authorization': key},
        json={'prompt': p, 'max_tokens': 200}, timeout=30)
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Tests Before Code

Characters: *Ananya (B.Tech CSE student), Professor Iyer, an AI coding assistant.*



SCENE 1

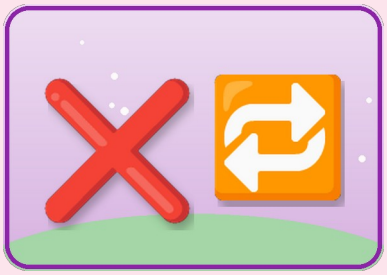
Professor Iyer asked Ananya to build a small API that returns grade statistics for her department. She typed 'build a marks API' into an AI assistant and received 200 lines of code. It ran, but she had no idea whether it was correct.

Pause & predict: What is risky about Ananya's first approach?

SCENE 2

She started again. This time she wrote a precise specification and eight pytest tests: normal marks, empty lists, invalid values above 100, and boundary cases like 39 and 40. Then she asked the assistant to write code that passed the tests.

Pause & predict: Why did Ananya include boundary cases?



SCENE 3

Six tests passed; two failed. The AI's code crashed on an empty list and treated 40 as a fail. She pasted the failing tests back, got a fix, and refactored the code to be cleaner, rerunning tests after every change. Everything turned green.

Pause & predict: How did the tests guide the AI's fixes?

SCENE 4

Before deploying, Professor Iyer spotted her API key hard-coded in the file. She moved it to an environment variable and added retry logic for rate limits. In her report she wrote: 'Code drafted with AI, verified by 8 tests.' He gave her full marks.

⏸ **Pause & predict:** Which professional habits did Ananya demonstrate?

🌟 The big idea

Tests turn AI-generated code from a guess into evidence.

💬 Discuss with your classmates

1. What would you test in a function that calculates attendance percentage?

2. Should all AI-assisted code be disclosed? Why?

SCENARIO: Program with AI

Read the scenario. Think about the choices made at each step.

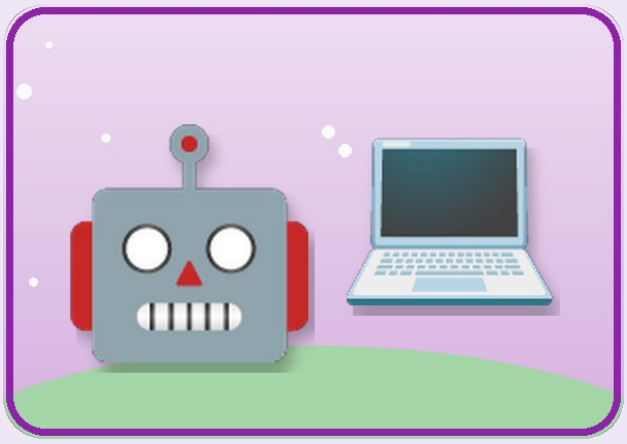
1

Ananya needs a marks-analysis API for her department.

2

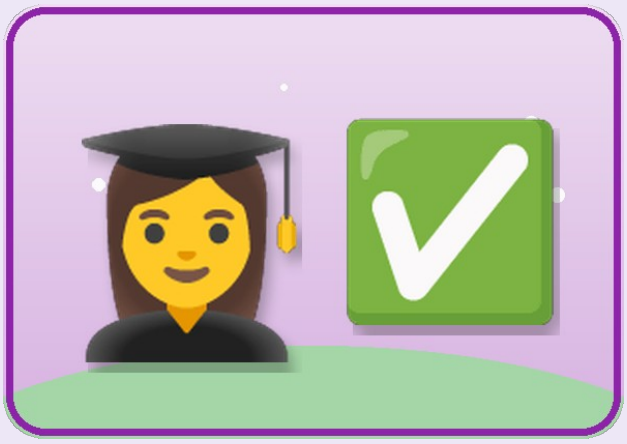
She writes 8 pytest tests describing correct behaviour.

3



AI drafts the code; 6 tests pass, 2 fail on edge cases.

4



She fixes them with AI's help; all green, then deploys.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: AI in the developer workflow

Code completion

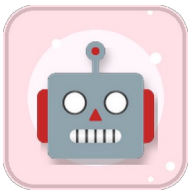
Test generation

Legacy migration

Docs & comments

Security scanning

Calling LLM APIs



BOLT SAYS:

Tell me exactly what 'done' means with tests, and I'll get there much faster!

FUN FACT!

Writing tests first (TDD) gives AI assistants a clear target to aim at!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Measuring AI Pair-Programming Speed

CASE FILE

In a controlled experiment published by GitHub researchers, developers using an AI coding assistant completed a task (writing an HTTP server in JavaScript) about 55% faster than those without it. Completion rates were also somewhat higher. Critics noted the task was small and well-defined, unlike much real-world software work.

🔍 Analyse the case

1. Why does a controlled experiment give stronger evidence than a survey?

2. Why might results differ for large, messy codebases?

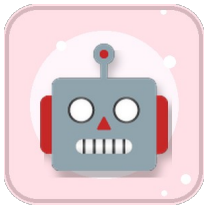
3. Design a fairer experiment for your college lab.

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Spec-First AI Coding

You need:

Laptops



AI assistant

Pairs

Steps:

1		Write a spec + 5 pytest tests for a grade calculator.
2		Ask the AI to write code that passes your tests.
3		Run tests; log failures and AI fix attempts.
4		Ask AI to refactor; re-run tests to confirm.

THINK & TALK

How did writing tests first change the AI's output?

Record your observations & results:

💻 CODE LAB: Testing an AI-Written Function

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  def grade(m):
2      if not 0 <= m <= 100: raise ValueError('bad marks')
3      if m >= 90: return 'A'
4      if m >= 75: return 'B'
5      return 'C' if m >= ____ else 'F'
6
7  def test_grade():
8      assert grade(90) == 'A' and grade(74) == 'C'
9      assert grade(39) == 'F'
```

COMPLETE **T1. Pass mark is 40. Fill the blank.**

PREDICT **T2. Do both asserts pass?**

DEBUG **T3. grade(100.5) — what happens? Is that right?**

EXTEND **T4. Write a pytest test that checks ValueError.**

WORKSHEET: Program with AI (Part A)

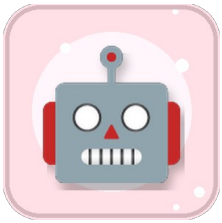
Name: _____ Date: _____

1. Test-driven AI coding. Write 1, 2, 3, 4.

Refactor safely ☐

Write failing tests ☐

Run tests until green ☐



Generate code with AI ☐

2. Safe or unsafe API practice? Tick ✓.

	Safe	Unsafe
Store API key in environment variable		
Commit API key to public GitHub		
Handle rate limits with retries		
No spending limit on paid API		

✏️ WORKSHEET: Program with AI (Part B)

Name: _____ Date: _____

1. API cost: tokens × price. Use the word box.

WORD BOX: ₹400 ✿ 800,000 ✿ ₹40

1,000 calls × 800 tokens = ____ tokens

800,000 × ₹0.0005 = ____

With 90% cached (only 10% billed): ____

2. Match the HTTP status to its meaning.

200 ●

401 ●

429 ●

500 ●

● **Too many requests (rate limit)**

● **Server error**

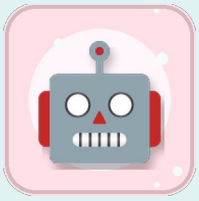
● **OK / success**

● **Unauthorised (bad key)**

PUZZLE & DEFINE: Program with AI

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

TDD

Meaning: _____

Example: _____

API key

Meaning: _____

Example: _____

Refactor

Meaning: _____

Example: _____

? QUIZ TIME: Program with AI

Circle the right answer!

Q1. TDD means writing tests...

a) First	b) Never	c) Randomly

Q2. API keys should be kept in...

a) Public code	b) Env variables	c) Emails

Q3. Improving structure, same behaviour:

a) Deleting	b) Styling	c) Refactoring

Q4. Too many API calls may hit a...

a) Rate limit	b) Wall	c) Bullseye

Q5. Tests turning green means...

a) Failing	b) Passing	c) Growing

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: Program with AI

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	TDD writes tests before the code.		
	Hard-coding API keys is safe in private repos.		
	HTTP 429 signals rate limiting.		
	Caching repeated calls can cut costs.		
	AI-generated code never needs refactoring.		
	Low temperature gives more consistent code.		

HOME CONNECT

Explain to a family member what an 'API' is using a restaurant waiter analogy.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Program with AI

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does writing tests first help when coding with AI?**

COMPARE **Q2. Compare temperature 0 and 0.8 for code generation.**

APPLY **Q3. 1,000 API calls × 800 tokens at ₹0.0005/token. Cost?**

EVALUATE **Q4. Should students submit AI-generated code in labs?**

CREATE **Q5. Write 4 test cases for a function grade(marks).**

📝 EXAM PREP: Program with AI

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is test-driven development?

2. Why store API keys in environment variables?

3. What does HTTP 429 mean?

4. Name two ways to reduce LLM API cost.

PART B • Long answer (5 marks)

5. Describe a safe, efficient workflow for building an application that calls an LLM API, covering testing, security and cost.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

Data into Decisions

LET'S LEARN

- Analytics: descriptive → diagnostic → predictive → prescriptive.
- SQL turns raw tables into business answers.
- A/B tests show cause, not just correlation.

Four levels of analytics

Descriptive

Diagnostic

Predictive

Prescriptive

KEY TERMS

SQL Query language for tables

A/B test Randomised comparison

KPI Key performance indicator

📖 CONCEPT EXPLAINED: Data into Decisions

💡 In this chapter

Organisations collect enormous amounts of data, but data only matters when it changes decisions. This chapter climbs the four levels of analytics, shows how SQL answers business questions, and explains A/B testing, the gold standard for discovering what actually works.

🪜 1. Four levels of analytics

Descriptive analytics asks what happened: total sales last month, attendance by department. Diagnostic analytics asks why it happened: sales dropped because delivery fees rose in one city. Predictive analytics asks what will happen: forecasting next month's demand. Prescriptive analytics asks what we should do: which price or stock level maximises profit.

Each level builds on the one below. You cannot diagnose without describing, and you cannot prescribe responsibly without predicting. Many organisations stay stuck at descriptive dashboards; moving up the ladder is where the real value lies.

🖐 TRY IT

For your college library, write one question for each analytics level.

🗄️ 2. SQL: asking questions of data

Most organisational data lives in relational databases, and SQL (Structured Query Language) is how we query it. SELECT chooses columns, FROM names the table, WHERE filters rows, GROUP BY aggregates rows by category, HAVING filters groups, ORDER BY sorts results and JOIN combines related tables.

SQL is declarative: you describe what you want, and the database decides how to get it efficiently. Even with AI tools that write SQL, you must understand queries well enough to check their logic, because a wrong JOIN can silently double-count results.

✏️ WORKED EXAMPLE

```
SELECT city, COUNT(*) AS orders FROM orders WHERE order_date >= '2026-09-01'
GROUP BY city HAVING COUNT(*) > 100 ORDER BY orders DESC;
```

🧪 3. A/B testing: finding causes

Correlation does not prove causation: ice-cream sales and drownings both rise in summer, but one does not cause the other. To learn what a change actually causes, organisations run randomised experiments called A/B tests. Users are randomly assigned to version A (control) or B (treatment), and a pre-chosen metric is compared.

Because assignment is random, the groups are similar on average in every way except the change, so a reliable difference can be attributed to the change. Good tests fix the metric and sample size in advance, run for full business cycles (e.g., whole weeks) and avoid 'peeking' and stopping early when results look exciting.

✏️ **WORKED EXAMPLE**

A: 120 of 2,000 users converted (6.0%). B: 150 of 2,000 (7.5%). Relative lift = (7.5 - 6) / 6 = 25%.

📏 4. Statistical vs practical significance

A significance test asks whether a difference is larger than random chance would usually produce. The p-value is the probability of seeing a difference at least this large if there were truly no effect; $p < 0.05$ is a common, though imperfect, threshold.

Statistically significant is not the same as important. With millions of users, a 0.1% lift may be 'significant' but not worth the engineering cost. Conversely, a promising result in a small test may simply need more data. Decisions should combine statistics, costs, risks and domain judgement.

🤔 **PAUSE & THINK**

A test shows a 0.2% lift with $p = 0.01$, but the change will cost ₹20 lakh to build. What else do you need to know before deciding?

✅ Key takeaways

- Analytics ladder: descriptive → diagnostic → predictive → prescriptive.
- SQL is the core language for querying organisational data.
- Randomised A/B tests reveal causal effects.
- Fix metrics and sample sizes in advance; don't peek.
- Weigh statistical significance against practical value.

? Quick check: answer in one or two lines.

1. Give an example question for each analytics level.

2. Differentiate WHERE and HAVING.

3. Why does randomisation allow causal conclusions?

🔬 CONCEPT DEEP DIVE: Data into Decisions

🔷 Descriptive (what), diagnostic (why), predictive (what next), prescriptive (what to do) analytics build on each other.

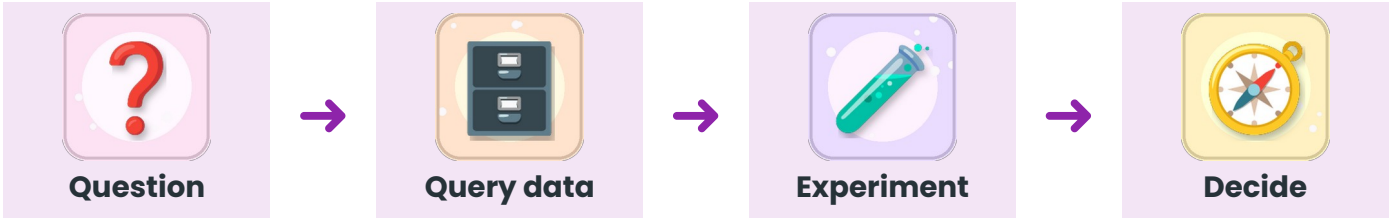
🔷 SQL essentials: SELECT, WHERE, GROUP BY, HAVING, JOIN, ORDER BY; window functions for running totals.

🔷 A/B testing: random assignment, one change, pre-set metric & sample size, then a significance test.

🔷 p-value: probability of a result at least this extreme if there were no real difference; p < 0.05 is a common threshold.

🔷 Beware Simpson's paradox, peeking early at results, and confusing statistical with practical significance.

🔄 How it works



💻 SQL for a business question

```
SELECT city, COUNT(*) AS orders, AVG(amount) AS avg_amt
FROM orders
WHERE order_date >= '2026-09-01'
GROUP BY city HAVING COUNT(*) > 100 ORDER BY orders DESC;
```

✍️ In your own words: explain this topic to a Class 10 student.

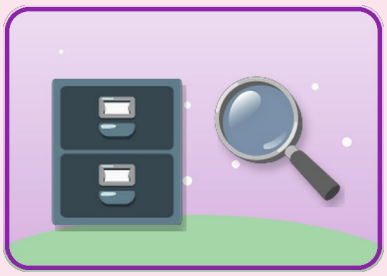
AI STORY: The Weekend Mystery

Characters: *Kiran (business analytics intern at a food-delivery start-up), Ms. Rao (product head).*

SCENE 1

Weekend orders at a food-delivery start-up dropped 15% in a month. Meetings were full of opinions: 'It's the rain', 'It's a competitor', 'It's the new logo'. Ms. Rao turned to the new intern, Kiran. 'Find out what actually happened.'

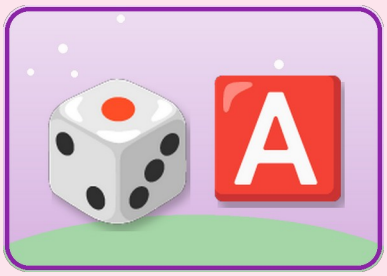
Pause & predict: Which analytics level is Ms. Rao asking for first?



SCENE 2

Kiran wrote SQL queries grouping orders by city and week. The drop appeared only in one city, starting the exact week delivery fees rose from ₹30 to ₹45. Other cities were stable. The data pointed to a cause, but Kiran knew correlation wasn't proof.

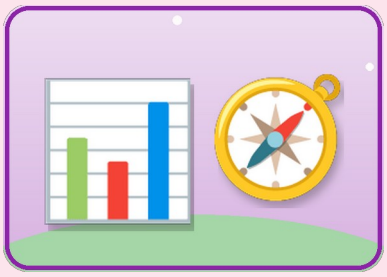
Pause & predict: Why couldn't Kiran conclude the fee caused the drop?



SCENE 3

He proposed an A/B test: randomly show half the users in that city the old fee and half the new fee, for two full weeks. He fixed the metric (profit per user) and sample size in advance and promised not to peek and stop early.

Pause & predict: Why did Kiran choose profit per user rather than number of orders?



SCENE 4

Results: the old fee produced more orders and slightly higher total profit. The difference was statistically significant and large enough to matter. The company reverted the fee in that city. Ms. Rao smiled: 'Data settled what an hour of arguing couldn't.'

⏸ **Pause & predict:** Which analytics level did the final decision represent?

🌟 The big idea

Descriptive data finds where to look; experiments show what actually works.

💬 Discuss with your classmates

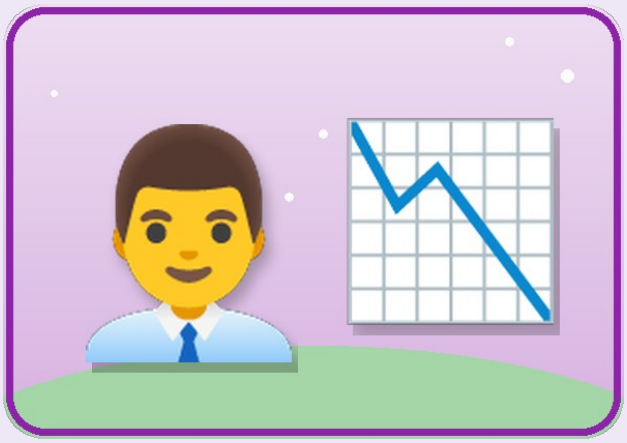
1. Design an A/B test for something at your college.

2. What ethical limits should A/B tests on users respect?

SCENARIO: Data into Decisions

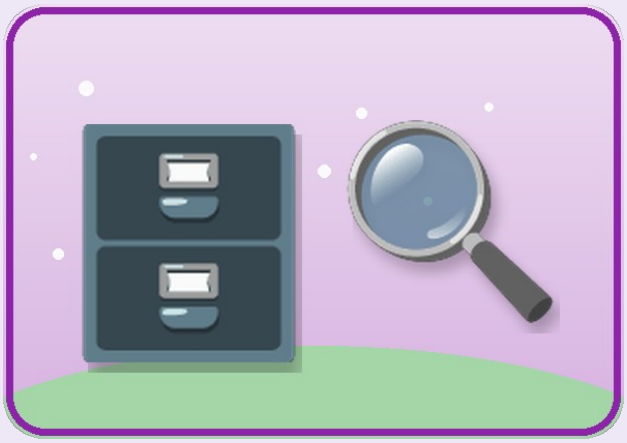
Read the scenario. Think about the choices made at each step.

1



A food app's weekend orders drop 15%.

2

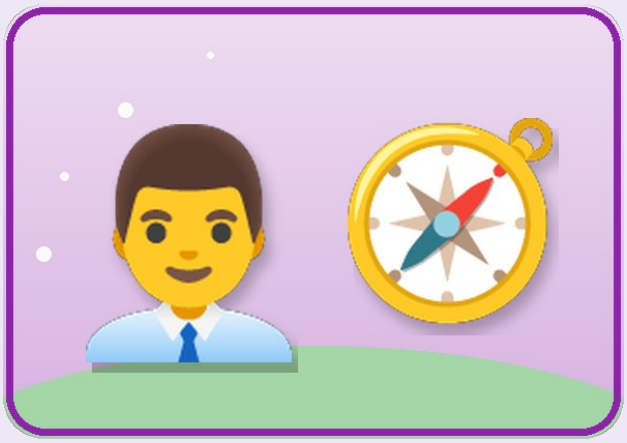


SQL shows the drop is only in one city after a fee change.

3

An A/B test compares old vs new delivery fees.

4



Data shows the old fee earns more overall. Decision made!

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Data-driven decisions

Pricing experiments

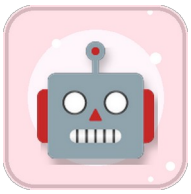
App feature A/B tests

Route planning

Bed allocation

Course redesign

Marketing spend



BOLT SAYS:

Opinions are cheap. A well-designed experiment tells you what actually works!

FUN FACT!

Large tech firms run thousands of A/B experiments every year to guide decisions!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: A Headline Tweak Worth Millions

CASE FILE

At Microsoft's Bing, an engineer proposed a small change to how ad headlines were displayed. The idea sat in the backlog for months as low priority. When finally A/B tested, it increased revenue by about 12%, worth over US$100 million a year in the US alone, a famous lesson that intuition about which ideas matter is often wrong.

🔍 Analyse the case

1. Why did the team's intuition underrate the idea?

2. Which A/B test safeguards make the 12% result trustworthy?

3. How could your college use A/B tests (ethically)?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Classroom A/B Test

You need:

2 posters

Sign-up sheets

Two sections

Steps:

1		Randomly assign sections to Poster A or Poster B.
2		Both invite students to a coding club.
3		Count sign-ups; compute conversion rates.
4		Decide which poster wins; discuss sample size.

THINK & TALK

Could the result be due to chance? How would you check?

Record your observations & results:

💻 CODE LAB: A/B Test Significance

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  from math import sqrt
2  na, ca, nb, cb = 2000, 120, 2000, 150
3  pa, pb = ca / na, cb / nb
4  p = (ca + cb) / (na + nb)
5  z = (pb - pa) / sqrt(p * (1 - p) * (1/na + 1/____))
6  print(round(pa, 3), round(pb, 3), round(z, 2))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Write pa and pb.**

EXPLAIN **T3. z ≈ 1.89. Is it significant at |z| > 1.96?**

EXTEND **T4. How many more users might make the test conclusive?**

WORKSHEET: Data into Decisions (Part A)

Name: _____ Date: _____

1. Match the analytics level to its question.

Descriptive ●

Diagnostic ●

Predictive ●

Prescriptive ●

● What will sales be next month?

● What happened last month?

● What should we do about it?

● Why did sales drop?

2. A/B conversion rates. Use the word box.

WORD BOX: 5% ✿ 8% ✿ +60%

A: 50 sign-ups / 1,000 views = ____

B: 80 sign-ups / 1,000 views = ____

Relative lift of B over A = ____

WORKSHEET: Data into Decisions (Part B)

Name: _____ Date: _____

1. A: 120/2,000. B: 150/2,000. Use the word box.

WORD BOX: 6% ✿ 7.5% ✿ 25%

Conversion A = ____

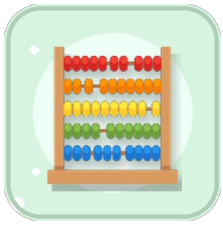
Conversion B = ____

Relative lift (7.5 - 6) / 6 = ____

2. Order the SQL clauses. Write 1, 2, 3, 4.

ORDER BY ☐

SELECT ... FROM ☐



GROUP BY ☐

WHERE ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Data into Decisions

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

SQL

Meaning: _____

Example: _____

A/B test

Meaning: _____

Example: _____

KPI

Meaning: _____

Example: _____

? QUIZ TIME: Data into Decisions

Circle the right answer!

Q1. 'Why did it happen?' is...

a) Diagnostic

b) Descriptive

c) Predictive

Q2. SQL is used to query...

a) Songs

b) Databases

c) Clouds

Q3. A/B tests can show...

a) Colours only

b) Only correlation

c) Causal effects

Q4. SELECT ... GROUP BY city gives...

a) Per-city results

b) One row

c) An error

Q5. 'What should we do?' is...

a) Descriptive

b) Prescriptive

c) Diagnostic

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Data into Decisions

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	A/B tests use random assignment.		
	Peeking and stopping early is good practice.		
	GROUP BY aggregates rows by category.		
	Descriptive analytics answers 'what happened'.		
	p < 0.05 proves a big business impact.		
	Simpson's paradox can reverse trends in aggregated data.		

HOME CONNECT

Run a tiny A/B test at home: two ways of reminding someone about a chore. Which works better?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Data into Decisions

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does random assignment allow causal conclusions in A/B tests?**

COMPARE **Q2. Compare WHERE and HAVING in SQL.**

APPLY **Q3. A: 120/2,000 convert, B: 150/2,000. Rates and relative lift?**

EVALUATE **Q4. p = 0.03 but lift is 0.1%. Ship it?**

CREATE **Q5. Design an A/B test for a library's reminder SMS.**

📝 EXAM PREP: Data into Decisions

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name the four levels of analytics.

2. Differentiate WHERE and HAVING.

3. What is a p-value?

4. Why randomise in A/B tests?

PART B • Long answer (5 marks)

5. Design and explain a complete A/B test for a business decision, including metric, sample size, analysis and pitfalls.

UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. HTTP 401 means...

a) Unauthorised | b) OK | c) Rate limit

Q2. Tests first, then code:

a) DDD | b) TDD | c) TV

Q3. HAVING filters...

a) Single rows | b) Colours | c) Groups

Q4. 6% → 7.5% is a lift of...

a) 25% | b) 1.5% | c) 75%

Q5. Stopping a test early when it looks good:

a) Best practice | b) Peeking (bad) | c) Fun

Q6. 'What should we do?' analytics:

a) Descriptive | b) Diagnostic | c) Prescriptive

WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

TDD

API KEY

REFACTOR

SQL

A/B TEST

KPI

U	R	H	D	K	P	I	S	E	I
H	E	G	B	O	B	B	M	M	F
H	F	A	/	B	T	E	S	T	A
C	A	D	C	A	P	I	K	E	Y
A	C	C	B	L	P	A	Q	M	I
S	T	O	E	I	Y	F	U	F	S
Q	O	H	A	F	T	N	E	U	B
L	R	O	H	T	E	E	R	B	A
M	W	R	D	D	S	K	Y	L	R
A	K	T	P	D	T	T	W	B	I

I finished AI CONNECT! Teacher's sign: _____

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI for Smarter Business

Topic 6

AI for Engineering Solutions

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI for Smarter Business

LET'S LEARN

- AI drives revenue, cuts cost and manages risk.
- Key metrics: churn, CLV, conversion, margin.
- Model errors have business costs: price them.

Business AI levers

Revenue growth

Cost reduction

Risk control

Customer experience

KEY TERMS

Churn Customers who leave

CLV Customer lifetime value

Cost matrix Price of each error

📖 CONCEPT EXPLAINED: AI for Smarter Business

💡 In this chapter

Businesses invest in AI to grow revenue, reduce costs and manage risk. But a model's value is measured in money and customer outcomes, not accuracy alone. This chapter introduces key business metrics, shows how to turn predictions into profitable decisions, and examines when business AI goes wrong.

💰 1. The language of business metrics

Every business AI project should connect to metrics leaders care about. Churn rate is the percentage of customers who leave in a period. Customer Lifetime Value (CLV) estimates the total profit a customer brings over their relationship. Customer Acquisition Cost (CAC) is what it costs to win a new customer. Conversion rate, average order value and margin complete the picture.

A healthy business keeps CLV comfortably above CAC. AI helps by predicting which customers might leave, which might buy more, and which actions improve these numbers.

✏️ WORKED EXAMPLE

Margin ₹250/month × expected 12 months = CLV ₹3,000. If CAC is ₹1,500, CLV ÷ CAC = 2.

⚖️ 2. From predictions to profitable decisions

A churn model outputs a probability for each customer. The business must decide where to set the threshold for action, such as offering a discount. Each outcome has a value: a true positive (saving a real churner) earns profit; a false positive (offering a discount to a loyal customer) wastes money; a false negative (missing a churner) loses future revenue.

A cost matrix assigns these values, and the best threshold is the one that maximises expected profit, not accuracy. Uplift models go further by estimating who will change behaviour because of an offer, avoiding waste on customers who would have stayed anyway.

🤔 PAUSE & THINK

Why might it be better to target 'persuadable' customers rather than those with the highest churn risk?

🏷️ 3. AI across business functions

Marketing uses AI for segmentation and personalised recommendations. Sales uses lead scoring. Operations uses demand forecasting and inventory optimisation. Finance uses fraud detection and credit scoring. Customer service uses chatbots and ticket routing. Retail uses association rules to find products often bought together.

Dynamic pricing adjusts prices based on demand, time and competition. It can be efficient, but it also raises fairness and regulatory concerns, especially for essential goods or during emergencies. Responsible businesses set clear limits on such systems.

🖐️ TRY IT

Pick a local business (pharmacy, bakery, mobile shop). Suggest one AI use for each of two business functions.

🏚️ 4. When business AI goes wrong

Acting on predictions at scale magnifies errors. Zillow, a US real-estate company, used algorithms to buy homes directly, expecting to resell them profitably. In 2021, its models failed to keep up with rapidly changing market conditions, and Zillow closed the business after large losses and staff cuts.

Lessons for any business AI: monitor for drift, start with limited exposure, stress-test models under unusual conditions, keep humans responsible for major decisions, and measure profit and risk, not just model accuracy.

🤔 PAUSE & THINK

What safeguard would you add before letting an algorithm make large purchases automatically?

✅ Key takeaways

- Track churn, CLV, CAC, conversion and margin.
- Choose thresholds with a cost matrix to maximise profit.
- Uplift models target customers who respond to action.
- Dynamic pricing needs fairness limits.
- Scale amplifies model errors: monitor, limit, stress-test.

❓ Quick check: answer in one or two lines.

1. Define CLV and CAC.

2. Why can maximising accuracy fail to maximise profit?

3. Give one lesson from the Zillow Offers case.

🔬 CONCEPT DEEP DIVE: AI for Smarter Business

🔷 Unit economics: CAC (acquisition cost), CLV (lifetime value), churn rate, margin; healthy firms keep CLV well above CAC.

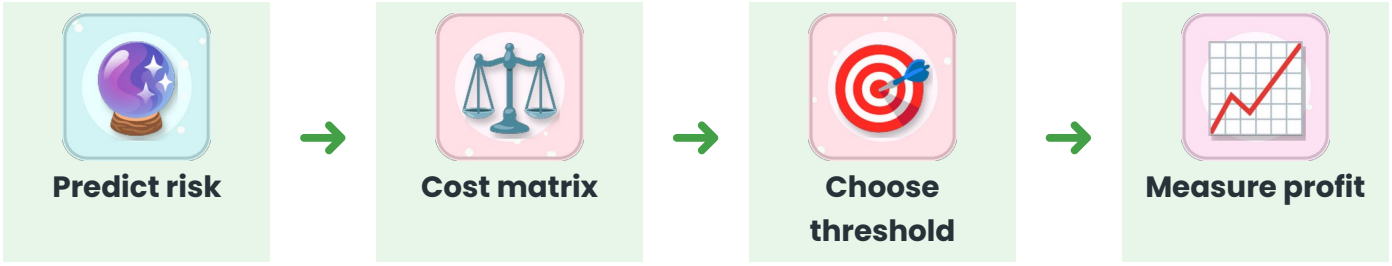
🔷 Churn models rank customers; uplift models go further and find those who respond to an intervention.

🔷 A cost matrix assigns money to TP, FP, FN, TN; choose the decision threshold that maximises expected profit.

🔷 Dynamic pricing uses demand elasticity; must avoid unfair or exploitative pricing and respect regulation.

🔷 Business AI must align with strategy, have an owner, measurable KPIs, and a plan for change management.

🔄 How it works



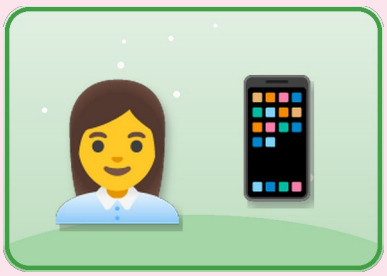
💻 Profit at a threshold

```
# saved +1800, wasted offer -200, missed churner -2000
tp, fp, fn = 60, 140, 40
profit = tp * 1800 + fp * (-200) + fn * (-2000)
print(profit)   # 108000 - 28000 - 80000 = 0
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Customers Who Almost Left

Characters: *Sana (MBA intern at a telecom firm), Mr. Das (marketing manager).*



SCENE 1

A telecom company was losing 3% of its customers every month. Mr. Das wanted to send a ₹200 discount to everyone. 'That costs crores,' said Sana, an analytics intern. 'Let's find who is actually likely to leave.'

Pause & predict: What is the problem with offering the discount to everyone?

SCENE 2

Sana built a churn model using usage drops, complaint counts and contract end dates. It ranked customers by risk. But she didn't stop at accuracy. She built a cost matrix: saving a churner earned ₹1,800 in future profit; a wasted offer cost ₹200; missing a churner lost ₹2,000.

Pause & predict: Why did Sana build a cost matrix?

SCENE 3

By testing different thresholds, she found that targeting the top 10% of risk scores maximised profit. Targeting more customers wasted offers; targeting fewer missed too many churners. Her plan cost a fraction of Mr. Das's idea.

Pause & predict: How did Sana decide on the 10% threshold?

SCENE 4

After three months, churn fell from 3% to 2.2% and profit rose even after offer costs. Next, Sana proposed an uplift model to find customers who would stay only because of the offer. 'Let's stop paying people who were never leaving,' she said.

⏸ **Pause & predict:** What extra value would an uplift model add?

🌟 The big idea

Business AI succeeds when predictions are turned into profitable, measured decisions.

💬 Discuss with your classmates

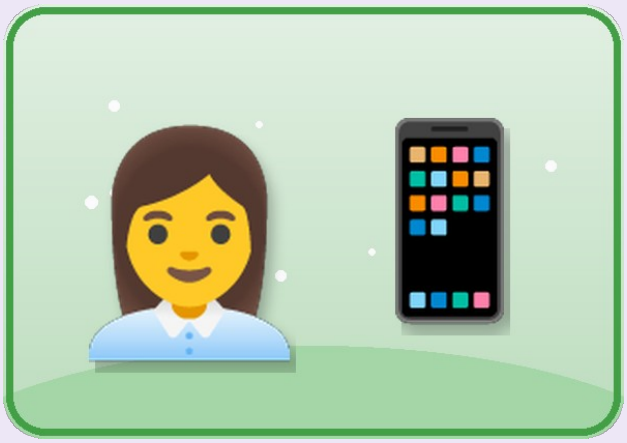
1. How would you calculate CLV for a mobile recharge customer?

2. Is it fair to give discounts only to customers likely to leave?

SCENARIO: AI for Smarter Business

Read the scenario. Think about the choices made at each step.

1

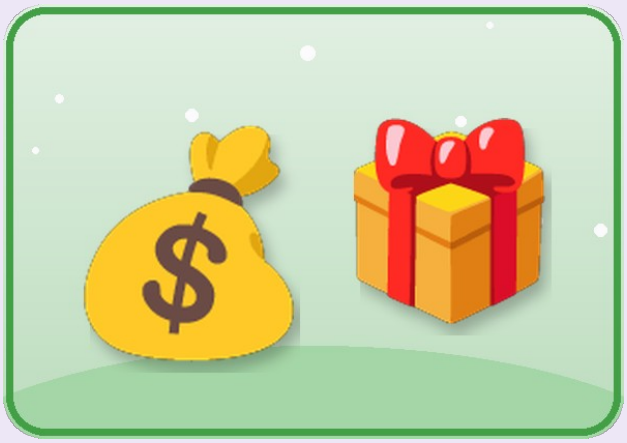


A telecom firm loses 3% of customers monthly.

2

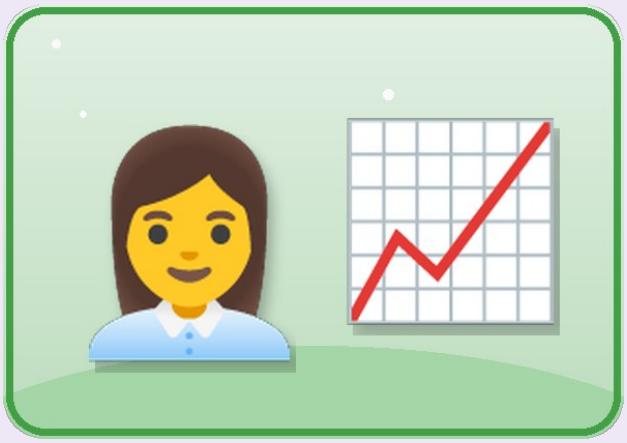
A churn model ranks customers by leaving risk.

3



Top 10% get retention offers; cost vs CLV is calculated.

4



Churn falls to 2.2%; profit rises despite offer costs.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: AI in business functions

Churn prediction

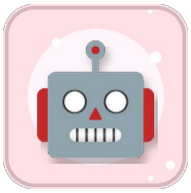
Dynamic pricing

Inventory optimisation

Targeted marketing

Fraud detection

Service automation



BOLT SAYS:

A model that is 90% accurate can still lose money if its mistakes are expensive!

FUN FACT!

Keeping an existing customer is often far cheaper than winning a new one!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Zillow's Algorithmic Home-Buying Exit

CASE FILE

Zillow Offers used algorithms to estimate home values and buy houses directly, planning to resell at a profit. In 2021, its models failed to anticipate rapid market shifts; Zillow bought many homes for more than it could sell them. It closed the business, wrote down hundreds of millions of dollars and cut about a quarter of its staff.

🔍 Analyse the case

1. Why is predicting prices riskier when you act on the predictions at scale?

2. Which business safeguards could have limited losses?

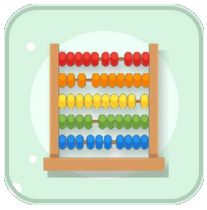
3. What does this teach about model error costs?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Churn War Room

You need:

Customer cards



Calculator

Teams of 4

Steps:

1		Each team gets 20 customer cards (usage, complaints).
2		Flag likely churners using 2 simple rules.
3		Flagged get a ₹200 offer; each save = ₹2,000 CLV.
4		Reveal truth; compute profit of your strategy.

THINK & TALK

Was it better to flag more customers or fewer? Why?

Record your observations & results:

💻 CODE LAB: Profit-Optimal Threshold

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
scores = [0.9, 0.8, 0.7, 0.6, 0.4, 0.3]
churn  = [1,   1,   0,   1,   0,   0]
def profit(t):
    tp = sum(1 for s, c in zip(scores, churn) if s >= t and c)
    fp = sum(1 for s, c in zip(scores, churn) if s >= t and not c)
    return tp * 1800 - fp * ____
print(profit(0.75), profit(0.55))
```

COMPLETE **T1. Each wasted offer costs 200. Fill the blank.**

PREDICT **T2. Compute profit(0.75) and profit(0.55).**

EXPLAIN **T3. Which threshold is better? Why?**

EXTEND **T4. Loop thresholds 0.1–0.9 and print the best.**

WORKSHEET: AI for Smarter Business (Part A)

Name: _____ Date: _____

1. CLV = monthly margin × expected months. Use the word box.

WORD BOX: ₹6,000 ✿ ₹3,600 ✿ ₹2,400

₹300 × 20 months = ____

₹150 × 24 months = ____

₹200 × 12 months = ____

2. Match the business problem to its AI method.

Churn ●

Pricing ●

Basket add-ons ●

Fraud ●

● **Association rules**

● **Classification**

● **Anomaly detection**

● **Demand elasticity models**

WORKSHEET: AI for Smarter Business (Part B)

Name: _____ Date: _____

1. Unit economics. Use the word box.

WORD BOX: ₹3,000 ✿ 2 ✿ 3%

CLV = ₹250 × 12 months = ____

CLV ÷ CAC (₹1,500) = ____

30 of 1,000 customers leave → churn = ____

2. Value of each outcome in a retention campaign. Tick ✓.

	Gain	Loss
TP: saved a real churner		
FP: offer to a loyal customer		
FN: missed a churner		
TN: correctly left alone		

PUZZLE & DEFINE: AI for Smarter Business

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Churn

Meaning: _____

Example: _____

CLV

Meaning: _____

Example: _____

Cost matrix

Meaning: _____

Example: _____

? QUIZ TIME: AI for Smarter Business

Circle the right answer!

Q1. Customers who stop buying:

a) Churn

b) Waves

c) Floaters

Q2. CLV measures a customer's...

a) Age

b) Lifetime value

c) Location

Q3. 'Bread → butter' patterns come from...

a) Dice

b) Radio

c) Association rules

Q4. Unusual transactions are found by...

a) Anomaly detection

b) Design

c) Music

Q5. Errors with different costs need a...

a) Map

b) Cost matrix

c) Calendar

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI for Smarter Business

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
📉	Churn is the rate customers leave.		
🎯	Maximising accuracy always maximises profit.		
💎	Healthy firms keep CLV above CAC.		
⚖️	A cost matrix prices each type of error.		
🏠	Zillow Offers was hugely profitable and expanded.		
🎁	Uplift models target persuadable customers.		

HOME CONNECT

Ask a shop owner how they keep regular customers. Map their tactics to churn & CLV ideas.

🔢 Self-assessment: tick one.

😀	🙂	😐
I know it well!	I know some.	I need practice.

CHALLENGE PROBLEMS: AI for Smarter Business

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why is accuracy a poor objective for churn campaigns?**

COMPARE **Q2. Compare churn models and uplift models.**

APPLY **Q3. CAC ₹1,500, margin ₹250/month, 12-month life. CLV and CLV/CAC?**

EVALUATE **Q4. Is dynamic pricing of medicines in a crisis ethical?**

CREATE **Q5. Design a KPI dashboard (5 metrics) for an AI retention programme.**

📝 EXAM PREP: AI for Smarter Business

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Define churn and CLV.

2. What is a cost matrix?

3. Differentiate churn and uplift models.

4. Give two business AI risks.

PART B • Long answer (5 marks)

5. Explain how to design a profit-maximising AI retention campaign, with a worked numeric example.

TOPIC 6 • MONTH 3 • AI APPLICATIONS

AI for Engineering Solutions

LET'S LEARN

- AI predicts failures, optimises designs, inspects quality.
- Digital twins mirror real systems in software.
- Physics + data together beat either alone.

Engineering AI toolkit

Predictive maintenance

Generative design

Visual inspection

Digital twin

KEY TERMS

Digital twin Live virtual replica

RUL Remaining useful life

Gen. design AI-explored shapes

📖 CONCEPT EXPLAINED: AI for Engineering Solutions

💡 In this chapter

Engineering has always relied on models: equations, simulations and safety factors. AI adds a powerful new kind of model that learns from sensor data and explores designs at superhuman speed. This chapter covers predictive maintenance, digital twins, generative design and the safety discipline engineering AI demands.

🔧 1. Predictive maintenance

Machines often show warning signs before failure: rising vibration, temperature or current draw. Predictive maintenance collects these signals from sensors, extracts features such as RMS vibration or temperature trend, and uses models to detect anomalies or estimate Remaining Useful Life (RUL).

Compared with reactive maintenance (fix after failure) and preventive maintenance (fix on a fixed schedule), predictive maintenance repairs equipment when its condition requires it. This reduces both unexpected breakdowns and unnecessary servicing.

✏️ WORKED EXAMPLE

Wear index 45, failure limit 90, rising 3 units per day → RUL ≈ (90 - 45) ÷ 3 = 15 days. Apply a safety margin: plan repair within about 12 days.

🪞 2. Digital twins

A digital twin is a virtual replica of a physical asset (a pump, a wind turbine, a factory line, even a city's water network) that is continuously updated with live sensor data. Engineers can test 'what if' scenarios on the twin, such as increasing load or changing maintenance schedules, before touching the real system.

Twins combine physics-based models, which encode known engineering laws, with data-driven models, which capture behaviour the equations miss. This hybrid approach is more accurate and trustworthy than either alone.

🤔 PAUSE & THINK

Why must a digital twin be updated with live data rather than relying only on the original design drawings?

🧬 3. Generative design and surrogate models

In generative design, engineers specify goals and constraints (loads, materials, weight limits, manufacturing method), and algorithms explore thousands of possible shapes. The results often look organic, like bones or branches, because they place material only where it is needed.

Surrogate models address another bottleneck: high-fidelity simulations such as computational fluid dynamics can take hours per run. An ML model trained on a set of simulation results can approximate new results in milliseconds, letting engineers explore many more design options quickly.

🖐️ **TRY IT**

List three constraints you would give a generative design tool for a bicycle frame.

🏗️ 4. Safety and responsibility

Engineering AI often controls physical systems where errors can injure people or damage infrastructure. Such systems require rigorous validation, estimates of uncertainty, fail-safe defaults, compliance with standards and certification, and clear human sign-off.

Professional engineers remain legally and ethically responsible for designs and decisions, even when AI assists. An AI suggestion is an input to engineering judgement, never a replacement for it.

🤔 **PAUSE & THINK**

A model predicts a bridge sensor reading is 'normal' with 70% confidence. Should engineers relax? Explain.

✅ Key takeaways

- Predictive maintenance acts on equipment condition, not calendars.
- RUL estimates guide repair timing; add safety margins.
- Digital twins combine physics and live data.
- Generative design and surrogates speed up exploration.
- Safety-critical AI needs validation and human sign-off.

❓ Quick check: answer in one or two lines.

1. Compare reactive, preventive and predictive maintenance.

2. What is a digital twin?

3. Compute RUL: current 60, limit 100, rising 5 per day.

🔬 CONCEPT DEEP DIVE: AI for Engineering Solutions

🔷 Predictive maintenance: sensors → features (RMS vibration, temperature trends) → anomaly or RUL models → work orders.

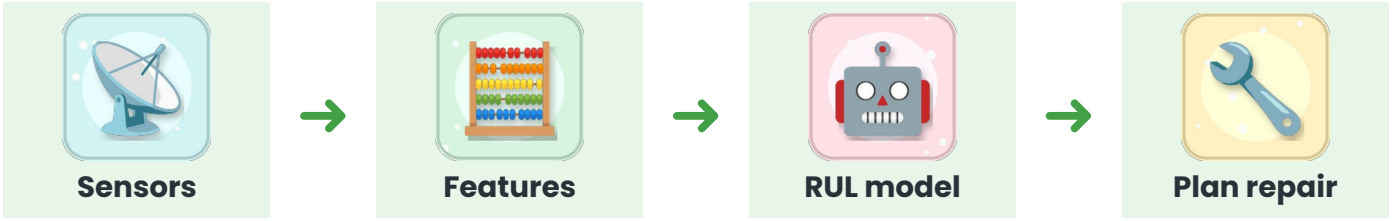
🔷 Digital twins combine physics models and live data to simulate 'what-if' scenarios before acting on real assets.

🔷 Physics-informed ML embeds known equations, needing less data and staying physically plausible.

🔷 Surrogate models replace slow simulations (e.g., CFD, FEA) with fast ML approximations for design exploration.

🔷 Safety-critical AI needs validation, uncertainty estimates, certification and human sign-off.

🔄 How it works



💻 RMS vibration feature

```
import numpy as np
signal = np.array([0.2, -0.4, 0.3, -0.1, 0.5])
rms = np.sqrt(np.mean(signal ** 2))
print(round(rms, 3))   # rising RMS → wear
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Fan That Spoke Too Late

Characters: *Suresh (B.E. Mechanical student), Mr. Gopal (plant maintenance engineer), a vibration sensor.*

SCENE 1

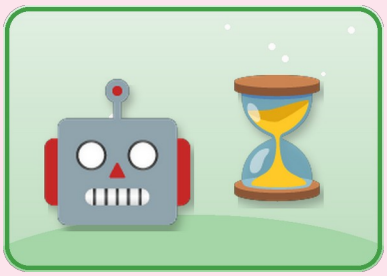
At a cement plant near Suresh's home, a giant kiln fan failed without warning. Production stopped for four days, costing lakhs. Mr. Gopal, the maintenance engineer, sighed. 'We service it every six months. It still broke.'

Pause & predict: What type of maintenance was the plant using?

SCENE 2

Suresh, an intern, installed low-cost vibration and temperature sensors on four motors. He calculated the RMS vibration daily and plotted the trends. One motor's vibration was rising steadily, though it still sounded normal to the workers.

Pause & predict: Why might a sensor notice problems before humans do?



SCENE 3

Using past failure records, Suresh fitted a simple model to estimate remaining useful life. The rising motor had about 12 days left. Mr. Gopal was sceptical but agreed to inspect it. Inside, they found a worn bearing close to failure.

Pause & predict: Why should Mr. Gopal still inspect the motor rather than trust the model blindly?

SCENE 4

The bearing was replaced during a planned weekend shutdown, with no lost production. The plant expanded the sensor system and started building a digital twin of the kiln line. Mr. Gopal told Suresh, 'The machines were always talking. Now we can listen.'

⏸ **Pause & predict:** How could a digital twin help the plant further?

🌟 The big idea

Sensors and AI let engineers act on a machine's condition, but human verification keeps it safe.

💬 Discuss with your classmates

1. Which machine at your college could benefit from predictive maintenance?

2. Who is responsible if an AI maintenance prediction is wrong?

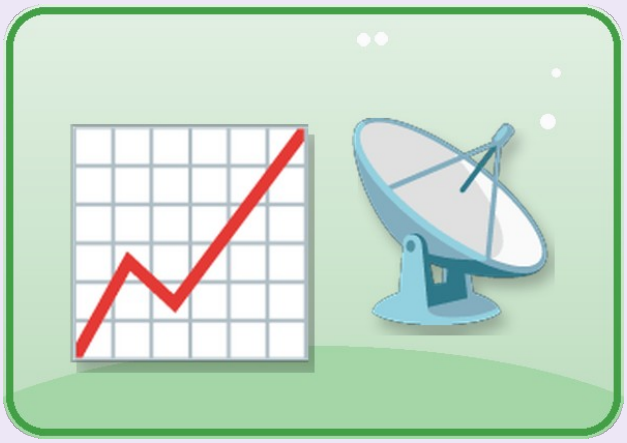
SCENARIO: AI for Engineering Solutions

Read the scenario. Think about the choices made at each step.

1

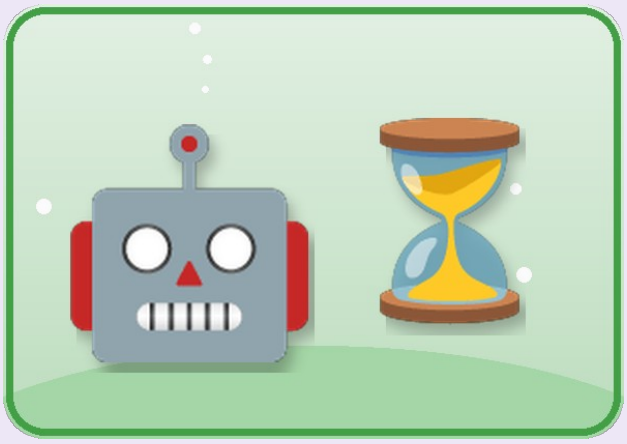
A cement plant's kiln fan fails without warning.

2



Engineers add vibration & temperature sensors.

3



An ML model estimates remaining useful life daily.

4

Repairs are planned; unplanned downtime drops sharply.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: AI across engineering

Jet-engine health

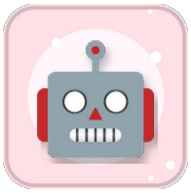
Bridge crack detection

Grid load balancing

Battery life prediction

Defect detection on lines

Water-leak localisation



BOLT SAYS:

Sensors give me vibrations and temperatures. I tell you which machine will fail next!

FUN FACT!

Generative design can produce lighter parts with shapes no human would sketch!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Generative Design of a Car Seat Bracket

CASE FILE

General Motors worked with Autodesk generative design software to redesign a seat-belt bracket. The AI explored many organic-looking options under strength and manufacturing constraints. The chosen 3D-printed design combined eight parts into one, and was reported to be about 40% lighter and 20% stronger than the original.

🔍 Analyse the case

1. Why can generative design find shapes humans miss?

2. Why must engineers still validate the AI design?

3. What manufacturing limits affect adoption?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Sensor Failure Forecast

You need:

Vibration data

Spreadsheet

Teams of 3

Steps:

1.
   Plot daily vibration readings for 4 motors.
2.
   Set a threshold where failures happened before.
3.
   Estimate days left (RUL) for each motor.
4.
   Schedule maintenance to minimise downtime.

THINK & TALK

What is the cost of a false alarm vs a missed failure?

Record your observations & results:

💻 CODE LAB: Remaining Useful Life

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import numpy as np
2  days = np.array([0, 1, 2, 3, 4])
3  wear = np.array([40, 43, 45, 49, 52])
4  slope, intercept = np.polyfit(days, wear, ____)
5  rul = (90 - wear[-1]) / slope
6  print(round(slope, 1), round(rul, 1))
```

COMPLETE **T1. Fit a straight line (degree?).**

PREDICT **T2. Slope is about? RUL about?**

EXPLAIN **T3. Why is a linear fit risky for wear?**

EXTEND **T4. Add a 20% safety margin to RUL.**

WORKSHEET: AI for Engineering Solutions (Part A)

Name: _____ Date: _____

1. Match the engineering field to its AI use.

Aerospace ●

● Load forecasting

Civil ●

● Engine health monitoring

Electrical ●

● Defect detection

Manufacturing ●

● Crack detection

2. RUL by linear trend. Use the word box.

WORD BOX: 10 ✿ 20 ✿ 5

Vibration 60, limit 80, +2 per day → days left: ____

Vibration 40, limit 80, +2 per day → ____

Vibration 70, limit 80, +2 per day → ____

WORKSHEET: AI for Engineering Solutions (Part B)

Name: _____ Date: _____

1. RUL = (limit - current) ÷ rate. Use the word box.

WORD BOX: 15 ✿ 8 ✿ 30

(90 - 45) ÷ 3 = ____ days

(100 - 60) ÷ 5 = ____ days

(80 - 20) ÷ 2 = ____ days

2. Preventive or predictive maintenance? Tick ✓.

	Prevent.	Predict.
Change oil every 6 months		
Replace bearing when vibration trend crosses limit		
Service lift every quarter		
Schedule repair from AI RUL estimate		

PUZZLE & DEFINE: AI for Engineering Solutions

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Digital twin

Meaning: _____

Example: _____

RUL

Meaning: _____

Example: _____

Gen. design

Meaning: _____

Example: _____

? QUIZ TIME: AI for Engineering Solutions

Circle the right answer!

Q1. A live virtual replica is a...

a) Digital twin

b) Real twin

c) Photo

Q2. RUL stands for remaining useful...

a) Length

b) Life

c) Loudness

Q3. AI-explored part shapes:

a) Cut design

b) Paint design

c) Generative design

Q4. Fixing before failure is...

a) Predictive maint.

b) Emergency repair

c) Luck

Q5. Best results often combine...

a) Luck + hope

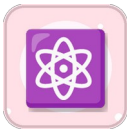
b) Physics + data

c) Music + art

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: AI for Engineering Solutions

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Digital twins mirror real systems with live data.		
Predictive maintenance follows a fixed calendar.		
RUL estimates remaining useful life.		
 Physics-informed ML embeds known equations.		
AI can replace engineers' legal sign-off.		
Generative design explores many shapes under constraints.		

HOME CONNECT

Spot one machine at home (fridge, fan, bike). What sensor signals could predict its failure?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI for Engineering Solutions

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do digital twins need live sensor data, not just CAD models?**

COMPARE **Q2. Compare preventive and predictive maintenance.**

APPLY **Q3. Wear index 45, limit 90, rising 3/day. Estimate RUL.**

EVALUATE **Q4. Should AI alone certify a bridge as safe?**

CREATE **Q5. Design a sensor + AI plan for a college water pump.**

📝 EXAM PREP: AI for Engineering Solutions

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is a digital twin?

2. Define RUL.

3. Preventive vs predictive maintenance?

4. What is a surrogate model?

PART B • Long answer (5 marks)

5. Describe an AI-based predictive-maintenance system for a factory, from sensors to decisions, and its safety requirements.

UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. ₹250 × 12 months CLV =

a) ₹3,000 b) ₹262 c) ₹2,500

Q2. Offer to a loyal customer is a...

a) TP b) FP c) FN

Q3. Zillow Offers was...

a) A big success b) Expanded c) Shut down

Q4. (90 - 45) ÷ 3 = RUL of...

a) 15 days b) 45 days c) 3 days

Q5. Condition-based maintenance:

a) Preventive b) Predictive c) Reactive

Q6. GM bracket was redesigned with...

a) Hand sketches b) Dice c) Generative design

WORD SEARCH: AI SOLVE

🔍 **Find these words. They go across → or down ↓.**

CHURN

CLV

COST MATRIX

RUL

GEN. DESIGN

TWIN

U	I	K	D	C	T	W	I	N	P
R	R	C	R	M	C	O	S	E	R
A	C	H	U	R	N	K	E	O	I
O	D	B	L	W	W	B	N	N	C
U	L	T	I	S	H	B	S	K	I
A	C	D	W	T	B	G	O	O	N
K	W	I	E	C	L	V	R	B	G
L	L	M	A	R	G	I	N	M	E
C	O	S	T	M	A	T	R	I	X
G	E	N	.	D	E	S	I	G	N

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

Pick Your AI Path

Topic 8

Portfolio to Profession

TOPIC 7 • MONTH 4 • AI & CAREER

Pick Your AI Path

LET'S LEARN

- Specialise: data, ML engineering, NLP, vision, MLOps, policy.
- Match path to strengths, interests and market demand.
- Electives, projects & internships signal your path.

AI specialisations

Data science

NLP

Computer vision

MLOps

KEY TERMS

NLP Language processing

Vision AI Image understanding

MLOps ML in production

📖 CONCEPT EXPLAINED: Pick Your AI Path

💡 In this chapter

By the second year, AI stops being one subject and becomes many possible careers. Natural language processing, computer vision, data science, MLOps, research and policy each demand different strengths. This chapter maps the main specialisations, gives you a structured way to choose, and shows how to signal your chosen path while keeping room to pivot.

🗺️ 1. The map of AI specialisations

Data science and analytics turn data into insights and predictions for decisions. Natural Language Processing (NLP) builds systems that understand and generate human language: chatbots, translation, sentiment analysis and speech. Computer vision teaches machines to interpret images and video: quality inspection, medical imaging, traffic monitoring.

MLOps engineers deploy, monitor and maintain models in production, using tools like Docker, CI/CD pipelines and monitoring dashboards. Research scientists invent new methods. AI product managers decide what to build. Governance and policy specialists ensure AI is safe, fair and legal. Increasingly, domain-AI roles combine a field like health, finance or agriculture with AI skills.

🤔 PAUSE & THINK

Which two specialisations excite you most right now? Write one reason for each.

🧰 2. What each path needs

Every path shares foundations: Python, mathematics (linear algebra, probability, statistics), data handling and core machine learning. On top of that, each adds a specialised stack. NLP needs tokenisation, embeddings and transformer models. Vision needs image processing, convolutional networks and libraries like OpenCV. MLOps needs software engineering, containers, cloud platforms and monitoring.

Analytics needs SQL, visualisation and business communication. Research needs deep mathematics, reading papers and running careful experiments. Policy needs law, ethics and the ability to explain technology to non-experts.

✏️ **WORKED EXAMPLE**

NLP starter stack: Python → text cleaning → TF-IDF baseline → word embeddings → fine-tuning a small transformer → evaluating with F1 score.

🧮 3. A structured way to choose

A simple decision framework rates each path on three factors from 1 to 5: interest (how much you enjoy it), strength (how good you are or could become), and demand (how many opportunities exist, judged from real job postings). Multiplying or weighting these factors gives a score that makes your reasoning visible.

Then add personal values: preferred location, salary expectations, desire for research or for building products, and the kind of impact you want. Scores are not destiny, but they turn vague worry into informed choice, and you can discuss them with mentors.

🖐️ **TRY IT**

Score three paths on interest, strength and demand (1–5 each). Multiply to rank them.

📣 4. Signalling your path and keeping options open

Once you choose a direction, signal it consistently: choose related electives, build two or three focused projects, seek an internship in that area, write about what you learn and join relevant communities or competitions. Recruiters look for this coherent story.

Your choice is not permanent. Strong foundations let you pivot: a vision engineer can move into robotics, an analyst into data science, an MLOps engineer into platform engineering. Choosing a path now is about direction and depth, not locking a door forever.

🤔 **PAUSE & THINK**

If your first-choice path becomes less popular in three years, how could you pivot using the same foundations?

✅ Key takeaways

- Specialisations: data science, NLP, vision, MLOps, research, product, policy.
- All share foundations; each adds a specialised stack.
- Score paths on interest × strength × demand, then add values.
- Signal your path with electives, projects and internships.
- Strong foundations make later pivots possible.

❓ Quick check: answer in one or two lines.

1. Name four AI specialisations and one tool for each.

2. Explain the interest × strength × demand framework.

3. List three ways to signal your chosen path.

🔬 CONCEPT DEEP DIVE: Pick Your AI Path

🔷 Common AI paths: data analyst/scientist, ML engineer, NLP engineer, computer-vision engineer, MLOps, research, AI product, policy.

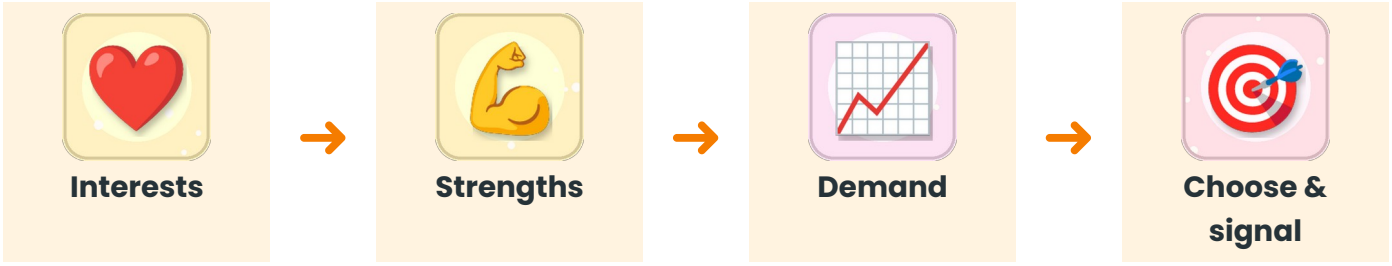
🔷 Each path has a core stack: NLP (transformers, tokenisers), CV (CNNs, OpenCV), MLOps (Docker, CI/CD, monitoring).

🔷 Decision framework: interest × strength × demand, plus values (impact, location, salary, learning curve).

🔷 Signal your path: electives, 2–3 focused projects, a relevant internship, and community participation.

🔷 Paths are not permanent; transferable foundations (maths, Python, data) let you pivot later.

🔄 How it works



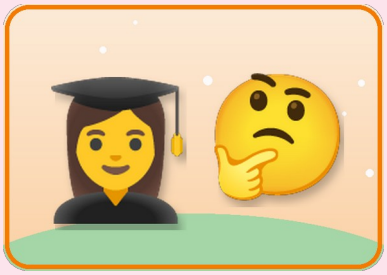
💻 Ranking paths in Python

```
paths = {'NLP': (5, 3, 4), 'Vision': (4, 4, 3),
         'MLOps': (3, 3, 3)}
score = {p: i * s * d for p, (i, s, d) in paths.items()}
best = sorted(score, key=score.get, reverse=True)
print(best, score)   # ['NLP', 'Vision', 'MLOps']
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Three Doors

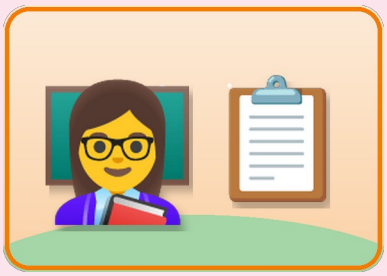
Characters: *Kavya (B.Tech student), her mentor Dr. Fatima, her grandmother Paati.*



SCENE 1

Kavya loved Tamil poetry, enjoyed designing posters and was good at coding. In her second year, everyone around her seemed to have chosen a path: data science, cloud, vision. Kavya felt lost. 'What if I choose wrong?' she asked Dr. Fatima.

Pause & predict: What advice would you give Kavya right now?



SCENE 2

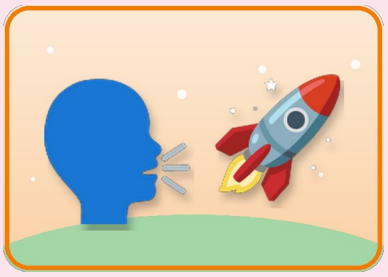
Dr. Fatima handed her a sheet with six paths. 'Score each on interest, strength and demand, from 1 to 5. Use real job posts for demand.' Kavya spent a weekend reading postings and scoring. NLP scored 60, computer vision 48, MLOps 27.

Pause & predict: What does the scoring framework help Kavya see?

SCENE 3

At home, Paati asked what NLP meant. 'Teaching computers to understand language, like Tamil,' Kavya said. Paati's eyes lit up. 'Then make one that can read my old letters aloud.' Kavya realised her path could connect her love of language with her skills.

Pause & predict: How did personal values influence Kavya's choice?



SCENE 4

Kavya chose NLP electives and built a Tamil sentiment classifier, then a text-to-speech demo for Paati's letters. She kept computer vision as a backup and kept improving her maths and Python. 'The door isn't locked behind me,' she told Dr. Fatima. 'I'm just walking in a direction.'

⏸ **Pause & predict:** Why did Kavya keep a backup path?

🌟 The big idea

Choose your AI path with evidence and heart, and keep your foundations strong so you can always pivot.

💬 Discuss with your classmates

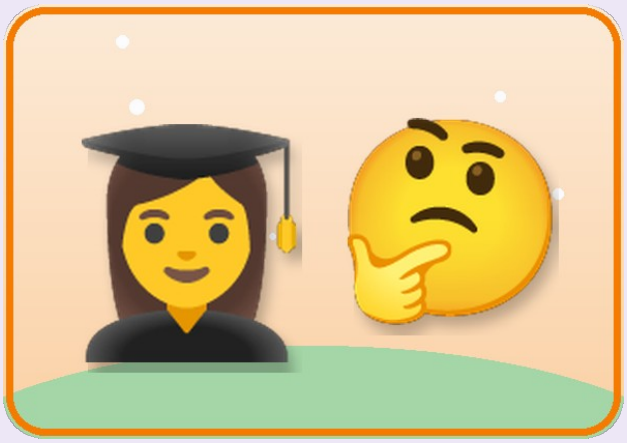
1. Score your top three AI paths. Which wins? Does the result feel right?

2. How could your personal interests shape your AI career?

SCENARIO: Pick Your AI Path

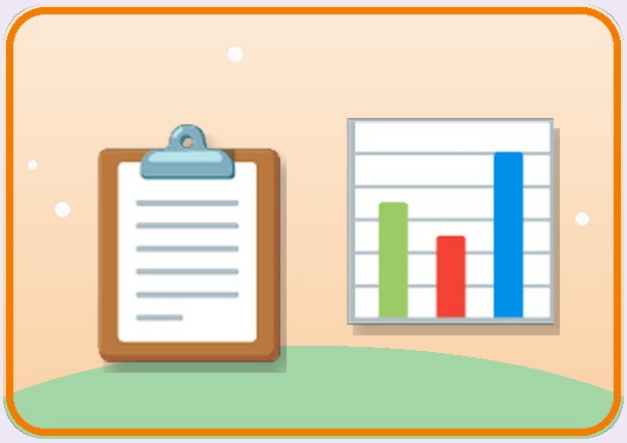
Read the scenario. Think about the choices made at each step.

1



Kavya likes language, design and coding. Which AI path?

2



She scores 6 paths on interest, strength & demand.

3

NLP wins; Computer Vision is her backup.

4

She picks NLP electives and a Tamil chatbot project.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

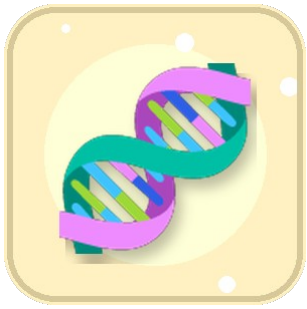
LOOK AROUND YOU: Specialists at work

Indian-language chatbots

Retail shelf vision

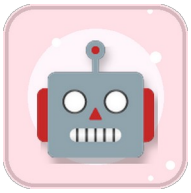
Model deployment

Business analytics



Health AI research

AI policy & audit



BOLT SAYS:

You don't need to master all of AI. Go deep in one area, stay curious about the rest!

FUN FACT!

Indian-language AI is a fast-growing area, with initiatives like Bhashini building speech and translation tools!

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: Pivoting from Mechanical to MLOps

CASE FILE

A mechanical engineering student enjoyed automating lab equipment with Python. In second year she took cloud and DevOps courses online, contributed to an open-source monitoring tool and built a pipeline that retrained a vibration model automatically. Her internship in MLOps came from a maintainer she met through open source.

Analyse the case

1. Which transferable skills enabled her pivot?

2. How did open source create a hiring channel?

3. Map a similar pivot route from your branch.

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Path-Finder Matrix

You need:

Matrix sheet

Job portals

Pairs

Steps:

1		Rate 6 specialisations on interest (1–5).
2		Rate your current strength in each (1–5).
3		Rate demand (1–5) from 10 job posts each.
4		Score = interest × strength × demand. Pick top 2.

THINK & TALK

Should demand outweigh interest? Discuss with your pair.

Record your observations & results:

💻 CODE LAB: Path Scoring

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
paths = {'NLP': (5, 3, 4), 'Vision': (4, 4, 3),
         'MLOps': (3, 3, 3), 'Analytics': (3, 5, 5)}
w = (0.5, 0.3, 0.2)   # weights: interest, strength, demand
score = {p: sum(a * b for a, b in zip(v, ____))
         for p, v in paths.items()}
print(max(score, key=score.get), score)
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Compute the NLP and Analytics scores.**

PREDICT **T3. Which path wins?**

MODIFY **T4. Change weights to (0.2, 0.3, 0.5). Winner?**

WORKSHEET: Pick Your AI Path (Part A)

Name: _____ Date: _____

1. Match the specialisation to a typical project.

NLP ●

● CI/CD pipeline for a model

Computer vision ●

● Tamil sentiment classifier

MLOps ●

● Sales driver analysis

Data science ●

● Helmet detection on CCTV

2. Path score = interest × strength × demand. Use the word box.

WORD BOX: 60 ✿ 48 ✿ 27

NLP: 5 × 3 × 4 = ____

Vision: 4 × 4 × 3 = ____

MLOps: 3 × 3 × 3 = ____

WORKSHEET: Pick Your AI Path (Part B)

Name: _____ Date: _____

1. Path scores (interest × strength × demand). Use the word box.

WORD BOX: 75 ✿ 48 ✿ Analytics

Vision: 4 × 4 × 3 = ____

Analytics: 3 × 5 × 5 = ____

Higher score: ____

2. Match the path to its core tool.

NLP ● ● **Docker & CI/CD**

Computer vision ● ● **SQL & dashboards**

MLOps ● ● **Tokenisers & transformers**

Analytics ● ● **OpenCV & CNNs**

PUZZLE & DEFINE: Pick Your AI Path

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

NLP

Meaning: _____

Example: _____

Vision AI

Meaning: _____

Example: _____

MLOps

Meaning: _____

Example: _____

? QUIZ TIME: Pick Your AI Path

Circle the right answer!

Q1. Language AI is called...

a) NLP

b) CV

c) MP3

Q2. Running ML in production:

a) DesignOps

b) MLOps

c) ChefOps

Q3. Detecting helmets in video is...

a) NLP

b) Accounting

c) Computer vision

Q4. Indian-language AI initiative:

a) Bhashini

b) Bollywood

c) IPL

Q5. Good path choice weighs interest, strength &...

a) Luck

b) Demand

c) Moon

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Pick Your AI Path

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	NLP works with human language.		
	Your AI path can never change later.		
	MLOps focuses on deploying and monitoring models.		
	Market demand is one factor in choosing a path.		
	Salary should be the only factor.		
	Open source can lead to internships.		

HOME CONNECT

Interview someone working in tech about how they chose their specialisation.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Pick Your AI Path

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN

Q1. Why are foundations (maths, Python, data) called 'transferable'?

COMPARE

Q2. Compare the daily work of an NLP engineer and an MLOps engineer.

APPLY

Q3. Scores: CV (4, 4, 3), Analytics (3, 5, 5). Which wins?

EVALUATE

Q4. Is choosing the highest-paying path always wise?

CREATE

Q5. Plan 3 semesters of electives + projects for your chosen path.

📝 EXAM PREP: Pick Your AI Path

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name four AI specialisations.

2. What is MLOps?

3. Why are foundations transferable?

4. List three ways to signal a chosen path.

PART B • Long answer (5 marks)

5. Using a decision framework, justify your choice of AI specialisation and plan the next three semesters.

TOPIC 8 • MONTH 4 • AI & CAREER

Portfolio to Profession

LET'S LEARN

- A portfolio proves skills; a profile gets it seen.
- Show problem → approach → result → impact.
- Interviews test fundamentals, projects & communication.

Portfolio pieces

GitHub repos

Case-study write-ups

Demo videos

Competition results

KEY TERMS

Case study
Project story write-up

Behavioural
Past-behaviour questions

Referral
Recommended by insider

📖 CONCEPT EXPLAINED: Portfolio to Profession

💡 In this chapter

Skills only open doors when others can see them. This chapter explains how to turn your projects into a portfolio that tells a clear story, how to build a professional profile, how to prepare for different interview rounds, and how to approach offers thoughtfully.

🗂️ 1. Quality over quantity

Recruiters spend only minutes on each candidate. A portfolio with twelve half-finished notebooks named 'final_v2' signals confusion. Three polished, relevant projects signal capability. Choose projects that match your target role, show a range of skills and solve problems someone cares about.

Each project should run reliably, have a clear README, show results with metrics compared to a baseline, and include a short demo, such as a two-minute video or a live web app.

🖐️ TRY IT

List your current projects. Which three best match the role you want? What does each still need?

📝 2. Writing a project case study

A case study tells the story of a project: the problem and why it mattered, the data and its challenges, your approach and the decisions you made, results with metrics, limitations and what you would do next, and your specific role if it was a team project.

Numbers make stories credible: 'Reduced forecast error from 15% to 8.4% MAPE, a 44% improvement over the baseline' is far stronger than 'built a good model'. Honesty matters too: describing limitations shows maturity, and claiming others' work as your own can end opportunities instantly.

✏️ WORKED EXAMPLE

Headline: 'Crop Price Forecaster — gradient boosting on 5 years of Agmarknet data; MAPE 8.4% vs 15% baseline; Streamlit demo; built the data pipeline and model.'

🎤 3. Preparing for interviews

AI-related hiring usually includes several rounds. Technical rounds test fundamentals: statistics, ML concepts, Python and sometimes data structures and SQL. Project deep-dives ask why you made each decision and what trade-offs you considered. Behavioural rounds ask about teamwork, conflict and failure, best answered with the STAR method (Situation, Task, Action, Result).

Some companies give take-home assignments or case studies. Practise by explaining your projects aloud, doing mock interviews with friends, and reviewing core concepts regularly rather than cramming.

🤔 PAUSE & THINK

Write a STAR answer outline for: 'Tell me about a time a project did not go as planned.'

🤝 4. Profiles, networks and offers

Your professional profile should have a clear headline ('Aspiring NLP Engineer | Tamil sentiment model, F1 0.86'), skills that match target roles, featured projects and honest descriptions. Networking is not about collecting contacts but about genuine conversations: asking thoughtful questions, sharing what you learn and helping others.

Referrals from people who have seen your work are among the strongest routes to interviews. When offers arrive, compare learning opportunities, mentorship, role fit, location and pay. Negotiation, done politely with evidence, is normal and professional.

🖐 TRY IT

Write a two-line professional headline for yourself.

✅ Key takeaways

- Three polished, relevant projects beat many unfinished ones.
- Case studies: problem, data, approach, results, limits, role.
- Use numbers and honesty to build credibility.
- Prepare for technical, project, behavioural and take-home rounds.
- Network genuinely; evaluate offers on more than salary.

❓ Quick check: answer in one or two lines.

1. What should a project README include?

2. Explain the STAR method.

3. Why are referrals powerful?

🔬 CONCEPT DEEP DIVE: Portfolio to Profession

🔷 Quality beats quantity: 3 polished, relevant projects outperform 12 unfinished ones.

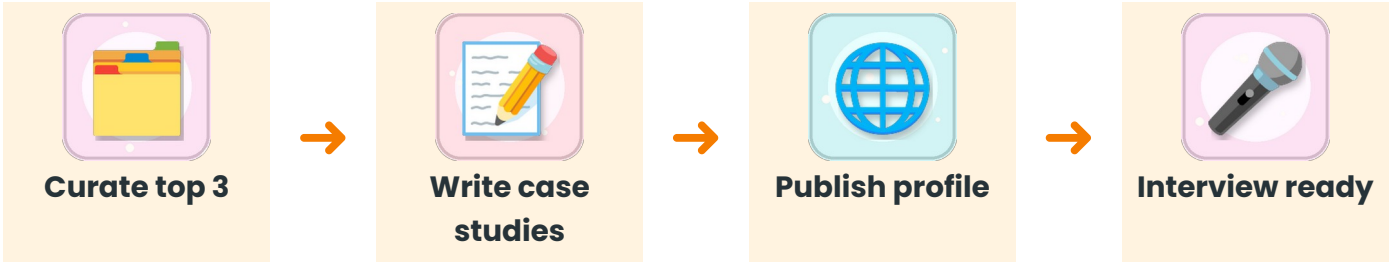
🔷 Each project page: problem, data, approach, results (with metrics), limitations, demo link, your role.

🔷 Professional profile: clear headline, skills matching target roles, featured projects, recommendations.

🔷 Interview prep: fundamentals (stats, ML, DSA basics), project deep-dives, behavioural (STAR), and a case question.

🔷 Offer stage: compare role, learning, mentorship, location and pay; negotiate politely with evidence.

🔄 How it works



💻 A strong project README

```
## Crop Price Forecaster
Problem: farmers sell at low prices | Data: Agmarknet 5 yrs
Model: gradient boosting | MAPE: 8.4% (baseline 15%)
Demo: <link> | My role: data pipeline + model + UI
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Twelve Repos, No Calls

Characters: *Vivek (B.Sc student), Anjali (his senior, now a data scientist).*

SCENE 1

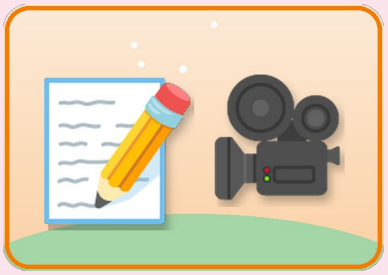
Vivek had twelve GitHub repositories and had applied to thirty internships. Not one reply. 'Maybe I need more projects,' he thought, and started a thirteenth.

Pause & predict: Is a thirteenth project likely to solve Vivek's problem? Why?

SCENE 2

Anjali reviewed his GitHub over coffee. 'I can't tell what any of these do,' she said gently. 'No READMEs, no results, files named final_final2. A recruiter will leave in thirty seconds.'

Pause & predict: What specific problems did Anjali spot?



SCENE 3

Vivek chose his three best projects. For each he wrote a case study: the problem, the data, his approach, results compared with a baseline, limitations and his role. He recorded two-minute demo videos and rewrote his profile headline: 'Aspiring Data Analyst | Crop price forecaster, 44% better than baseline.'

Pause & predict: Which change do you think made the biggest difference? Why?

SCENE 4

Within a month, three recruiters contacted him, and Anjali referred him to her company. In the interview, he explained every decision in his project confidently. He received an internship offer and politely negotiated a start date that fitted his exams.

⏸ **Pause & predict:** How did the polished portfolio help during the interview?

🌟 The big idea

A portfolio is a story about your skills. Make it clear, honest and easy to see.

💬 Discuss with your classmates

1. Which of your projects would you polish first? What does it need?

2. Write a one-line headline for your professional profile.

SCENARIO: Portfolio to Profession

Read the scenario. Think about the choices made at each step.

1

Vivek has 12 GitHub repos but no interview calls.

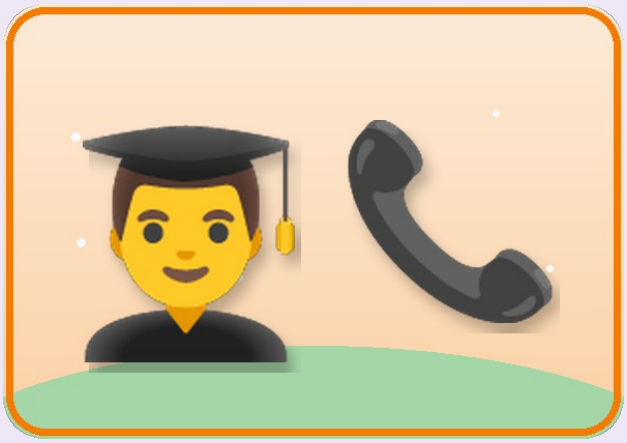
2

A mentor finds: no READMEs, no results, no demos.

3

He polishes his top 3 with case studies and demo videos.

4



Within a month, 3 recruiters reach out!

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: From portfolio to offer

Recruiter outreach

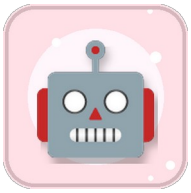
Technical interviews

Take-home assignments

Referrals

Hackathon hires

Tech talks & blogs



BOLT SAYS:

A recruiter spends minutes on your portfolio. Make the first screen tell the story!

FUN FACT!

Referrals from people who know your work are among the strongest hiring channels!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Hired Through a Hackathon

CASE FILE

At a national student hackathon, a four-member team built a speech-to-text tool for a regional language in 36 hours. A sponsoring company's engineers judged the event. Two team members were invited to interview directly after the demo, and both received internship offers, citing the live demo and clear problem framing.

🔍 Analyse the case

1. What made the live demo a strong hiring signal?

2. How could the other two members leverage this project?

3. Plan how to present a hackathon project in your portfolio.

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Mock Interview Circuit

You need:

Question cards

Timer

Trios

Steps:

1.
   Roles: candidate, interviewer, observer (rotate).
2.
   Round 1: 2 fundamentals questions (5 min).
3.
   Round 2: explain one project with STAR (5 min).
4.
   Observer scores clarity, depth, honesty (1–5).

THINK & TALK

Which question type was hardest? How will you practise?

Record your observations & results:

💻 CODE LAB: Portfolio Metrics

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  baseline_mape, model_mape = 15.0, 8.4
2  gain = (baseline_mape - model_mape) / ____ * 100
3  repos = {'crop': 12, 'chat': 3, 'vision': 7}   # stars
4  top = sorted(repos, key=repos.get, reverse=True)[:2]
5  print(round(gain), top)
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. What is printed?**

EXPLAIN **T3. Is GitHub stars a good quality metric? Why?**

EXTEND **T4. Write a README line using the gain value.**

WORKSHEET: Portfolio to Profession (Part A)

Name: _____ Date: _____

1. Project case-study structure. Write 1, 2, 3, 4.

Results & impact ☐

Problem & context ☐

Approach & model ☐

Data & challenges ☐

2. Strong or weak portfolio signal? Tick ✓.

	Strong	Weak
README with problem, results, demo link		
Repo with only 'final_final2.ipynb'		
2-min demo video		
Forked tutorial with no changes		

WORKSHEET: Portfolio to Profession (Part B)

Name: _____ Date: _____

1. Improvement over baseline. Use the word box.

WORD BOX: **44%** ✿ **6.6** ✿ **25%**

Error reduction: 15 - 8.4 = ____ points

Relative improvement: 6.6 ÷ 15 = ____

Accuracy 80% → 85%? Error 20% → 15%: error cut = ____

2. Match the interview round to its focus.

Technical ●

Project deep-dive ●

Behavioural ●

Take-home ●

● **Teamwork & conflict (STAR)**

● **Build a small task in days**

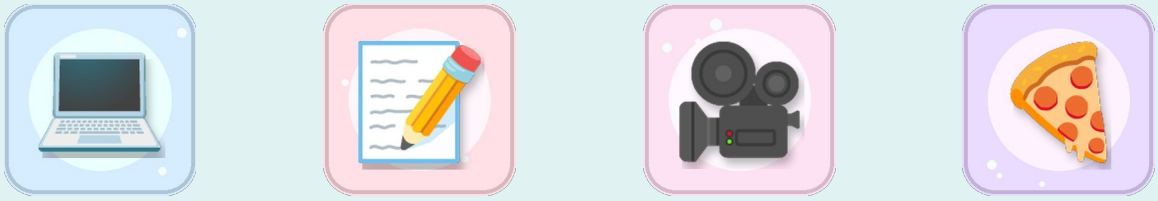
● **Stats, ML & coding basics**

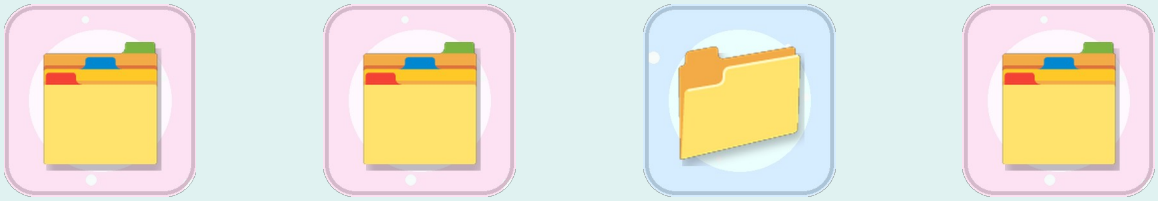
● **Your decisions & trade-offs**

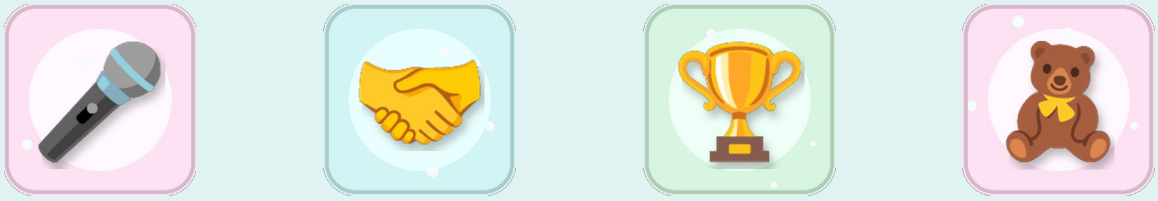
PUZZLE & DEFINE: Portfolio to Profession

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A 

B 

C 

2. Define it! Write each term in your own words with an example.

Case study

Meaning: _____

Example: _____

Behavioural

Meaning: _____

Example: _____

Referral

Meaning: _____

Example: _____

? QUIZ TIME: Portfolio to Profession

Circle the right answer!

Q1. Best first section of a case study:

a) Problem

b) Logo

c) Phone

Q2. 'Tell me about a conflict' is...

a) Technical

b) Behavioural

c) Random

Q3. Being recommended by an insider:

a) Gift

b) Spam

c) Referral

Q4. Unchanged tutorial forks are...

a) Weak signals

b) Strong

c) Rare

Q5. Short demos are best as...

a) Long essays

b) 2-min videos

c) Radio

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Portfolio to Profession

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🗂️	Three polished projects can beat twelve unfinished ones.		
📦	READMEs are unnecessary if code is good.		
🎥	Demo videos help recruiters understand projects.		
🤝	Referrals are a strong hiring channel.		
👥	Claiming a team project as solely yours is fine.		
💬	Polite, evidence-based negotiation is acceptable.		

HOME CONNECT

Ask a senior or relative to review your best project README and give one honest suggestion.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Portfolio to Profession

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do recruiters value READMEs with metrics?**

COMPARE **Q2. Compare a resume line and a project case study.**

APPLY **Q3. Baseline MAPE 15%, your model 8.4%. Relative improvement?**

EVALUATE **Q4. Should you list group projects as solely yours?**

CREATE **Q5. Write a 2-line professional headline for your target role.**

📝 EXAM PREP: Portfolio to Profession

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What should a project case study include?

2. Why do READMEs matter?

3. Name four interview round types.

4. What is a referral?

PART B • Long answer (5 marks)

5. Explain how to convert a portfolio into job offers, covering curation, profile, outreach, interviews and negotiation.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. Path score 3 × 5 × 5 =

a) 75 b) 13 c) 35

Q2. Docker & CI/CD belong to...

a) NLP b) MLOps c) Design

Q3. Bhashini focuses on...

a) Cricket b) Films c) Indian languages

Q4. MAPE 15 → 8.4 improvement ≈

a) 44% b) 6.6% c) 84%

Q5. 'Tell me about a failure' is...

a) Technical b) Behavioural c) Take-home

Q6. Claim a team project as solo?

a) Yes b) Sometimes c) No, state your role

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

NLP	VISION AI	MLOPS	CASE STUDY	REFERRAL	PATH

P	I	P	R	O	F	I	L	E	S
N	C	A	S	E	S	T	U	D	Y
L	D	O	C	K	E	R	T	R	L
P	E	G	C	O	G	P	I	F	M
O	U	L	T	G	L	P	A	T	H
D	R	E	F	E	R	R	A	L	B
H	I	U	W	M	L	O	P	S	H
D	O	F	F	E	R	L	F	K	P
A	V	I	S	I	O	N	A	I	B
M	C	K	L	A	L	K	L	E	O

🏅 **I finished AI RISE! Teacher's sign: _____**

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

AI Data Studio

Topic 10

Smart Automation Tools

TOPIC 9 • MONTH 5 • AI TOOLS

AI Data Studio

LET'S LEARN

- Visualise before you model: plots reveal what stats hide.
- Choose the right chart for the question.
- Dashboards turn analysis into ongoing decisions.

Right chart, right question

Trend → line

Compare → bar

Relation → scatter

Spread → box/hist

KEY TERMS

Dashboard Live visual summary

Encoding Data → visual property

Heatmap Colour-coded grid

📖 CONCEPT EXPLAINED: AI Data Studio

💡 In this chapter

Visualisation is where data meets human understanding. A good chart can reveal a pattern that statistics hide; a bad chart can mislead thousands. This chapter teaches how perception works, how to choose the right chart, how to design honestly, and how to build dashboards people actually use.

👁️ 1. How we perceive charts

Research in visual perception shows that people judge some visual encodings more accurately than others. Position along a common scale is the most accurate, followed by length, then angle and area, with colour intensity among the least precise for quantities. This is why bar charts usually beat pie charts for comparisons.

Colour is excellent for categories and highlighting, but should be used carefully: choose colour-blind-safe palettes, use sequential colours for ordered data and avoid rainbow scales that create false boundaries.

🤔 PAUSE & THINK

Five departments' placement rates differ by 2–3%. Would a pie chart or a bar chart show the differences better? Why?

📐 2. Choosing the right chart

Start with the question, not the chart. Trends over time suit line charts. Comparisons across categories suit bar charts. Distributions suit histograms, box plots or violin plots. Relationships between two numbers suit scatter plots. Composition (parts of a whole) suits stacked bars, or a simple pie with very few slices. Geography suits maps.

Always plot your data. Anscombe's quartet, published in 1973, contains four datasets with nearly identical means, variances, correlations and regression lines, yet their scatter plots look completely different: one linear, one curved, one with an outlier, one almost vertical.

✏️ WORKED EXAMPLE

Question: 'Do students who attend more score higher?' → scatter plot of attendance (x) vs marks (y), with a trend line and points coloured by department.

✅ 3. Designing honest charts

Honest charts start bar axes at zero, label axes with units, cite data sources and avoid 3D effects that distort size. Titles should state the main insight ('Section B leads by 9 points') rather than just the variable names.

Common tricks to avoid include truncated axes that exaggerate small differences, cherry-picked time ranges, dual axes that suggest false relationships, and maps that make large, sparsely populated areas look more important than dense cities.

🖐 TRY IT

Find a chart in a newspaper or on social media. List anything that could mislead a reader.

🖥 4. Building useful dashboards

A dashboard is a live visual summary for ongoing decisions. Good dashboards have one clear purpose and audience, show the most important KPIs at the top, allow drilling down with filters, use consistent colours and layout, and state when data was last refreshed.

Tools include Python libraries (Matplotlib, Seaborn, Plotly, Streamlit) and business tools (Power BI, Tableau, Looker Studio). Before building, sketch the dashboard on paper and test it with real users; a dashboard nobody understands creates no value.

🤔 PAUSE & THINK

List three KPIs a hostel warden would want on a dashboard.

✅ Key takeaways

- Position and length are judged most accurately; prefer bars to pies.
- Choose charts based on the question being asked.
- Anscombe's quartet: always plot your data.
- Honest design: zero baselines, labels, sources, no distortion.
- Dashboards need a clear purpose, key KPIs and user testing.

? Quick check: answer in one or two lines.

1. Which chart suits a trend over time?

2. What does Anscombe's quartet demonstrate?

3. Give two ways charts can mislead.

🔬 CONCEPT DEEP DIVE: AI Data Studio

🔷 Visual encodings rank by accuracy: position > length > angle > area > colour; prefer bars over pies for comparisons.

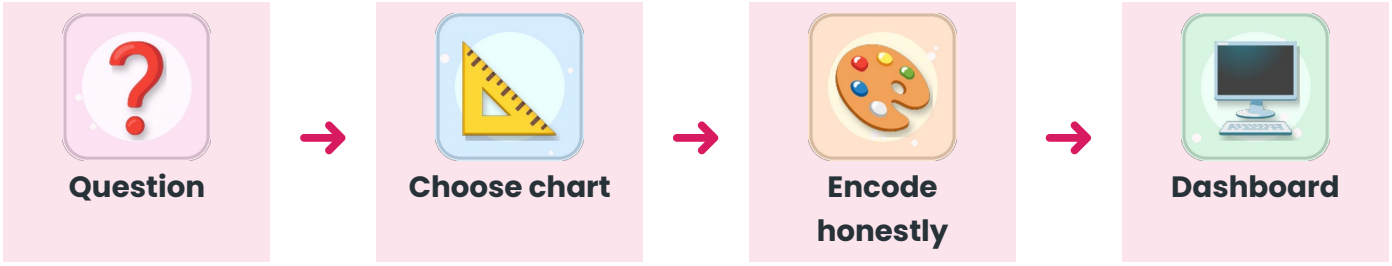
🔷 Chart choice follows the question: trend, comparison, distribution, relationship, composition, geography.

🔷 Honest charts: zero-based bars, labelled axes & units, colour-blind-safe palettes, no chart-junk or 3D distortion.

🔷 Dashboards: one purpose, key KPIs at top, filters, consistent colours, and a clear data-refresh schedule.

🔷 Tools: Matplotlib/Seaborn/Plotly in Python; Power BI, Tableau, Looker Studio for business dashboards.

🔄 How it works



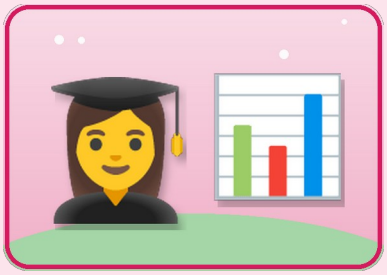
💻 A clear chart with Matplotlib

```
import matplotlib.pyplot as plt
plt.bar(['A', 'B', 'C'], [62, 71, 58])
plt.ylim(0, 100); plt.ylabel('Avg marks (%)')
plt.title('Section B leads by 9 points'); plt.show()
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Equal Averages, Unequal Stories

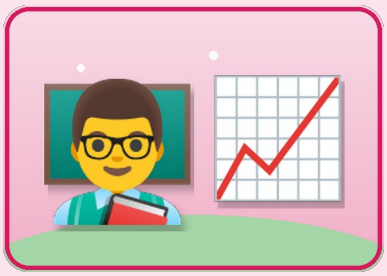
Characters: *Sneha (B.Sc Statistics student), Professor Raman, the Head of Department.*



SCENE 1

The Head of Department asked Sneha to compare two sections' exam results. She calculated the averages: 62% and 62%. 'Both sections performed equally,' she reported confidently.

Pause & predict: What might the averages be hiding?



SCENE 2

Professor Raman smiled. 'Plot the distributions.' Sneha made box plots and histograms. Section A's marks clustered between 55 and 70. Section B was split: many students above 80, and many below 40. Same average, very different stories.

Pause & predict: What different actions might each section need?

SCENE 3

Sneha built a dashboard: KPIs at the top (pass rate, median, top-10 share), box plots by section, a trend line across tests and filters for subjects. She started all bar axes at zero and wrote titles that stated insights, like 'Section B has twice as many failures.'

Pause & predict: Which honest-design rules did Sneha follow?

SCENE 4

The department arranged remedial classes for struggling Section B students and advanced projects for its toppers. Months later, the failure rate dropped. The Head of Department now asks every report to include at least one plot. Sneha calls it 'the Anscombe rule'.

⏸ **Pause & predict:** Why is 'always plot your data' important for decisions?

🌟 The big idea

Averages can hide the truth. Honest, well-chosen charts reveal it.

💬 Discuss with your classmates

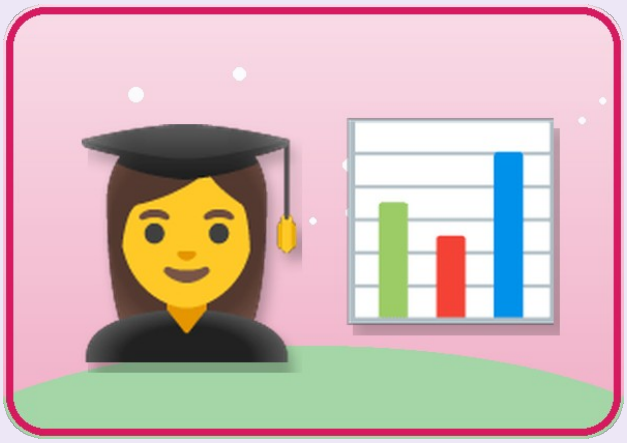
1. Find a situation where an average could mislead your college.

2. What would you put on a dashboard for your class representative?

SCENARIO: AI Data Studio

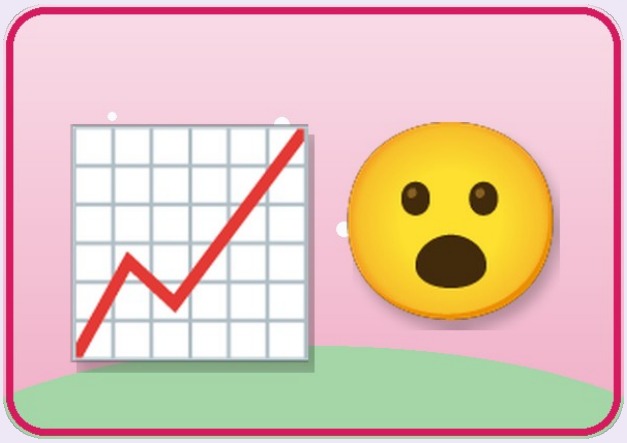
Read the scenario. Think about the choices made at each step.

1



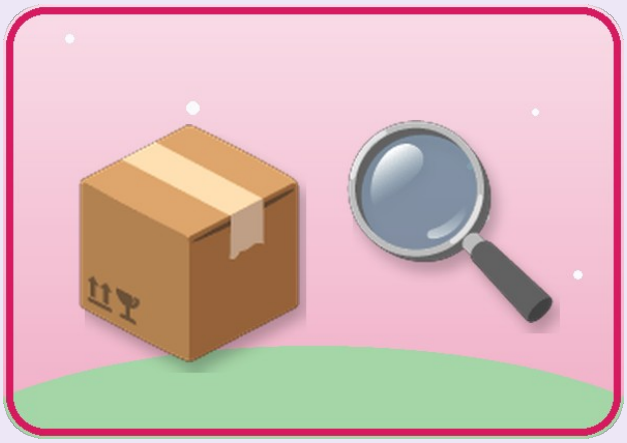
Sneha reports: 'average marks are equal across sections'.

2



Her professor asks her to plot the distributions.

3



Box plots reveal one section with many fails and many toppers!

4

She builds a dashboard showing spread, not just averages.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Data studios in action

Public health dashboards

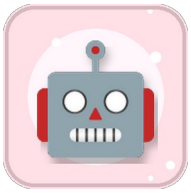
Sales cockpits

District crop maps

Placement analytics

Traffic heatmaps

Energy monitors



BOLT SAYS:

Same mean, same correlation, totally different stories. Always plot your data!

FUN FACT!

Anscombe's quartet shows four datasets with identical statistics but very different plots!

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: Anscombe's Quartet

CASE FILE

In 1973, statistician Francis Anscombe created four small datasets. Each had nearly identical means, variances, correlation (about 0.82) and the same best-fit line. When plotted, they looked completely different: one linear, one curved, one with a single outlier, one vertical with a single influential point. The lesson: always visualise.

Analyse the case

1. Why did summary statistics fail to reveal the differences?

2. Which chart type exposes these differences best?

3. Find a modern dataset where plots changed a conclusion.

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Chart Makeover Studio

You need:

3 bad charts

Plotting tool

Pairs

Steps:

1		Critique 3 misleading charts (axes, 3D, colours).
2		Redesign each with the right chart type.
3		Add a clear title stating the main insight.
4		Class votes: which makeover is clearest?

THINK & TALK

How can a truncated y-axis mislead a reader?

Record your observations & results:

💻 CODE LAB: Reproducing Anscombe

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import seaborn as sns
2  df = sns.load_dataset('anscombe')
3  print(df.groupby('dataset')[['x', 'y']].____().round(2))
4  r = df.groupby('dataset').apply(lambda g: g.x.corr(g.y))
5  print(r.round(2))
6  sns.lmplot(data=df, x='x', y='y', col='dataset', col_wrap=2)
```

COMPLETE **T1. Fill the blank to get means.**

PREDICT **T2. What do you expect the 4 correlations to be?**

EXPLAIN **T3. What does the lmplot reveal?**

EXTEND **T4. Add a box plot of y per dataset.**

WORKSHEET: AI Data Studio (Part A)

Name: _____ Date: _____

1. Match the question to the best chart.

Monthly sales trend ●

Sales by region ●

Ads vs sales relation ●

Spread of exam marks ●

● Scatter plot

● Line chart

● Histogram / box plot

● Bar chart

2. Good or misleading chart practice? Tick ✓.

	Good	Misleading
Bar chart y-axis starts at zero		
3D pie chart with 9 slices		
Title states the key insight		
Rainbow colours for ordered data		

WORKSHEET: AI Data Studio (Part B)

Name: _____ Date: _____

1. Visual perception ranking. Use the word box.

WORD BOX: Position ✿ Angle ✿ Colour

Most accurate encoding: ____

Pie charts rely on ____

Least precise for quantities: ____

2. Circle signs of an honest chart.

Axis labels with units

3D exploded pie

Bars start at zero

Colour-blind-safe palette

Truncated bar axis

Source cited

PUZZLE & DEFINE: AI Data Studio

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Dashboard

Meaning: _____

Example: _____

Encoding

Meaning: _____

Example: _____

Heatmap

Meaning: _____

Example: _____

QUIZ TIME: AI Data Studio

Circle the right answer!

Q1. Best chart for trends over time:

a) Line chart

b) Pie chart

c) Map

Q2. Anscombe's quartet teaches us to...

a) Skip plots

b) Always plot data

c) Guess

Q3. A live visual summary is a...

a) Scroll

b) Painting

c) Dashboard

Q4. Bar charts should start at...

a) Zero

b) Ten

c) Anywhere

Q5. Relation between two numbers:

a) Pie

b) Scatter

c) Stacked bar

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: AI Data Studio

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
📈	Line charts suit trends over time.		
🥧	Pie charts are best for precise comparisons.		
📊	Anscombe's quartet shows the value of plotting.		
0️⃣	Bar charts should start at zero.		
🌈	Rainbow palettes are ideal for ordered data.		
💻	Dashboards should focus on key KPIs.		

HOME CONNECT

Find a chart in a newspaper. Is it honest? Redraw it better and explain your changes.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI Data Studio

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why are bar charts usually better than pie charts?**

COMPARE **Q2. Compare a histogram and a box plot.**

APPLY **Q3. Bars 62, 71, 58 drawn with y-axis from 55. Why misleading?**

EVALUATE **Q4. Is a dashboard with 25 charts effective?**

CREATE **Q5. Sketch a placement-cell dashboard: 4 tiles and 2 filters.**

📝 EXAM PREP: AI Data Studio

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Rank visual encodings by accuracy.

2. Bar vs pie: which is better for comparison?

3. What is Anscombe's quartet?

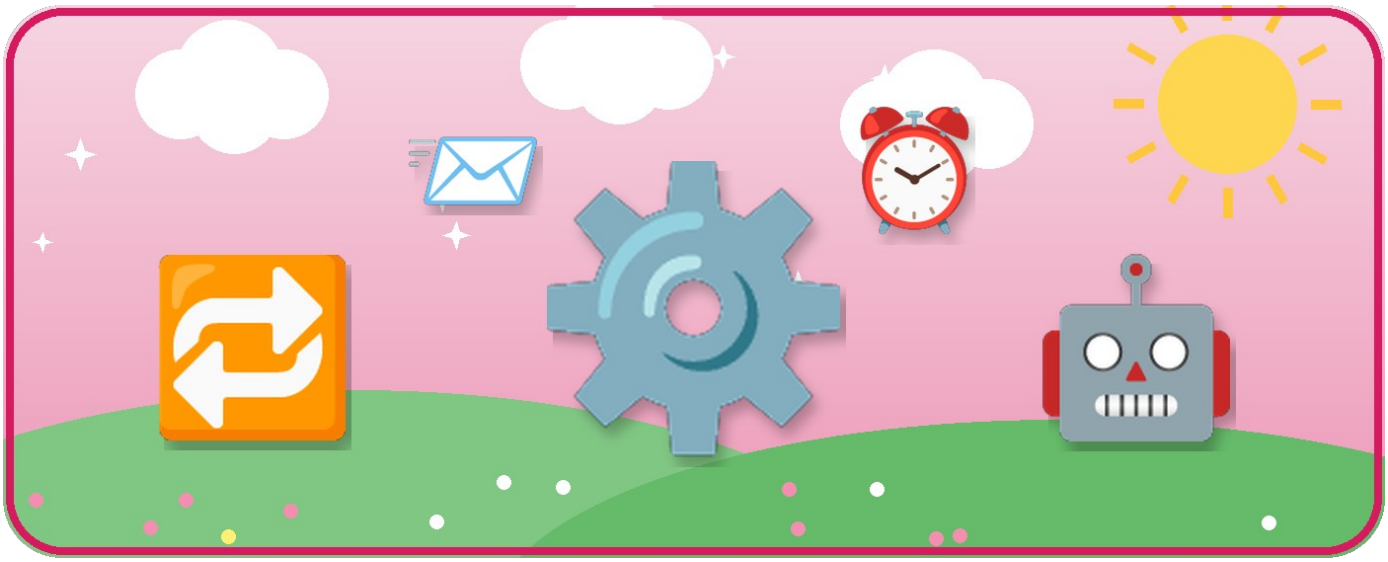
4. List three honest-chart rules.

PART B • Long answer (5 marks)

5. Design a dashboard for a college placement cell, justifying chart choices, layout and interactivity.

TOPIC 10 • MONTH 5 • AI TOOLS

Smart Automation Tools



LET'S LEARN

- Automate repeatable, rule-based, high-volume tasks.
- Tools: Python scripts, schedulers, RPA, no-code flows.
- Add logging, alerts and kill-switches for safety.

Automation building blocks

Scheduler / trigger

Script / flow

APIs

Monitoring

KEY TERMS

RPA Robotic process automation

Cron Time-based scheduler

Kill switch Emergency stop

📖 CONCEPT EXPLAINED: Smart Automation Tools

💡 In this chapter

Much office and technical work is repetitive: copying data between systems, generating reports, sending reminders, organising files. Automation tools can do this work faster and with fewer errors, freeing people for judgement and creativity. But automation can also fail spectacularly. This chapter shows how to automate wisely and safely.

🔁 1. What should be automated?

Good automation candidates are high-volume, repetitive, rule-based, stable over time, digital and error-prone when done by hand. Examples include copying invoice data into ledgers, generating weekly attendance reports and sending fee reminders.

Poor candidates involve judgement, empathy or frequently changing rules: counselling students, negotiating contracts, handling unusual complaints. Even good candidates often need humans to handle exceptions.

✏️ WORKED EXAMPLE

Task: 5 hours/week of manual report generation. Monthly saving ≈ 20 hours × ₹500/hour = ₹10,000. If building the automation costs ₹20,000, payback ≈ 2 months.

🧰 2. The automation toolbox

Python scripts can read files, call APIs, transform data and send emails. Schedulers like cron (Linux) or Task Scheduler (Windows) run scripts at set times: '0 9 * * 1-5' means 9:00 every weekday. Robotic Process Automation (RPA) tools mimic a person clicking through software screens, useful for old systems without APIs.

No-code automation platforms connect apps through triggers and actions ('when a form is submitted, add a row and send an email'). AI agents add flexibility by deciding steps themselves, but they also add unpredictability, so they need tighter controls.

🖐 TRY IT

Write the cron expression for 'every Friday at 6 pm'.

3. Building reliable automations

Reliable automations log what they do, alert humans when something unusual happens, retry temporary failures with increasing delays (back-off), and stop when errors pile up. They are designed to be idempotent: running a step twice should not create duplicate payments or emails.

Every automation should have limits (maximum emails per hour, maximum transaction size) and a kill switch that a human can use to stop it immediately. Test in a sandbox, release to a small pilot, then monitor.

PAUSE & THINK

Why is it dangerous if an automated payment step is not idempotent?

4. Lessons from automation failures

In August 2012, Knight Capital, a US trading firm, deployed new software, but old unused code was accidentally activated on one server. Its automated system sent millions of unintended orders in about 45 minutes, losing roughly 440 million US dollars. Weak deployment checks and the lack of an effective kill switch turned a software mistake into a near-collapse.

Good governance assigns an owner to every automation, controls who can change it, keeps change logs and reviews outcomes regularly. Automation multiplies whatever it does, good or bad, so safeguards must scale with its power.

PAUSE & THINK

Write three rules your college should follow before deploying any automated system.

✅ Key takeaways

- Automate repetitive, rule-based, stable, high-volume tasks.
- Toolbox: scripts + schedulers, RPA, no-code flows, AI agents.
- Reliability: logging, alerts, retries, idempotency, limits.
- Always include a kill switch and staged rollout.
- Automation multiplies errors; governance must scale too.

❓ Quick check: answer in one or two lines.

1. List three features of a good automation candidate.

2. What does the cron expression '0 9 * * 1-5' mean?

3. Name two safeguards missing in the Knight Capital case.

🔬 CONCEPT DEEP DIVE: Smart Automation Tools

🔷 Automation candidates: high-volume, rule-based, stable, digital inputs, measurable error rates.

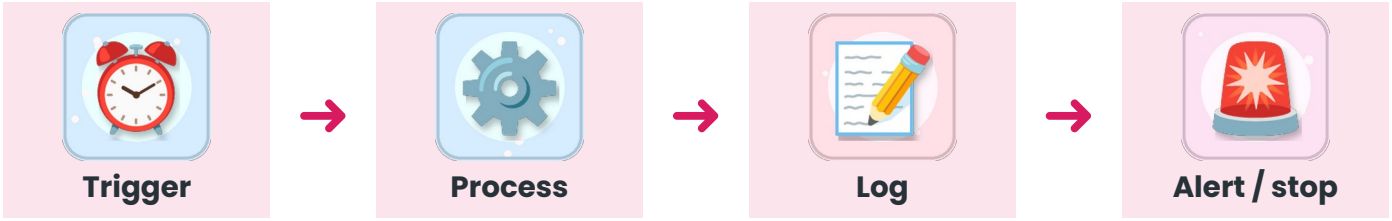
🔷 Tool spectrum: Python scripts + cron/Task Scheduler; RPA (UI bots); no-code flows (trigger → actions); AI agents.

🔷 APIs beat screen-scraping: more reliable and secure; RPA is useful for legacy systems without APIs.

🔷 Reliability: idempotent steps, retries with back-off, logging, alerts, rate limits and a kill switch.

🔷 Governance: owner for each bot, access controls, change logs and periodic review of outcomes.

🔄 How it works



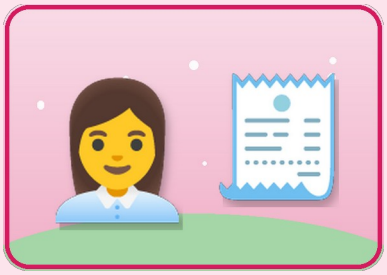
💻 A scheduled Python job

```
# crontab: run every weekday at 7:30
# 30 7 * * 1-5  python3 daily_report.py
import logging; logging.basicConfig(filename='bot.log')
logging.info('Report sent to %d staff', n)
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Bot With a Brake

Characters: *Mrs. Latha (college office superintendent), Arvind (B.Sc CS student volunteer), a receipt bot.*



SCENE 1

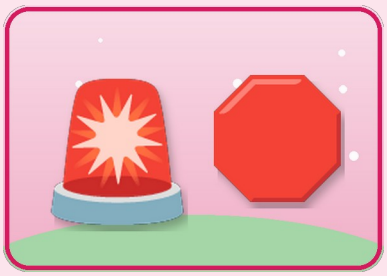
Every month, Mrs. Latha's office typed 600 fee receipts into a ledger by hand. It took 25 hours and produced frequent typing errors. Arvind, a student volunteer, offered to automate it.

Pause & predict: What makes this task a good automation candidate?

SCENE 2

Arvind wrote a Python script that read the receipt PDFs, extracted the data and filled a spreadsheet. A scheduler ran it every evening at 7 pm. In testing, it processed 590 receipts perfectly but stumbled on 10 blurry scans.

Pause & predict: How should the bot handle receipts it cannot read?



SCENE 3

Instead of guessing, the bot logged each unreadable receipt and emailed Mrs. Latha a list for manual entry. Arvind also added a limit: if more than 20 errors appeared in one run, the bot stopped completely. 'A brake,' he explained. 'Like the one Knight Capital didn't have.'

Pause & predict: Why did Arvind add an error limit and a stop?

SCENE 4

Months later, the scanner was replaced and its new format confused the bot. It hit the error limit and stopped safely, with no wrong entries in the ledger. Arvind updated the script in an hour. Mrs. Latha's team now spends their saved time helping students with scholarship forms.

⏸ **Pause & predict:** What would have happened without the brake?

🌟 The big idea

Good automation saves time; safe automation also knows when to stop.

💬 Discuss with your classmates

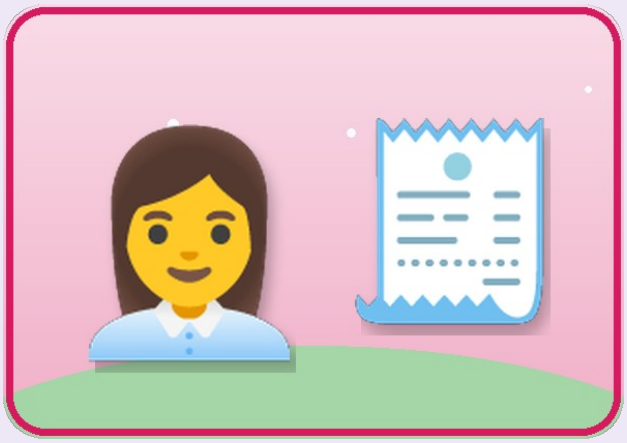
1. Which task at your college would you automate first? What safeguards would it need?

2. Who should own and maintain automations in an organisation?

SCENARIO: Smart Automation Tools

Read the scenario. Think about the choices made at each step.

1



The college office types 600 fee receipts into a ledger monthly.

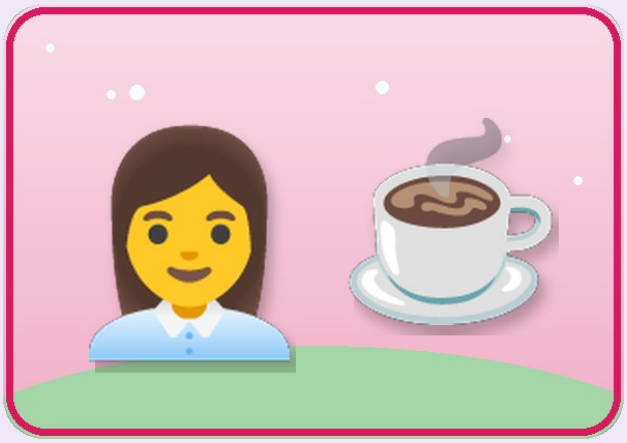
2

A Python script + scheduler reads PDFs and fills a sheet.

3

Logs and alerts flag 12 unreadable receipts for humans.

4



Staff save 25 hours a month and focus on students.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Automation at work

Email triage

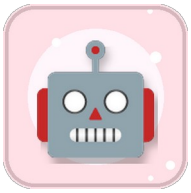
Invoice processing

Daily report generation

File organisation

Social posting

Server alerts



BOLT SAYS:

Automate the boring stuff, but always keep a big red STOP button!

FUN FACT!

RPA bots can click through software screens just like a human clerk!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Knight Capital's 45-Minute Meltdown

CASE FILE

In August 2012, Knight Capital deployed new trading software, but old, unused code was accidentally activated on one server. Its automated system sent millions of unintended orders in about 45 minutes, losing roughly US$440 million. The firm had no effective kill switch, and it was soon acquired by a rival.

🔍 Analyse the case

1. Which automation safeguards were missing?

2. Explain how a kill switch and limits could have helped.

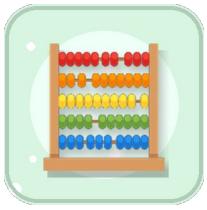
3. Write 3 deployment rules for any automated system.

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Automation Opportunity Scan

You need:

Task log



Calculator

Teams of 3

Steps:

1.
   List 8 tasks a college office does weekly.
2.
   Score each: frequency, rules-based?, error-prone?
3.
   Estimate hours saved per month if automated.
4.
   Design one automation with logging & a stop rule.

THINK & TALK

Which task should NOT be automated even if possible?

Record your observations & results:

CODE LAB: A Safe Automation Loop

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
import logging, time
logging.basicConfig(level=logging.INFO)
MAX_ERRORS, errors = 3, 0
for rec in records:
    try: process(rec)
    except Exception as e:
        errors += 1; logging.error('Failed %s: %s', rec, e)
        if errors >= ____: logging.critical('Kill switch!');
break
```

COMPLETE **T1. Fill the blank.**

EXPLAIN **T2. Why stop after 3 errors instead of continuing?**

DEBUG **T3. What if process() hangs forever?**

EXTEND **T4. Add retry with back-off (1 s, 2 s, 4 s).**

WORKSHEET: Smart Automation Tools (Part A)

Name: _____ Date: _____

1. Automation payback. Use the word box.

WORD BOX: 20 ✿ ₹10,000 ✿ 2

Saves 5 h/week → hours per month (4 weeks): ____

20 h × ₹500/h = monthly saving: ____

Build cost ₹20,000 → payback months: ____

2. Good automation candidate? Tick ✓.

	Good	Poor
Copy 500 invoices into a ledger daily		
Negotiating a contract		
Weekly attendance report		
Counselling a stressed student		

WORKSHEET: Smart Automation Tools (Part B)

Name: _____ Date: _____

1. Match the cron expression to its schedule.

0 9 * * * ●

●

Every 15 minutes

0 18 * * 5 ●

●

Daily at 9:00

0 0 1 * * ●

●

1st of every month, midnight

*/15 * * * * ●

●

Fridays at 18:00

2. Safe bot deployment. Write 1, 2, 3, 4.

Monitor & review weekly

☐

Test in a sandbox

☐

Add limits & kill switch

☐

Release to a small pilot

☐

PUZZLE & DEFINE: Smart Automation Tools

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

RPA

Meaning: _____

Example: _____

Cron

Meaning: _____

Example: _____

Kill switch

Meaning: _____

Example: _____

? QUIZ TIME: Smart Automation Tools

Circle the right answer!

Q1. Software bots clicking screens:

a) RPA

b) RPG

c) FM

Q2. Time-based Linux scheduler:

a) Melody

b) Cron

c) Bread

Q3. Best automation candidates are...

a) Creative

b) Emotional

c) Repetitive

Q4. Emergency stop for a bot:

a) Kill switch

b) Party

c) Volume

Q5. Automations need logging to...

a) Decorate

b) Audit & debug

c) Slow down

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Smart Automation Tools

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Rule-based repetitive tasks suit automation.		
	Kill switches are unnecessary for trading bots.		
	Cron schedules jobs by time.		
	APIs are often more reliable than screen-scraping.		
	Logs are only needed when bots fail.		
	Knight Capital lost about US$440M in 45 minutes.		

HOME CONNECT

Automate one personal task (e.g., a phone reminder routine). Note time saved in a week.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Smart Automation Tools

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why are APIs preferred to RPA screen-clicking where available?**

COMPARE **Q2. Compare cron scripts and no-code automation tools.**

APPLY **Q3. Cron '0 18 * * 5' runs when?**

EVALUATE **Q4. Should a bot auto-send fee-default notices to parents?**

CREATE **Q5. Design an automation with trigger, 3 steps, log and kill switch.**

📝 EXAM PREP: Smart Automation Tools

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. List three features of good automation candidates.

2. What is RPA?

3. What does '0 9 * * 1-5' mean in cron?

4. Why add a kill switch?

PART B • Long answer (5 marks)

5. Using the Knight Capital case, explain reliability and governance practices for automated systems.

UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. Most accurate visual encoding:

a) Position | b) Colour | c) Angle

Q2. Anscombe's correlation ≈

a) 0.10 | b) 0.82 | c) 1.00

Q3. Spread of marks → chart:

a) Pie | b) Line | c) Box plot

Q4. '0 18 * * 5' runs on...

a) Fridays 18:00 | b) Daily 5 pm | c) 5th of month

Q5. Knight Capital lacked a...

a) Logo | b) Kill switch | c) Coffee

Q6. UI-clicking software bots:

a) APIs | b) Cron | c) RPA

WORD SEARCH: AI CREATE

🔍 **Find these words. They go across → or down ↓.**

DASHBOARD

ENCODING

HEATMAP

RPA

CRON

KILL SWITCH

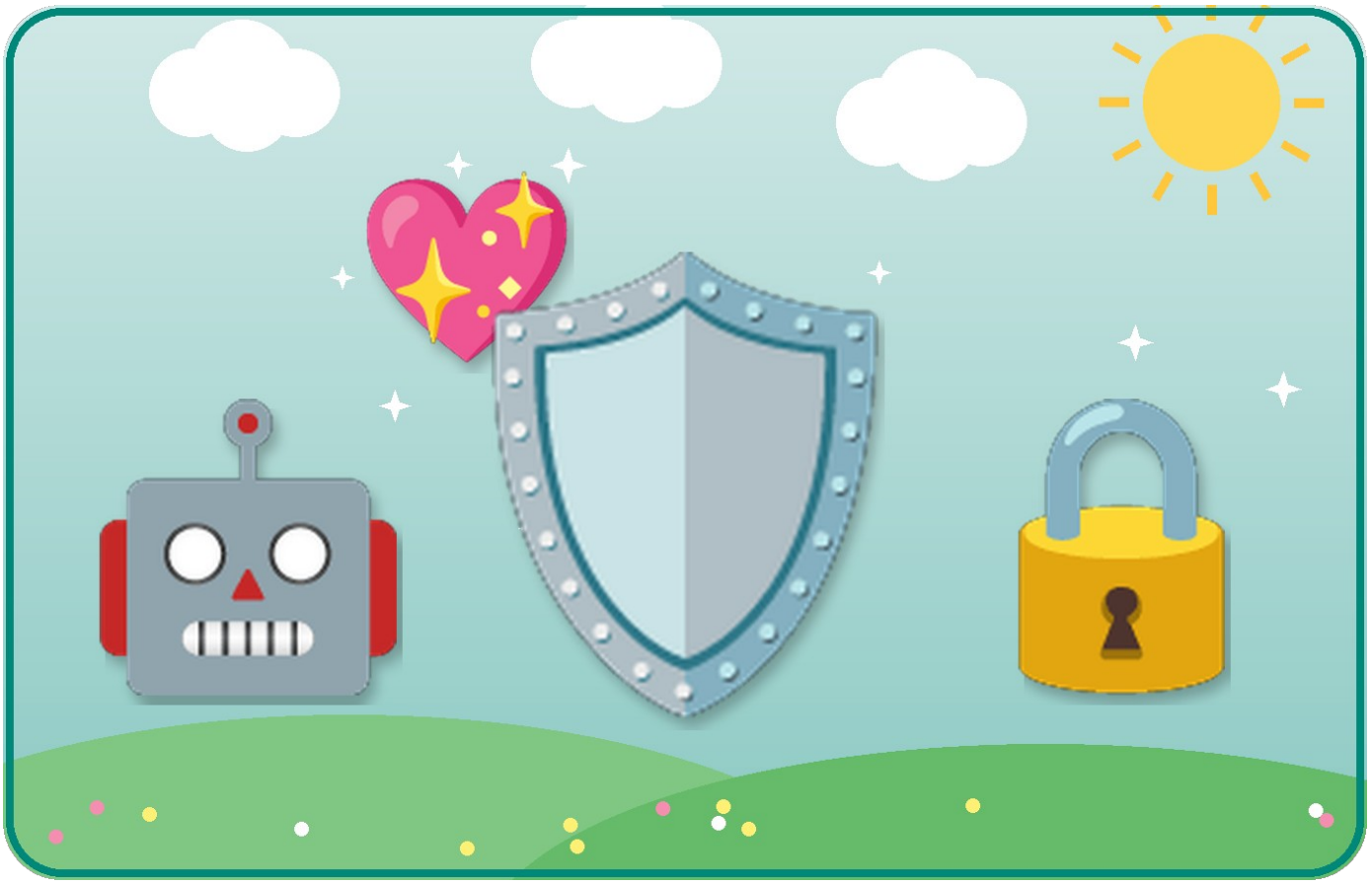
N	D	P	A	T	B	O	T	K	C
O	A	I	M	N	U	A	P	I	R
B	S	P	F	W	G	D	R	L	O
H	H	E	A	T	M	A	P	L	N
P	B	L	P	L	O	T	A	S	M
S	O	I	C	H	A	R	T	W	M
S	A	N	I	P	D	U	M	I	K
B	R	E	O	C	G	L	S	T	W
M	D	E	L	I	T	C	K	C	L
E	N	C	O	D	I	N	G	H	K

I finished AI CREATE! Teacher's sign: _____

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

Fair AI, Fair Future

Topic 12

Ownership in the AI Age

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Fair AI, Fair Future

LET'S LEARN

- Fairness can be measured in several ways.
- Fairness metrics can conflict: choose with care.
- Mitigate bias before, during and after training.

Fairness metrics

Demographic parity

Equal opportunity

Equalised odds

Calibration

KEY TERMS

Equal opp. Equal TPR across groups

Calibration Scores mean same risk

Mitigation Reduce unfairness

📖 CONCEPT EXPLAINED: Fair AI, Fair Future

💡 In this chapter

In your first year you learnt that AI can be biased. In the second year we go further: how is fairness measured, why do different fairness definitions conflict, and what can engineers actually do to reduce unfairness? This chapter combines mathematics with ethics, because fairness is both.

📊 1. Measuring fairness

Group fairness metrics compare outcomes across groups. Demographic parity requires equal selection rates: the same percentage of each group receives loans. Equal opportunity requires equal true positive rates (TPR): among people who truly deserve a loan, each group is approved at the same rate. Equalised odds requires both equal TPR and equal false positive rates (FPR).

Calibration requires that a score means the same thing for everyone: among people given a 70% risk score, about 70% should actually have the outcome, in every group.

✏️ WORKED EXAMPLE

Group A: TP 40, FN 10 → TPR 0.80. Group B: TP 30, FN 20 → TPR 0.60. The gap of 0.20 violates equal opportunity.

📐 2. Why fairness definitions conflict

In 2016, ProPublica analysed the COMPAS risk-assessment tool used in US courts and found that Black defendants who did not reoffend were almost twice as likely to be labelled high-risk as white defendants who did not reoffend: unequal false positive rates. The developer responded that the scores were equally calibrated across groups.

Researchers then proved that when groups have different underlying rates of the outcome (base rates), calibration and equal error rates generally cannot all be satisfied at the same time. Choosing which fairness definition to prioritise is therefore a values decision, not just a technical one.

🤔 PAUSE & THINK

In a bail decision, which error is worse: wrongly labelling someone high-risk, or missing someone who will reoffend? Who should decide?

🔧 3. Mitigating unfairness

Pre-processing methods change the data before training: collecting more representative data, reweighting under-represented groups or removing problematic labels. In-processing methods change training itself, for example by adding a fairness penalty to the loss function. Post-processing methods adjust outputs, such as using group-specific decision thresholds.

Each method has trade-offs. Mitigation may slightly reduce overall accuracy, and some methods raise legal questions about treating groups differently. Testing several approaches and documenting the trade-offs transparently is good practice.

🖐 TRY IT

For a scholarship-selection model, suggest one pre-, one in- and one post-processing mitigation.

🤝 4. Fairness as a process

Metrics alone cannot guarantee fairness. Fair AI requires involving affected communities, conducting impact assessments before deployment, explaining decisions, offering appeals and human review, and monitoring outcomes over time because data and populations change.

Fairness also extends beyond individual decisions: who benefits from AI, who bears its risks, whose languages and needs are represented, and who has the power to shape these systems. As future builders, your choices influence these questions.

🤔 PAUSE & THINK

Who should be consulted before a city deploys an AI system that allocates housing benefits?

✅ Key takeaways

- Metrics: demographic parity, equal opportunity, equalised odds, calibration.
- With different base rates, some fairness criteria conflict.
- Choosing a fairness metric is a values decision.
- Mitigation: pre-, in- and post-processing, each with trade-offs.
- Fairness needs process: participation, appeals and monitoring.

❓ Quick check: answer in one or two lines.

1. Define equal opportunity in one line.

2. What did ProPublica find about COMPAS?

3. Name one pre-processing mitigation method.

🔬 CONCEPT DEEP DIVE: Fair AI, Fair Future

🔷 Group fairness metrics: demographic parity (equal selection), equal opportunity (equal TPR), equalised odds (TPR & FPR).

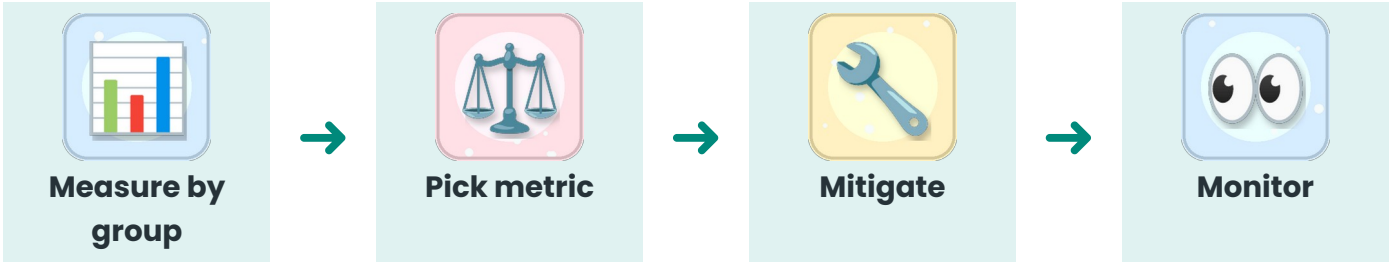
🔷 Calibration: among people scored p, about p truly have the outcome, in every group.

🔷 Impossibility results: with different base rates, calibration and equal error rates generally cannot all hold.

🔷 Mitigation: pre-processing (reweighting), in-processing (fairness constraints), post-processing (group thresholds).

🔷 Fairness also needs process: stakeholder input, impact assessments, appeals and ongoing monitoring.

🔄 How it works



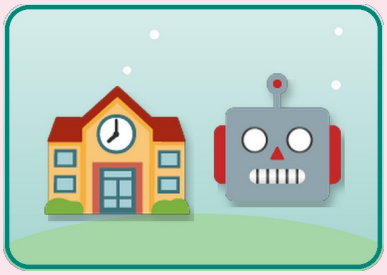
💻 Group TPR and FPR

```
def rates(tp, fp, fn, tn):
    return tp / (tp + fn), fp / (fp + tn)   # TPR, FPR
print(rates(40, 20, 10, 130))   # (0.8, 0.133)
print(rates(30, 45, 20, 105))   # (0.6, 0.3)
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Scholarship Algorithm

Characters: *Priya (B.A. Economics student), Rahul (B.Tech student), the college scholarship committee.*



SCENE 1

The college trialled an AI model to shortlist scholarship applicants. Priya, from a rural school, noticed that very few rural students had been shortlisted. She teamed up with Rahul to investigate the model's results.

Pause & predict: What data would Priya and Rahul need to check fairness?

SCENE 2

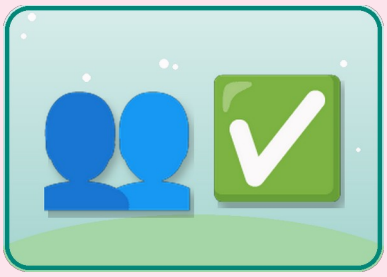
They computed metrics by group. Among students who truly deserved scholarships (according to later results), 80% of urban students were shortlisted but only 60% of rural students. The true positive rates were unequal, violating equal opportunity.

Pause & predict: Which fairness metric did they use? What did it show?

SCENE 3

Digging deeper, Rahul found the cause: the model relied heavily on 'number of certificates', and rural schools offered fewer certificate courses. The feature acted as a proxy for location, not merit. They proposed reweighting the training data and replacing the feature with grade improvement over time.

Pause & predict: Why was 'number of certificates' an unfair proxy?



SCENE 4

The committee tested the revised model: TPR for both groups rose above 75%, with only a small change in overall accuracy. The committee also added human review for borderline cases and an appeal process. Priya presented their audit at a national student conference.

⏸ **Pause & predict:** Why did the committee add human review and appeals as well?

🌟 The big idea

Fairness must be measured, its causes understood, and its fixes tested, with humans accountable for the outcome.

💬 Discuss with your classmates

1. Which fairness metric would you prioritise for scholarships? Why?

2. What other proxies could hide unfairness in college admissions?

SCENARIO: Fair AI, Fair Future

Read the scenario. Think about the choices made at each step.

1

A court uses a risk score to inform bail decisions.

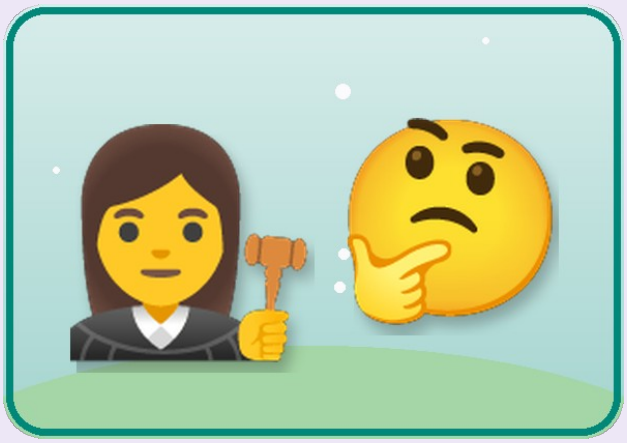
2

Scores are calibrated: a '7' means similar risk in both groups.

3

But false-positive rates differ widely between groups!

4



Which fairness matters more? A values debate begins.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Fairness in high-stakes AI

Risk scores in courts

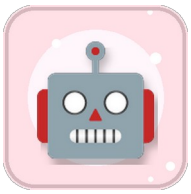
Loan approvals

Hiring filters

Clinical risk tools

Admissions screening

Housing ads



BOLT SAYS:

'Fair' has more than one maths definition. Choosing one is a values decision!

FUN FACT!

Researchers proved some fairness definitions cannot all hold at once when base rates differ!

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: The COMPAS Risk-Score Debate

CASE FILE

In 2016, ProPublica analysed the COMPAS recidivism tool used in US courts. It found Black defendants who did not reoffend were almost twice as likely to be labelled high-risk as white defendants who did not reoffend. The developer responded that scores were equally calibrated across groups. Researchers then showed both properties cannot generally hold together.

Analyse the case

1. Which fairness metric did ProPublica emphasise? Which did the developer?

2. Why can't both be satisfied when base rates differ?

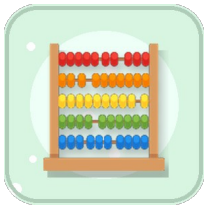
3. Who should decide which fairness definition a court uses?

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Fairness Metric Face-Off

You need:

Data tables



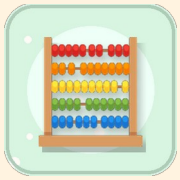
Calculator

Teams of 4

Steps:

1

Get confusion matrices for Group A and Group B.

2 

Compute selection rate, TPR and FPR for each.

3

Which fairness metrics pass? Which fail?

4

Debate: which metric should a bank prioritise?

THINK & TALK

Can you satisfy every fairness metric at once here?

Record your observations & results:

💻 CODE LAB: Fairness Metrics by Group

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
# (tp, fp, fn, tn) per group
groups = {'A': (40, 20, 10, 130), 'B': (30, 45, 20, 105)}
for g, (tp, fp, fn, tn) in groups.items():
    sel = (tp + fp) / (tp + fp + fn + tn)
    tpr, fpr = tp / (tp + fn), fp / (fp + ____)
    print(g, round(sel, 2), round(tpr, 2), round(fpr, 2))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Write sel, TPR, FPR for A and B.**

EXPLAIN **T3. Which fairness metrics fail?**

EXTEND **T4. Compute the demographic parity ratio.**

WORKSHEET: Fair AI, Fair Future (Part A)

Name: _____ Date: _____

1. Group A: TP 40, FN 10. Group B: TP 30, FN 20. Use the word box.

WORD BOX: **0.8** ✿ **0.6** ✿ **Fails**

TPR(A) = 40 / 50 = ____

TPR(B) = 30 / 50 = ____

Equal opportunity (gap ≤ 0.05)? ____

2. Match the metric to its condition.

Demographic parity ●

Equal opportunity ●

Equalised odds ●

Calibration ●

● **Equal TPR and FPR**

● **Equal selection rates**

● **Same score = same actual risk**

● **Equal TPR only**

WORKSHEET: Fair AI, Fair Future (Part B)

Name: _____ Date: _____

1. FPR = FP ÷ (FP + TN). Use the word box.

WORD BOX: 0.13 ✿ 0.30 ✿ Equalised odds

FPR(A) = 20 ÷ 150 ≈ ____

FPR(B) = 45 ÷ 150 = ____

Equal TPR and FPR is called ____

2. Pre-, in- or post-processing? (Pre/In = left, Post = right) Tick ✓.

	Pre/In	Post
Reweight under-represented samples		
Use group-specific thresholds after training		
Add fairness penalty to loss		
Adjust output scores after prediction		

PUZZLE & DEFINE: Fair AI, Fair Future

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Equal opp.

Meaning: _____

Example: _____

Calibration

Meaning: _____

Example: _____

Mitigation

Meaning: _____

Example: _____

? QUIZ TIME: Fair AI, Fair Future

Circle the right answer!

Q1. Equal selection rates across groups:

a) Demographic parity

b) Randomness

c) Calibration

Q2. Equal opportunity equalises...

a) Salaries

b) True positive rates

c) Ages

Q3. All fairness metrics together are...

a) Always possible

b) Easy

c) Often impossible

Q4. Reweighting training data is...

a) Pre-processing

b) Painting

c) Launching

Q5. Choosing a fairness metric is a...

a) Coin toss

b) Values decision

c) Speed test

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Fair AI, Fair Future

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🎯	Equal opportunity compares true positive rates.		
✅	All fairness metrics can always be satisfied together.		
📏	Calibration means scores reflect actual risk.		
🔧	Reweighting data is a pre-processing method.		
🙈	Fairness is purely a technical problem.		
⚖️	COMPAS sparked a major fairness debate.		

HOME CONNECT

Discuss with family: is 'equal treatment' always the same as 'fair treatment'? Give an example.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Fair AI, Fair Future

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why is calibration alone not sufficient for fairness?**

COMPARE **Q2. Compare pre-, in- and post-processing mitigation.**

APPLY **Q3. A: TP 40, FN 10, FP 20, TN 130. B: TP 30, FN 20, FP 45, TN 105. TPR & FPR?**

EVALUATE **Q4. Should different groups get different thresholds?**

CREATE **Q5. Draft a fairness impact assessment outline for a scholarship AI.**

EXAM PREP: Fair AI, Fair Future

Name: _____ Date: _____

Time: 30 min Max marks: 13 Answer all questions.

PART A • Short answers (2 marks each)

1. Define demographic parity.

2. Define equalised odds.

3. What is calibration?

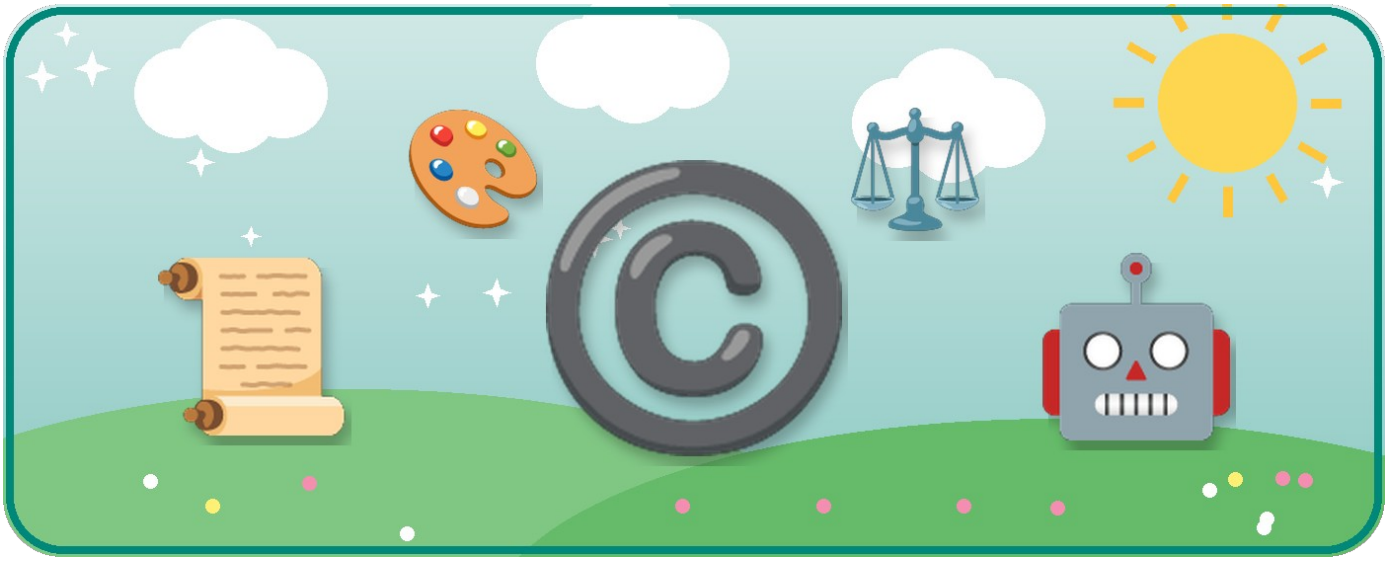
4. Name three mitigation stages.

PART B • Long answer (5 marks)

5. Explain the COMPAS debate and the impossibility of satisfying all fairness metrics. How should institutions decide?

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

Ownership in the AI Age



LET'S LEARN

- Who owns AI outputs, data and models is contested.
- Many laws require human authorship for copyright.
- Respect licences: data, code, models and content.

What can be owned?

Training data

Model weights

AI outputs

Code

KEY TERMS

Copyright Right over creative work

Licence Permission to use

Open source Code shared openly

📖 CONCEPT EXPLAINED: Ownership in the AI Age

💡 In this chapter

Generative AI raises new questions about ownership. Who owns an image generated by AI? Can AI companies train on books and art without permission? What licences govern datasets, code and models? Can someone clone your voice? This chapter introduces copyright, licensing and personality rights, and offers practical guidance for students and creators.

© 1. Copyright and human authorship

Copyright protects original creative expression, such as writing, art, music and code, giving creators rights over copying and distribution. Many legal systems require human authorship. In the United States, the Copyright Office refused to register an image that Stephen Thaler listed as created solely by his AI system, and courts upheld that human authorship is required.

When humans make significant creative contributions, such as selecting, arranging and substantially editing AI outputs, those human contributions may be protected. Laws differ across countries and are evolving rapidly, so creators should document their creative process.

🤔 PAUSE & THINK

You generate an image with one prompt and make no edits. Do you think you should own it? Argue your view.

📚 2. Training data disputes

AI models are trained on vast amounts of text, images, music and code, much of it copyrighted. Authors, artists, news publishers and music companies have filed lawsuits against AI developers in several countries, arguing that training without permission or payment infringes their rights. Developers often argue that training is a fair or transformative use.

Courts have reached different conclusions in different cases, and new licensing deals between AI firms and publishers are emerging. Whatever the legal outcome, the ethical questions of consent, credit and compensation for creators remain important.

TRY IT

List two arguments for and two against allowing AI training on publicly available art.

3. Licences: permission to use

A licence grants permission to use something under conditions. Creative Commons licences combine elements: BY (give credit), SA (share alike under the same licence), NC (non-commercial only) and ND (no derivatives). Software licences range from permissive (MIT, Apache 2.0: reuse freely, keep notices) to copyleft (GPL: derivatives must also be open under GPL).

AI models and datasets often come with their own licences, sometimes restricting commercial use or certain applications. Before using any asset in a project, especially a commercial one, read its licence and record it.

WORKED EXAMPLE

Correct attribution: '"Sunrise" by R. Devi, licensed under CC BY 4.0, via [source link].'

4. Personality rights and responsible creation

Voice cloning and deepfakes make it easy to imitate real people. Personality rights protect a person's name, image, voice and likeness from unauthorised commercial use or misuse. Courts in India have granted protection to well-known personalities against unauthorised AI imitations of their voices and likenesses.

As a student creator, follow simple rules: disclose AI use, use properly licensed assets, credit creators, never clone real people's voices or faces without consent, and respect your institution's policies. Ownership in the AI age is ultimately about respect for others' work and identity.

PAUSE & THINK

A friend wants to make a funny video using an AI clone of a teacher's voice. What would you advise?

✅ Key takeaways

- Copyright generally requires human authorship.
- Training-data lawsuits raise unresolved legal and ethical questions.
- Licences (CC, MIT, GPL, model licences) define permitted use.
- Personality rights protect voice, face and name.
- Disclose AI use, credit creators and get consent.

❓ Quick check: answer in one or two lines.

1. Why was Thaler's AI-only image refused copyright?

2. What do CC BY and CC NC mean?

3. Differentiate MIT and GPL licences.

🔬 CONCEPT DEEP DIVE: Ownership in the AI Age

🔷 Copyright protects original human expression; many jurisdictions require meaningful human authorship.

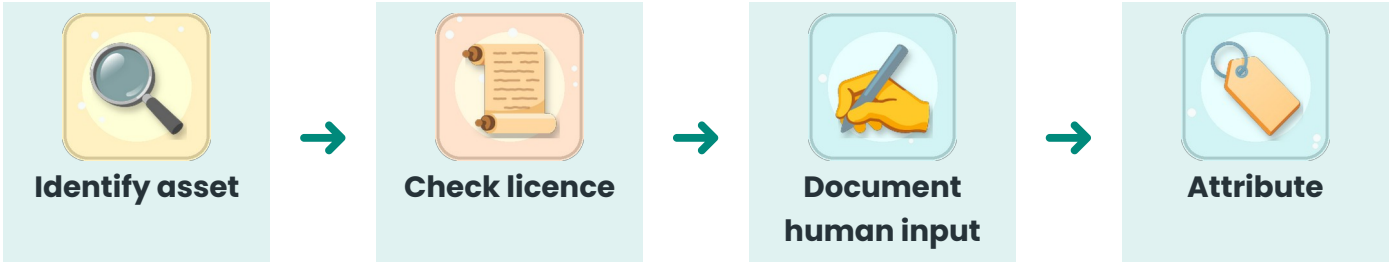
🔷 AI-assisted works may be protected for the human contributions (selection, arrangement, edits), not raw AI output.

🔷 Training-data disputes: authors, artists and news publishers have sued AI firms over use of their works; outcomes vary.

🔷 Licences govern reuse: CC (BY, SA, NC, ND), software (MIT, Apache, GPL), and model licences with use restrictions.

🔷 Personality rights: courts (including in India) have protected celebrities' names, voices and likenesses from misuse.

🔄 How it works



💻 Checking licences in a project

```
assets = {'logo.png': 'CC-BY', 'model': 'Apache-2.0',
          'song.mp3': 'proprietary'}
risky = [a for a, l in assets.items() if l == 'proprietary']
print('Need permission for:', risky)
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Who Owns the Poster?

Characters: *Arun (B.Des student), Meena (law student), a company representative.*

SCENE 1

Arun created a stunning festival poster using an AI image generator, then spent hours editing colours, composing elements and adding hand-drawn lettering. A company saw it and offered ₹50,000 to use it in an advertising campaign.

Pause & predict: Who do you think owns the poster? Why?

SCENE 2

His friend Meena, a law student, advised caution. They read the AI tool's terms, which gave users rights to outputs but no guarantee of copyright. Meena explained that many legal systems require human authorship, so Arun's own edits and arrangement mattered.

Pause & predict: Why did Arun's human edits matter legally?

SCENE 3

They checked every element. The background music for the video version came from a song Arun had downloaded without a licence; it had to go. A font was licensed for personal use only. Arun replaced both with properly licensed alternatives and documented his creative steps.

Pause & predict: What could have happened if Arun had ignored the licences?

SCENE 4

Arun signed a clear contract with the company, disclosing that AI assisted the work and listing his human contributions. The campaign launched successfully. Arun now keeps a 'creation log' for every project and teaches his juniors to check licences first.

⏸ **Pause & predict:** Why is a creation log useful for creators using AI?

🌟 The big idea

In the AI age, owning your work means documenting your contribution and respecting everyone else's.

💬 Discuss with your classmates

1. Should AI-generated art receive copyright protection? Argue your view.

2. What rules should your college fest follow for AI-made content?

SCENARIO: Ownership in the AI Age

Read the scenario. Think about the choices made at each step.

1

Arun generates a poster with AI for a college fest.

2

A company wants to buy it. Who owns it?

3

He checks the tool's terms and copyright law.

4

He adds major human edits and documents his creative steps.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Ownership questions today

AI art copyright

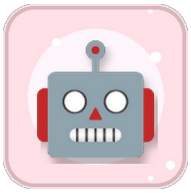
Books used for training

Voice & music clones

Open-source licences

Trademarks in prompts

Artists' consent



BOLT SAYS:

Just because content is online doesn't mean it's free to use. Check the licence!

FUN FACT!

US courts have held that works created solely by AI without human authorship can't be copyrighted!

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: Can an AI Be an Author?

CASE FILE

Computer scientist Stephen Thaler sought US copyright for an image, 'A Recent Entrance to Paradise', listing his AI system as the author. The US Copyright Office refused, and in 2023 a federal court upheld the refusal, stating that human authorship is a bedrock requirement of copyright. The ruling was upheld on appeal in 2025.

🔍 Analyse the case

1. Why did the court insist on human authorship?

2. How might the outcome differ if Thaler had heavily edited the image?

3. Should AI-generated works get any protection? Argue.

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Licence Detective

You need:

Asset list

Licence pages

Pairs

Steps:

1		Get 8 assets: images, datasets, code, a model, a song.
2		Find each licence (CC-BY, MIT, GPL, proprietary…).
3		Mark: free to use? credit needed? commercial OK?
4		Write a correct attribution line for 3 assets.

THINK & TALK

Which asset was hardest to verify? Why?

Record your observations & results:

💻 CODE LAB: Licence Checker

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
assets = [('logo.png', 'CC-BY'), ('font.ttf', 'OFL'),
          ('song.mp3', 'proprietary'), ('img.jpg', 'CC-BY-NC')]
commercial_ok = {'CC-BY', 'OFL', 'MIT', 'Apache-2.0'}
flag = [n for n, l in assets if l not in ____]
print('Check before commercial use:', flag)
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Which assets are flagged?**

EXPLAIN **T3. Why is CC-BY-NC flagged?**

EXTEND **T4. Print an attribution line for each CC-BY asset.**

WORKSHEET: Ownership in the AI Age (Part A)

Name: _____ Date: _____

1. Match the licence to its key rule.

CC-BY ●

MIT ●

GPL ●

Proprietary ●

● Use only with paid / explicit permission

● Reuse with credit

● Derivatives must stay open (copyleft)

● Very permissive; keep notice

2. Likely OK or likely infringing? Tick ✓.

	OK	Risky
Using a CC-BY image with credit		
Cloning a singer's voice for an ad without consent		
Using MIT code and keeping its notice		
Selling copies of a textbook made with AI rewording		

WORKSHEET: Ownership in the AI Age (Part B)

Name: _____ Date: _____

1. Match the CC element to its meaning.

BY ●

● No derivatives / changes

SA ●

● Give credit

NC ●

● Share alike (same licence)

ND ●

● Non-commercial only

2. Circle actions that respect ownership.

Credit a CC-BY photo

Use a hit song in a paid ad without licence

Read a model's licence before commercial use

Clone a celebrity's voice for memes

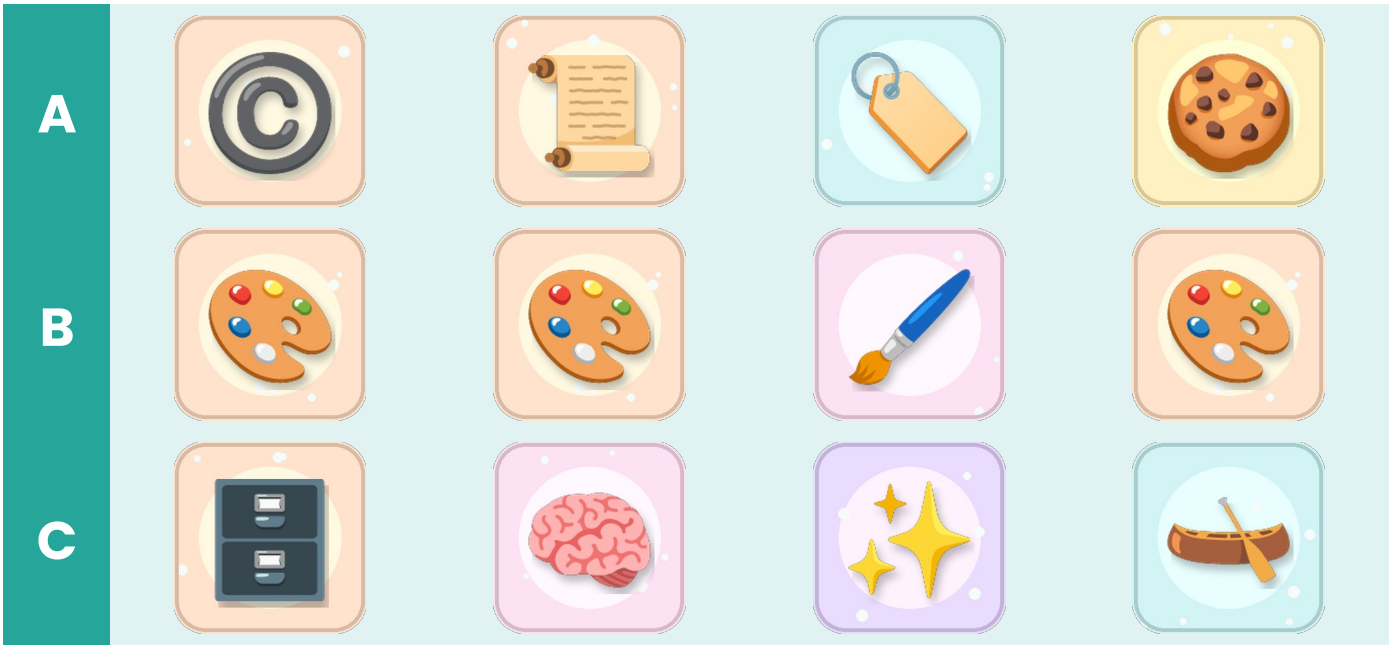
Document your human edits

Keep MIT licence notice in code

PUZZLE & DEFINE: Ownership in the AI Age

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

Copyright

Meaning: _____

Example: _____

Licence

Meaning: _____

Example: _____

Open source

Meaning: _____

Example: _____

? QUIZ TIME: Ownership in the AI Age

Circle the right answer!

Q1. Right over creative work:

a) Copyright

b) Licence plate

c) Ticket

Q2. Copyleft licence example:

a) MIT

b) GPL

c) Proprietary

Q3. CC-BY requires...

a) Payment

b) Nothing

c) Attribution

Q4. Purely AI-made works (US) are...

a) Not copyrightable

b) Auto-copyrighted

c) Patented

Q5. Using a voice clone needs...

a) Luck

b) Consent

c) Volume

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Ownership in the AI Age

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Many laws require human authorship for copyright.		
Anything online is free to reuse.		
MIT licence requires keeping the notice.		
CC-NC allows any commercial use.		
Personality rights can protect a person's voice.		
Documenting human contributions can support ownership claims.		

HOME CONNECT

Check the licence of one app, font or image your family uses. What does it allow?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Ownership in the AI Age

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why can using a CC-BY-NC image in a paid ad be a problem?**

COMPARE **Q2. Compare MIT and GPL licences for a student project.**

APPLY **Q3. Write a CC-BY attribution for 'Sunrise' by R. Devi.**

EVALUATE **Q4. Should artists be paid when AI trains on their work?**

CREATE **Q5. Draft an AI-use & IP policy (4 rules) for a student fest.**

📝 EXAM PREP: Ownership in the AI Age

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Why do many laws require human authorship?

2. Differentiate MIT and GPL licences.

3. What do CC BY and NC mean?

4. What are personality rights?

PART B • Long answer (5 marks)

5. Discuss ownership questions raised by generative AI: training data, outputs, models and personal likeness, with cases.

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. Equal TPR and FPR =

a) Equalised odds | b) Parity | c) Calibration

Q2. FPR = FP ÷ ...

a) (TP + FN) | b) (FP + TN) | c) (TP + FP)

Q3. Group-specific thresholds are...

a) Pre-processing | b) In-processing | c) Post-processing

Q4. Thaler's AI-only image was...

a) Refused copyright | b) Copyrighted | c) Patented

Q5. CC 'SA' means...

a) Stop all | b) Share alike | c) Sell always

Q6. GPL is a ... licence.

a) Proprietary | b) Public-domain | c) Copyleft

WORD SEARCH: AI CARE

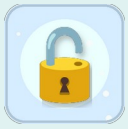
🔍 **Find these words. They go across → or down ↓.**

EQUAL OPP.

MITIGATION

COPYRIGHT

LICENCE


OPEN SOURCE

PARITY

O	N	R	K	P	L	M	M	F	W
P	N	W	I	L	E	N	I	C	R
E	F	K	T	A	Q	C	T	O	L
N	P	A	W	P	U	O	I	P	I
S	A	U	C	B	A	N	G	Y	C
O	R	D	R	F	L	S	A	R	E
U	I	I	E	A	O	E	T	I	N
R	T	T	D	W	P	N	I	G	C
C	Y	U	I	D	P	T	O	H	E
E	M	M	T	R	.	U	N	T	B

I finished AI CARE! Teacher's sign: _____

AI GLOSSARY (1/2)

A/B test
Randomised comparison

API key
Secret access token

Behavioural
Past-behaviour questions

Bias–variance
Underfit vs overfit

Boosting
Fix previous errors

Calibration
Scores mean same risk

Case study
Project story write-up

Churn
Customers who leave

CLV
Customer lifetime value

Copyright
Right over creative work

Cost matrix
Price of each error

Cron
Time-based scheduler

CV
Cross-validation folds

Dashboard
Live visual summary

Digital twin
Live virtual replica

EDA
Exploratory data analysis

Encoding
Data → visual property

Equal opp.
Equal TPR across groups

AI GLOSSARY (2/2)

Gen. design
AI-explored shapes

Heatmap
Colour-coded grid

Kill switch
Emergency stop

KPI
Key performance indicator

Licence
Permission to use

Mitigation
Reduce unfairness

MLOps
ML in production

NLP
Language processing

Open source
Code shared openly

Random forest
Many voting trees

Refactor
Improve code structure

Referral
Recommended by insider

Regularisation
Penalty on complexity

RPA
Robotic process automation

RUL
Remaining useful life

SQL
Query language for tables

TDD
Test-driven development

Vision AI
Image understanding

❓ OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. Imputing before splitting can cause...

a) Leakage

b) Faster CV

c) Better tests

Q2. Google Flu Trends overestimated due to...

a) Rain

b) Data drift

c) Power cuts

Q3. Votes S, S, N, S, N → majority:

a) Tie

b) Not spam

c) Spam

Q4. Boosting mainly reduces...

a) Bias

b) Colour

c) Noise volume

Q5. 1,000 × 800 tokens at ₹0.0005/token:

a) ₹4,000

b) ₹400

c) ₹40

❓ OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. z = 1.89 at 5% (needs |z| > 1.96) is...

a) Significant

b) Huge

c) Not significant

Q7. WHERE runs ... GROUP BY.

a) Before

b) After

c) Instead of

Q8. Bing headline change raised revenue ~

a) 1%

b) 12%

c) 50%

Q9. tp 3, fp 1; gain 1800, cost 200. Profit:

a) 5,600

b) 3,600

c) 5,200

Q10. CLV ₹3,000, CAC ₹1,500. Ratio:

a) 2

b) 0.5

c) 4,500

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. Wear 52, limit 90, +3/day → RUL ≈

a) 30 days

b) 12.7 days

c) 3 days

Q12. GM's bracket: 8 parts became...

a) 16

b) 4

c) 1

Q13. NLP 5 × 3 × 4 path score:

a) 60

b) 12

c) 45

Q14. Strongest portfolio signal:

a) Unchanged forks

b) README + demo

c) 50 empty repos

Q15. Bars 62, 71, 58 with y from 55:

a) Honest

b) Pretty

c) Misleading

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. '*/15 * * * *' runs every...

a) 15 minutes

b) 15th day

c) 15 hours

Q17. Knight Capital lost about...

a) US$4.4M

b) US$440M

c) US$44B

Q18. TPR 0.8 vs 0.6 violates...

a) Calibration only

b) Style

c) Equal opportunity

Q19. ProPublica emphasised unequal...

a) False-positive rates

b) Salaries

c) Calibration

Q20. Using CC-BY-NC art in a paid ad is...

a) Allowed

b) Needs permission

c) Required

MY SCORE: I got ____ / 20. Colour a star for every 4 right!

CERTIFICATE

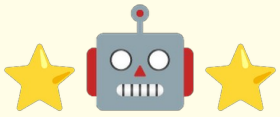
of AI Explorer

This is to say that

_____

of UG 2nd Year finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. Learning from Data	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Missing - Impute or drop; Duplicates - drop_duplicates(); Scales - Standardise; Outliers - Investigate, cap or remove **Part A 2:** Train 62%, validation 60%: Bias; Train 99%, validation 71%: Variance; Linear model on curved data: Bias; Very deep unpruned tree: Variance **Part B 1:** 0.80, 0.84, 0.08 **Part B 2:** Circle: Scaling on full data before split, Using 'discharge date' to predict admission, Same patient in train and test **Challenge hints:** Q1: Scaler learns test statistics; evaluation becomes optimistic. • Q2: Hold-out: fast, one estimate, noisy; CV: 5 estimates, more reliable, slower. • Q3: Mean 0.80; spread ±0.04 → reasonably stable. • Q4: No: distorts variance & relationships; consider median, model-based, indicators. • Q5: Shape, types, missing, duplicates, distributions, correlations/outliers. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: median; ages [21.0, 22.0, 23.0, 22.0, 22.0]; duplicates: 0 (no two rows identical); df['z'] = (df.marks - df.marks.mean()) / df.marks.std(). **Exam prep key points:** 1: Exploratory analysis: summary stats, missing values, distributions, correlations. • 2: Simple models underfit (bias); complex overfit (variance); balance. • 3: Split into k folds; train k times, each fold validates once; average. • 4: Info from test/future in training, e.g., scaling before split. • 5: EDA, cleaning, imputation, encoding, scaling in pipeline, CV, test set, avoid leakage.
2. Making Machines Smarter	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-F 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Bagging - Bootstrap & vote; Boosting - Fix previous errors; L2 - Penalise large weights; Grid search - Try every combo **Part A 2:** Spam, Not spam, 3 of 5 **Part B 1:** 6, 5, 30 **Part B 2:** Can set some weights exactly to zero: L1 Lasso; Shrinks all weights smoothly: L2 Ridge; Useful for feature selection: L1 Lasso; Handles correlated features gently: L2 Ridge **Challenge hints:** Q1: Averaging many de-correlated trees cancels individual errors. • Q2: Bagging: parallel, reduces variance; boosting: sequential, reduces bias. • Q3: 2 × 3 = 6 combos × 5 folds = 30 fits. • Q4: Depends on business value, latency, maintenance; often not. • Q5: Attendance trend, fee delay days, marks change, distance to campus. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: 1; prints spam 0.57; then 3 spam vs 4 ham → ham 0.57; weighted: sum weights per class, pick max. **Exam prep key points:** 1: Bagging: parallel bootstraps, reduces variance; boosting: sequential, reduces bias. • 2: Penalty on complexity; L1 Lasso, L2 Ridge. • 3: Set before training: max_depth, learning rate, n_estimators. • 4: Better inputs expose patterns; often bigger gains than algorithms. • 5: Bootstrap + feature randomness vs sequential residual fitting; CV grid/random search.

3. Program with AI

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 1-Write failing tests, 2-Generate code with AI, 3-Run tests until green, 4-Refactor safely
Part A 2: Store API key in environment variable: Safe; Commit API key to public GitHub: Unsafe; Handle rate limits with retries: Safe; No spending limit on paid API: Unsafe
Part B 1: 800,000, ₹400, ₹40
Part B 2: 200 - OK; 401 - Unauthorised; 429 - Rate limit; 500 - Server error
Challenge hints: Q1: Clear success criteria; catches hallucinations & edge-case bugs. • Q2: 0: deterministic, consistent; 0.8: varied, more bugs; prefer low. • Q3: 800,000 × 0.0005 = ₹400. • Q4: Only per policy with disclosure; must explain & test it. • Q5: 90→'A', 75→'B', -1→error, 100→'A' (boundary).
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 40; yes both pass; raises ValueError — correct; with pytest.raises(ValueError): grade(101).
Exam prep key points: 1: Write tests first, then code to pass them, then refactor. • 2: Prevents leaks in code repositories. • 3: Too many requests: rate limited. • 4: Caching, shorter prompts, batching, smaller models. • 5: Spec & tests, env keys, input validation, error handling, retries, caching, logging, budget.

4. Data into Decisions

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Descriptive - What happened; Diagnostic - Why it dropped; Predictive - Next month; Prescriptive - What to do
Part A 2: 5%, 8%, +60%
Part B 1: 6%, 7.5%, 25%
Part B 2: 1-SELECT ... FROM, 2-WHERE, 3-GROUP BY, 4-ORDER BY
Challenge hints: Q1: Groups are equal on average; only the change differs. • Q2: WHERE filters rows before grouping; HAVING filters groups after. • Q3: 6% vs 7.5%; lift = 25%. • Q4: Statistically significant but maybe practically trivial; weigh costs. • Q5: Random students, A: plain, B: personalised; metric: on-time returns; n fixed.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: nb; pa 0.06, pb 0.075; z ≈ 1.89 < 1.96 → not significant at 5%; larger samples (e.g., ~3,000+ per arm) to detect this lift.
Exam prep key points: 1: Descriptive, diagnostic, predictive, prescriptive. • 2: WHERE filters rows; HAVING filters groups. • 3: Probability of data this extreme if no true effect. • 4: Balances groups so differences are caused by the change. • 5: Hypothesis, random split, primary KPI, fixed n, significance test, avoid peeking, practical significance.

5. AI for Smarter Business

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: ₹6,000, ₹3,600, ₹2,400
Part A 2: Churn - Classification; Pricing - Demand elasticity; Basket - Association rules; Fraud - Anomaly detection
Part B 1: ₹3,000, 2, 3%
Part B 2: TP: saved a real churner: Gain; FP: offer to a loyal customer: Loss; FN: missed a churner: Loss; TN: correctly left alone: Gain
Challenge hints: Q1: Error costs differ; profit depends on TP gains vs FP & FN costs. • Q2: Churn: who will leave; uplift: whose behaviour changes if treated. • Q3: CLV ₹3,000; ratio 2. • Q4: Generally not; exploitation risk; regulation & fairness concerns. • Q5: Churn rate, saves, offer cost, net profit, NPS.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 200; t=0.75: tp 2, fp 0 → 3600; t=0.55: tp 3, fp 1 → 5200; 0.55 better (more saves outweigh one wasted offer); loop with max().
Exam prep key points: 1: Churn: rate customers leave; CLV: total value of a customer over lifetime. • 2: Assigns money values to TP, FP, FN, TN. • 3: Churn predicts leaving; uplift predicts response to treatment. • 4: Error costs at scale, unfair pricing, poor adoption, drift. • 5: Scores, cost matrix, threshold choice, expected profit, monitoring; numbers.

6. AI for Engineering Solutions

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Aerospace - Engine health; Civil - Crack detection; Electrical - Load forecasting; Manufacturing - Defect detection
Part A 2: 10, 20, 5
Part B 1: 15, 8, 30
Part B 2: Change oil every 6 months: Prevent.; Replace bearing when vibration trend crosses limit: Predict.; Service lift every quarter: Prevent.; Schedule repair from AI RUL estimate: Predict.
Challenge hints: Q1: Real assets drift from design; live data keeps the twin accurate. • Q2: Preventive: fixed schedule; predictive: condition-based, fewer wasted repairs. • Q3: (90 - 45) / 3 = 15 days. • Q4: No: legal & safety duty; engineers must verify & sign off. • Q5: Flow, pressure, vibration sensors → anomaly model → SMS to staff.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 1; slope ≈ 3.0, RUL ≈ (90 - 52)/3 ≈ 12.7 days; wear often accelerates (non-linear); safe = 0.8 × RUL ≈ 10.1.
Exam prep key points: 1: Live virtual replica of a physical asset using real data. • 2: Remaining useful life of a component. • 3: Schedule-based vs condition-based. • 4: Fast ML approximation of a slow simulation. • 5: Sensors, features, anomaly/RUL models, alerts, work orders, validation, human sign-off.

7. Pick Your AI Path

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: NLP - Tamil sentiment; Vision - Helmet detection; MLOps - CI/CD pipeline; Data science - Sales drivers
Part A 2: 60, 48, 27
Part B 1: 48, 75, Analytics
Part B 2: NLP - Tokenisers & transformers; Vision - OpenCV & CNNs; MLOps - Docker & CI/CD; Analytics - SQL & dashboards
Challenge hints: Q1: They support every AI path, enabling later pivots. • Q2: NLP: models for text; MLOps: deploy, monitor, automate pipelines. • Q3: CV 48 vs Analytics 75 → Analytics. • Q4: No: interest & strength drive persistence and long-term success. • Q5: Sem 3 foundations, Sem 4 specialised elective + project, Sem 5 internship.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: w; NLP 4.2, Vision 3.8, MLOps 3.0, Analytics 4.0; winner NLP; with (0.2, 0.3, 0.5): Analytics 4.6 beats NLP 3.9.
Exam prep key points: 1: Data science, NLP, computer vision, MLOps, research, policy. • 2: Practices to deploy, monitor and maintain ML in production. • 3: Maths, Python, data underpin every AI path. • 4: Electives, focused projects, internships, communities. • 5: Interest × strength × demand, values, electives, projects, internship, pivot plan.

8. Portfolio to Profession

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 1-Problem & context, 2-Data & challenges, 3-Approach & model, 4-Results & impact
Part A 2: README with problem, results, demo link: Strong; Repo with only 'final_final2.ipynb': Weak; 2-min demo video: Strong; Forked tutorial with no changes: Weak
Part B 1: 6.6, 44%, 25%
Part B 2: Technical - Stats/ML/coding; Project - Decisions & trade-offs; Behavioural - STAR; Take-home - Small task
Challenge hints: Q1: Quick evidence of impact & clarity; saves their time. • Q2: Resume: 1-line summary; case study: full story, decisions, metrics. • Q3: (15 - 8.4) / 15 = 44%. • Q4: No: state your role honestly. • Q5: E.g., 'Aspiring NLP Engineer | Built Tamil sentiment model (F1 0.86)'.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: baseline_mape; prints 44 ['crop', 'vision']; stars reflect popularity, not quality; 'Reduced forecast error by 44% vs baseline'.
Exam prep key points: 1: Problem, data, approach, results, limitations, role, demo. • 2: Quick, clear evidence for reviewers. • 3: Technical, project, behavioural, take-home/case. • 4: Recommendation by someone inside the company. • 5: Top 3 projects, case studies, profile, networking, prep plan, offer comparison.

9. AI Data Studio	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Trend - Line; Region - Bar; Relation - Scatter; Spread - Histogram / box plot **Part A 2:** Bar chart y-axis starts at zero: Good; 3D pie chart with 9 slices: Misleading; Title states the key insight: Good; Rainbow colours for ordered data: Misleading **Part B 1:** Position, Angle, Colour **Part B 2:** Circle: Axis labels with units, Bars start at zero, Colour-blind-safe palette, Source cited **Challenge hints:** Q1: Position/length are judged more accurately than angles/areas. • Q2: Histogram: full shape; box plot: median, quartiles, outliers compactly. • Q3: Truncation exaggerates differences visually. • Q4: Usually not: overload; focus on key KPIs & questions. • Q5: Tiles: placed %, median CTC, top recruiters, trend; filters: branch, year. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: mean; all ≈ 0.82 (and means x 9.0, y ≈ 7.5); plots differ: linear, curve, outlier, vertical cluster; sns.boxplot(data=df, x='dataset', y='y'). **Exam prep key points:** 1: Position > length > angle > area > colour. • 2: Bar: length judged more accurately. • 3: Four datasets, same stats, different plots. • 4: Zero baseline for bars, labelled axes, no 3D distortion. • 5: Purpose, KPIs, chart types, filters, colours, refresh, audience.
10. Smart Automation Tools	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 20, ₹10,000, 2 **Part A 2:** Copy 500 invoices into a ledger daily: Good; Negotiating a contract: Poor; Weekly attendance report: Good; Counselling a stressed student: Poor **Part B 1:** 0 9 * * * - Daily 9:00; 0 18 * * 5 - Fridays 18:00; 0 0 1 * * - Monthly 1st; */15 - Every 15 min **Part B 2:** 1-Test in a sandbox, 2-Add limits & kill switch, 3-Release to a small pilot, 4-Monitor & review weekly **Challenge hints:** Q1: More stable, faster, secure; UI changes break RPA bots. • Q2: Cron: flexible, needs coding; no-code: quick, limited, vendor lock-in. • Q3: Every Friday at 18:00. • Q4: Risky: errors harm trust; require human review. • Q5: New form → validate → update sheet → email; log; stop on 3 errors. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: MAX_ERRORS; prevents cascading damage; add timeouts; for d in [1,2,4]: try … except: time.sleep(d). **Exam prep key points:** 1: Repetitive, rule-based, high-volume, stable, digital. • 2: Software bots mimicking user actions on screens. • 3: 9:00 on weekdays. • 4: Stop runaway automation quickly to limit damage. • 5: Testing, staged rollout, limits, kill switch, logging, alerts, ownership, reviews.

11. Fair AI, Fair Future

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 0.8, 0.6, Fails
Part A 2: Parity - Equal selection rates; Equal opp. - Equal TPR; Equalised odds - Equal TPR & FPR; Calibration - Same score, same risk
Part B 1: 0.13, 0.30, Equalised odds
Part B 2: Reweight under-represented samples: Pre/In; Use group-specific thresholds after training: Post; Add fairness penalty to loss: Pre/In; Adjust output scores after prediction: Post
Challenge hints: Q1: Groups can still have very different false-positive rates. • Q2: Pre: fix data; in: constrain training; post: adjust thresholds. • Q3: A: 0.8, 0.13; B: 0.6, 0.30. • Q4: Debated: can equalise errors but raises legal & fairness-of-treatment issues. • Q5: Purpose, groups, metrics, data, risks, mitigation, appeals, review.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: tn; A: 0.30, 0.80, 0.13; B: 0.375, 0.60, 0.30; equal opportunity & equalised odds fail; parity ratio 0.30/0.375 = 0.8.
Exam prep key points: 1: Equal selection rates across groups. • 2: Equal TPR and FPR across groups. • 3: Scores mean same actual risk across groups. • 4: Pre-, in-, post-processing. • 5: ProPublica FPR vs calibration; base rates; values, stakeholders, transparency, appeals.

12. Ownership in the AI Age

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-F 5-T 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: CC-BY - Reuse with credit; MIT - Permissive, keep notice; GPL - Copyleft; Proprietary - Explicit permission
Part A 2: Using a CC-BY image with credit: OK; Cloning a singer's voice for an ad without consent: Risky; Using MIT code and keeping its notice: OK; Selling copies of a textbook made with AI rewording: Risky
Part B 1: BY - Give credit; SA - Share alike; NC - Non-commercial; ND - No derivatives
Part B 2: Circle: Credit a CC-BY photo, Read a model's licence before commercial use, Document your human edits, Keep MIT licence notice in code
Challenge hints: Q1: NC prohibits commercial use without permission. • Q2: MIT: permissive; GPL: derivatives must be GPL (copyleft). • Q3: 'Sunrise' by R. Devi, licensed under CC BY 4.0 (link). • Q4: Arguments: consent & compensation vs fair use/innovation; evolving law. • Q5: Disclose AI use, use licensed assets, credit creators, no voice/face clones.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: commercial_ok; flagged ['song.mp3', 'img.jpg']; NC = non-commercial only; for n, l in assets: if l == 'CC-BY': print(f'{n} — CC BY, credit author').
Exam prep key points: 1: Copyright protects human creativity; AI isn't a legal person. • 2: MIT permissive; GPL copyleft. • 3: BY: give credit; NC: non-commercial only. • 4: Protection of a person's name, voice, likeness. • 5: Thaler, training lawsuits, licences, human contribution, personality rights; policy views.

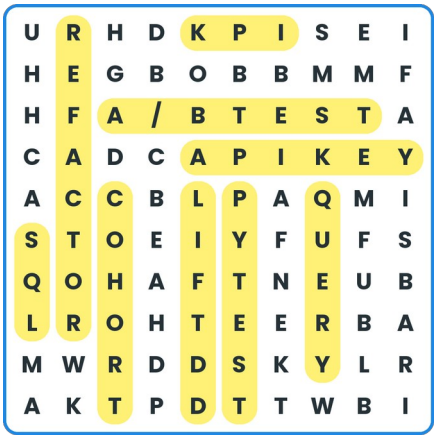
Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

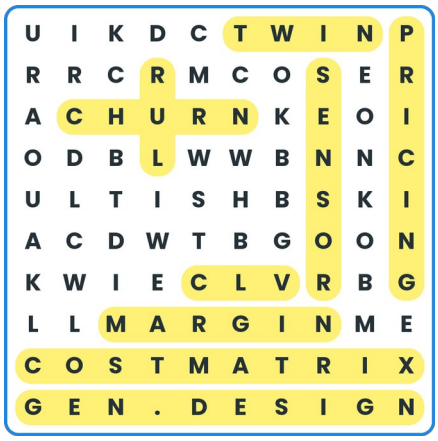
Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

WORD SEARCH ANSWERS

AI DISCOVER

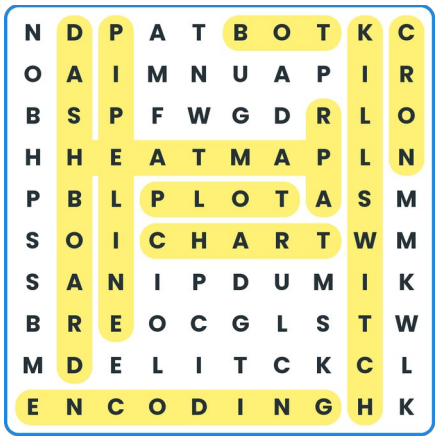


AI CONNECT

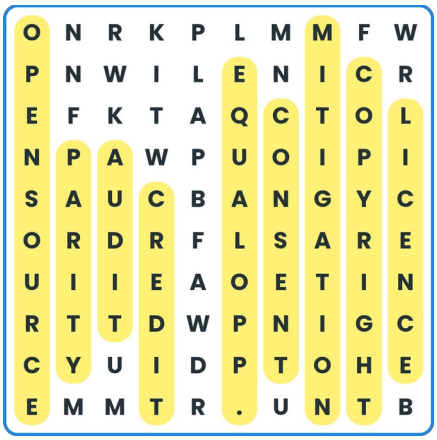


AI SOLVE

AI RISE



AI CREATE

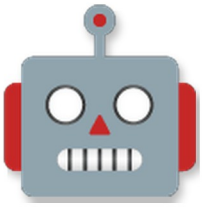


AI CARE

MAMMA ADA CENTRE

GAIO • Global AI Olympiad

From Data to Decisions for a Smarter Future



About this book

A university-level AI book for UG 2nd Year students: concepts, code labs, stories & exam prep.

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- 🐍 Python Code Labs & lab activities
- 📚 AI Stories & real-world case studies
- 🧠 Challenge problems & exam prep
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